



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R17)

I/IV B.TECH

(With effect from 2017-2018 Admitted Batch onwards)

Under Choice Based Credit System

GROUP-B (CSE, ECE & IT)

I-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/ Week	Internal Marks	External Marks	Total Marks
B17 BS 1101	English – I*	3	3	1	--	4	30	70	100
B17 BS 1102	Mathematics – I*	3	3	1	--	4	30	70	100
B17 BS 1103	Mathematics-II	3	3	1	--	4	30	70	100
B17 BS 1104	Engineering Physics	3	3	1	--	4	30	70	100
B17 CS 1101	Computer Programming Using C	3	3	1	--	4	30	70	100
B17 CE 1101	Environmental Studies *	2	2	1	--	3	30	70	100
B17 BS 1106	Engineering Physics Lab	2	--	--	3	3	50	50	100
B17 BS 1108	English Communication Skills Lab – I *	2	--	--	3	3	50	50	100
# DL1	Department Lab	2	--	--	3	3	50	50	100
B17 BS 1110	Engineering Physics Virtual Labs-Assignments	--	--	--	2	2	--	--	--
B17 BS 1112	NCC	--	--	--	2	2	--	--	--
Total		23	17	6	13	36	330	570	900

* Common to both Group - A and Group - B

#DL 1	CSE & IT	B17 CS 1102	C Programming Lab & Hardware Fundamentals
	ECE	B17 CS 1103	C Programming Lab

Code: B17 BS 1101

ENGLISH - I
(Common to all Branches)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To improve the language proficiency of the students in English with emphasis on LSRW skills.
2. To enable the students to study and comprehend the prescribed lessons and subjects more effectively relating to their theoretical and practical components.
3. To develop the communication skills of the students in both the formal and informal situations.
4. To expose the students to various forms of writing in formal settings in order to build confidence in standard grammar
5. To introduce the students various types of reading comprehension practices.

Course Outcomes: At the end of the Course, Student will be able to:

1. Understand the rudiments of LSRW Skills, comprehension and fluency of speech.
2. Gain confidence and competency in vocabulary and grammar.
3. Listen, speak, read and write effectively in both the academic and non- academic environment.
4. Extend his/her reading skills towards literature.
5. Strengthen his/her analytical and compositional skills.

SYLLABUS

Life through Language: An Effective Learning Experience

Life through Language has a systematic structure that builds up communicative ability progressively through the chapters. It will enable the learner to manage confusion; frame question for themselves and others; develop new ideas; support ideas with evidence; express themselves with poise and clarity; and think critically. Acquisition of skill leads to confidence.

UNIT-I

People and Places:- Word search - Ask yourself-Self-assessment-I -Self-assessment-II - Sentence and its types- Describing people, places and events-Writing sentences-Self-awareness-Self-motivation, Dialogue writing.

UNIT-II

Personality and Lifestyle:- Word quiz – Verbs-Adverbs-Negotiations-Proving yourself-Meeting Carl Jung- Describing yourself- Living in the 21st century- Using your dictionary-Communication-Adaptability.

UNIT-III

Media and Environment: - A list of 100 basic words – Nouns- Pronouns- Adjectives-News report- Magazine article- User's Manual for new iPod- A documentary on the big cat- Why we need to save our tigers: A dialogue- Global warming- Paragraph Writing-Arguing a case-Motivation- Problem solving.

UNIT-IV

Entertainment and Employment:- One word substitutes- Parts of speech- Gerunds and infinitives- An excerpt from a short story an excerpt from a biography- A consultant interviewing employees- Your first interview- Reality TV- Writing an essay-Correcting sentences- Integrity Sense of humor.

UNIT-V

Work and Business:- A list of 100 difficult words- Articles, Quantifiers- Punctuation - Open letter to the Prime Minister Business dilemmas: An email exchange- A review of *IPL: The Inside Story*, Mark Zuckerberg: World's Youngest Billionaire- A conversation about a business idea- Pair work: Setting up a new business- Recession- Formal letters-Emails- Reports- Professionalism-Ethics, Fill in the blanks.

Text Book:

1. *Life through Language: A Holistic Approach to Language Learning*. Board of Editors, Pearson Publishers, India. 2013.

Reference Books:

1. Basic Vocabulary. Edgar Thorpe, Showick Thorpe. Pearson P. 2008.
2. Quick Solutions to Common Errors in English, Angela Bunt. MacMillan P. 2008.
3. Know Your English (Volume 1&2), by Dr. S. Upendra, Universities Press, India 2012
4. Business Communication Strategies. Mathukutty Monippally. Tata Mc Grahill P. 2009.

MATHEMATICS - I
(Common to all Branches)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. The course is designed to equip the students with the necessary mathematical skills and techniques that is essential for an engineering course.
2. The skills derived from the course will help the student form a necessary base to develop analytic and design concepts.
3. Learn about linear first order and higher order ordinary differential equations and their applications.
4. Acquire knowledge of Laplace transform, partial differentiation and their applications.
5. Learn certain first order and higher order partial differential equations.

Course Outcomes: At the end of the Course, Student will be able to:

1. Solve linear ordinary differential equations of first order and first degree. Also will be able to apply the knowledge in simple applications such as Newton's law of cooling, orthogonal trajectories and simple electrical circuits.
2. Solve linear ordinary differential equations of second order and higher order. Also will be able to apply the knowledge in simple applications such as LCR circuits and Simple harmonic motion.
3. Determine Laplace transform and inverse Laplace transform of various functions.
4. Use Laplace transforms to solve a linear ODE.
5. Calculate total derivative, Jacobian and maxima/minima of functions of two variables.
6. Form partial differential equations and solve some standard types of first order PDEs. Find complimentary function and particular integral of linear higher order homogeneous and non-homogeneous PDEs.

SYLLABUS

UNIT I: Differential equations of first order and first degree:

Linear, Bernoulli, Exact, Reducible to exact types.

Applications: Newton's Law of cooling, Law of natural growth and decay, Orthogonal trajectories, Simple electrical circuits, Chemical reactions.

UNIT II: Linear differential equations of higher order:

Non-homogeneous equations of higher order with constant coefficients with RHS term

of the type e^{ax} , $\sin ax$, $\cos ax$, polynomials in x , $e^{ax}V(x)$, $xV(x)$, Method of Variation of parameters.

Applications: LCR circuit, Simple Harmonic motion.

UNIT III: Laplace transforms:

Laplace transforms of standard functions, transforms of $tf(t)$, $f(t)/t$, properties, transforms of derivatives and integrals, transforms of unit step function, Dirac delta function, Inverse Laplace transforms, convolution theorem (without proof).

Applications: Solving ordinary differential equations (initial value problems) using Laplace transforms.

UNIT IV: Partial differentiation:

Introduction, Homogeneous functions, Euler's theorem, Total derivative, Chain rule, which variable is to be treated as constant, Functional dependence, Jacobians, Taylor series for a function of two variables, Leibnitz rules for differentiation under the integral sign.

Applications: Errors and Approximations, Maxima and Minima of functions of two variables without constraints, Lagrange's method (with constraints)

UNIT V: First order and higher order partial differential equations:

Formation of partial differential equations by elimination of arbitrary constants and arbitrary functions, solutions of first order Lagrange linear equation and nonlinear equations of standard types (excluding Charpit's method).

Solutions of Linear homogeneous and non-homogeneous Partial differential equations with constant coefficients - RHS terms of the type e^{ax+by} , $\sin(ax+by)$, $\cos(ax+by)$, $x^m y^n$.

Text Books:

1. B.S.Grewal, Higher Engineering Mathematics, 43rd Edition, Khanna Publishers.
2. N.P.Bali & Manish Goyal, A Text book of Engineering Mathematics, Lakshmi Publications.

Reference Books:

1. Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, Wiley-India.
2. Michael Greenberg, Advanced Engineering Mathematics, 9th edition, Pearson.
3. Dean G. Duffy, Advanced engineering mathematics with MATLAB, CRC Press.
4. Peter O'Neil, Advanced Engineering Mathematics, Cengage Learning.
5. Srimanta Pal, Subodh C.Bhunia, Engineering Mathematics, Oxford University Press.
6. Dass H.K., Rajnish Verma. Er., Higher Engineering Mathematics, S. Chand Co. Pvt. Ltd, New Delhi.

MATHEMATICS – II
(Common to CSE, ECE& IT)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. The course is designed to equip the students with the necessary mathematical skills and techniques that are essential for an engineering course.
2. The skills derived from the course will help the student form a necessary base to develop analytic and design concepts.
3. Understand some basic numerical methods to solve algebraic and transcendental equations.
4. Learn interpolation methods for equally spaced and unequally spaced data points. Learn methods for numerical evaluation of integrals and for solving first order ODEs.
5. Learn concepts of a Fourier series and Fourier transform.

Course Outcomes: At the end of the Course, Student will be able to:

1. Find a real root of algebraic and transcendental equations using different methods.
2. Know the relation between the finite difference operators. Determine interpolation polynomial for a given data.
3. Evaluate numerically certain definite integrals applying Trapezoidal and Simpson's rules.
4. Solve a first order ordinary differential equation by Euler and RK methods.
5. Find Fourier series of a given function satisfying Dirichlet conditions. Find half range cosine and sine series for appropriate functions.
6. Find Fourier transforms, Fourier cosine and sine transforms of appropriate functions and evaluate certain integrals using inverse transforms and Fourier integral.

SYLLABUS

UNIT I: Solution of Algebraic and Transcendental Equations:

Introduction, Bisection method, Method of false position, Iteration method, Newton- Raphson method (One variable and simultaneous Equations).

UNIT II: Interpolation:

Introduction, Errors in polynomial interpolation, Finite differences, Forward differences, Backward differences, Central differences and Symbolic relations between the operators, Differences of a polynomial, Newton's formulae for interpolation, Interpolation with unequal intervals, Lagrange's interpolation formula.

UNIT III: Numerical Integration and solution of Ordinary Differential equations:

Trapezoidal rule, Simpson's $1/3^{\text{rd}}$ and $3/8^{\text{th}}$ rules, Solution of ordinary differential equations by Taylor series method, Picard's method of successive approximations, Euler's method, Runge-Kutta methods (second order and fourth order).

UNIT IV: Fourier Series:

Introduction, Periodic functions, Fourier series of a periodic function, Dirichlet's conditions, Even and odd functions, Change of interval, Half-range sine and cosine series, Parseval's formula.

UNIT V: Fourier Transforms:

Fourier integral theorem (without proof), Complex form of Fourier integral, Fourier sine and cosine integrals, Fourier transform, Fourier sine and cosine transforms, properties, inverse transforms, Parseval's identities, Finite Fourier transforms.

Text Books:

1. B.S.Grewal, Higher Engineering Mathematics, 43rd Edition, Khanna Publishers.
2. N.P.Bali & Manish Goyal, A Text book of Engineering Mathematics, Lakshmi Publications.

Reference Books:

1. Dean G. Duffy, Advanced engineering mathematics with MATLAB, CRC Press.
2. V.Ravindranath and P. Vijayalakshmi, Mathematical Methods, Himalaya Publishing House.
3. Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, Wiley-India.
4. David Kincaid, Ward Cheney, Numerical Analysis-Mathematics of Scientific Computing, 3rd Edition, Universities Press.
5. Srimanta Pal, Subodh C. Bhunia, Engineering Mathematics, Oxford University Press.
6. Dass H.K., Rajnish Verma. Er., Higher Engineering Mathematics, S. Chand Co. Pvt. Ltd, New Delhi.

ENGINEERING PHYSICS
(Common to CSE, ECE& IT)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

To re-orient Physics curriculum to the needs of the circuit and non circuit branches of Engineering / Technology courses offered by SRKREC(A) Bhimavaram that serves to understand the general and branch specific basic topics and introduce the related advanced technologies to the stake holders.

1. To impart the knowledge of physical optics phenomena like Interference and Diffraction required to understand the design and working of optical instruments with high resolution.
2. To understand the concept of coherence and generation of high intensity light sources and propagation of light waves in fibers for technological applications.
3. To study and analyze the behavior of electric and magnetic fields and their mutual interaction, and to understand the science of ultrasonics.
4. To understand the basic concepts of Quantum mechanics to know the behavior of fundamental particles in materials and to understand the basic nature of materials in general and classify them.
5. To study the structure property relationship of crystalline solids and understand basics of nanotechnology.

Course Outcomes: At the end of the course the students-

1. Learn the basic concepts of interference and diffraction of light and its applications.
2. Understand the science of producing high intensity light beams for technological applications and also understand the propagation of light waves in optical fibers in various applications.
3. Understand the inter relationship of electric and magnetic fields and learn ultrasonic's as a tool for technological applications
4. Learn the behavior of particles at the very microscopic level by using wave nature of particles and understand the behavior of materials and be able to classify them using the band theory of solids.
5. Learn the basics of structures of solid materials and nano material preparation Techniques/methods.

SYLLABUS

UNIT I: Interference and Diffraction

Principle of superposition-coherence-interference in thin films (reflected system) – Wedge shaped film-Newton's rings-Michelson's interferometer. Fraunhofer's diffraction at single slit, Diffraction grating-Resolving power of a grating.

UNIT- II: Lasers and Optical Fibers

Introduction, Spontaneous emission and Stimulated emission – Einstein's relation – Requirements of Laser device- Ruby laser- He-Ne gas laser- Characteristics of laser- Applications.

Description of optical fiber, Principle of light propagation- Optical fiber –Acceptance angle- Numerical aperture of optical fiber- Modes of propagation- Classification of fibers- Applications of fiber.

UNIT- III: Electro Magnetic Fields and Ultrasonics

Concept of Electromagnetic induction , Faraday's law, Lenz's law, Electric fields due to time varying magnetic fields, Magnetic fields due to time varying electric fields, Displacement current, Modified Ampere's law, Maxwell's equations and their significance (without derivation).

Definition of Ultrasonics-Methods of Producing Ultrasonics- Detection of Ultrasonics- Applications of Ultrasonics.

UNIT- IV: Quantum Mechanics and Band Theory of Solids

Introduction, de Broglie matter waves- properties-Experimental confirmation, wave function-significance- Schrodinger's time dependent and time independent wave equations- Eigen values and functions, Particle in a box.

Band theory of Solids- Introduction- Kronig Penney model (Qualitative)- Energy bands of crystalline solids- Distinction between Conductors, Semi conductors and insulators.

UNIT-V: Crystallography and Nano Materials

Basis and Lattice, Crystal systems, Bravais lattice, Unit cell Coordination number – Packing fraction for SC ,FCC, and BCC lattices, Miller indices- Diffraction of X rays from crystals- Bragg's law.

Introduction to Nanomaterials – Synthesis methods : Condensation, ball milling, sol-gel, chemical vapour deposition methods, properties and applications.

Text Books:

1. Physics by Resnick & Halliday. Wiley - Eastern (India) Ltd.
2. Engineering Physics By M.N. Avadhanulu & PG Kshirasagar, S.Chand & Co.
3. Engineering Physics by V. Rajendran, Mc Graw Hill Education (India) Pvt.Ltd.

Reference Books:

1. Engineering Physics by MR Srinivasan, New International Publishers.
2. Solid State Physics – A.J. Dekkar, MacMillan (India) Ltd.
3. Engineering Physics by Gaur Gupta, Dhanpat Rai Publications, Meerut, India.
4. Engineering Physics by P k Palanisamy, Scitech Publications (India) Pvt.Ltd

(Note: Assignment Marks of **Engineering Physics** are to be considered from the Internal marks of **Engineering Physics-- Virtual Labs – Assignments B17 BS 1110**)

COMPUTER PROGRAMMING USING C
(Common to CSE, ECE & IT)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

Formulating algorithmic solutions to problems and implementing algorithms in C

1. Notion of Operation of a CPU, Notion of an algorithm and computational procedure, editing and executing programs in Linux
2. understanding branching, iteration and data representation using arrays
3. Modular programming and recursive solution formulation
4. Understanding pointers and dynamic memory allocation
5. Understanding miscellaneous aspects of C
6. Comprehension of file operations

Course Outcomes:

1. Understand the basic terminology used in computer programming
2. Write, compile and debug programs in C language.
3. Use different data types in a computer program.
4. Design programs involving decision structures, loops and functions.
5. Explain the difference between call by value and call by reference
6. Understand the dynamics of memory by the use of pointers
7. Use different data structures and create/update basic data files.

SYLLABUS

UNIT I:

Unit objective: Notion of Operation of a CPU, Notion of an algorithm and computational procedure, editing and executing programs in Linux

Introduction: Computer systems, Hardware and Software Concepts.

Problem Solving: Algorithm / Pseudo code, flowchart, program development steps, computer languages: machine, symbolic and highlevel languages, Creating and Running Programs: Writing, Editing(vi/emacs editor), Compiling(gcc), Linking and Executing in under Linux.

BASICS OF C: Structure of a c program, identifiers, basic data types and sizes. Constants, Variables, Arithmetic , relational and logical operators, increment and decrement operators, conditional operator, assignment operator, expressions, type conversions, Conditional Expressions, precedence and order of evaluation, Sample Programs.

UNIT II:

Unit objective: understanding branching, iteration and data representation using arrays

SELECTION – MAKING DECISION: TWO WAY SELECTION: if-else, null else, nested if, examples, Multi-way selection: switch, else-if, examples.

ITERATIVE: loops- while, do-while and for statements , break, continue, initialization and updating, event and counter controlled loops, Looping applications: Summation, powers, smallest and largest.

ARRAYS: Arrays- concepts, declaration, definition, accessing elements, storing elements, Strings and String Manipulations, 1-D arrays, 2-D arrays and character arrays, string manipulations, Multidimensional arrays, array applications: Matrix operations, checking the symmetricity of a Matrix. **STRINGS: concepts, c strings.**

UNIT III:

Objective: Modular programming and recursive solution formulation

FUNCTIONS- MODULAR PROGRAMMING: functions, basics, parameter passing, storage classes extern, auto, register, static, scope rules, block structure, user defined functions, standard library functions, recursive functions, Recursive solutions for fibonacci series, towers of Hanoi, header files, C Preprocessor, example c programs, Passing 1-D arrays, 2-D arrays to functions.

UNIT IV:

Objective: Understanding pointers and dynamic memory allocation

POINTERS: pointers- concepts, initialization of pointer variables, pointers and function arguments, passing by address- dangling memory, address arithmetic, character pointers and functions, pointers to pointers, pointers and multi-dimensional arrays, dynamic memory management functions, command line arguments

UNIT V:

Objective: Understanding miscellaneous aspects of C

ENUMERATED, STRUCTURE AND UNION TYPES: Derived types- structures declaration, definition and initialization of structures, accessing structures, nested structures, arrays of structures, structures and functions, pointers to structures, self referential structures, unions, typedef, bit-fields, program applications

BIT-WISE OPERATORS: logical, shift, rotation, masks.

Objective: Comprehension of file operations

FILEHANDLING: Input and output- concept of a file, text files and binary files, Formatted I/O, File I/O operations, example programs

Text Books:

1. Problem Solving and Program Design in C, Hanly, Koffman, 7thed, PERSON
2. Programming in C, Second Edition PradipDey and Manas Ghosh, OXFORD Higher Education
3. Programming in C, A practical approach Ajay Mittal PEARSON
4. The C programming Language by Dennis Richie and Brian Kernighan
5. Programming in C, B. L. Juneja, Anith Seth, Cengage Learning.

Reference Books:

1. C Programming, A Problem Solving Approach, Forouzan, Gilberg, Prasad, CENGAGE
2. Programming with C, Bichkar, Universities Press
3. Programming in C, ReemaThareja, OXFORD
4. C by Example, Noel Kalicharan, Cambridge

ENVIRONMENTAL STUDIES
(Common to all Branches)

Lecture	: 2 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 2

Course Objectives:

1. Developing an awareness and sensitivity to the total environment and its related problems.
2. Motivating students for active participation in environmental protection and improvement.
3. Developing skills for active identification and development of solutions to environmental problems
4. Evaluation of environment programmes in terms of social, economic, ecological and aesthetic factors.
5. Creating a “CONCERN AND RESPECT FOR THE ENVIRONMENT”.

Course Outcomes:

1. To bring awareness among the students about the nature and natural ecosystems
2. Sustainable utilization of natural resources like water, land, energy and air
3. Resource pollution and over exploitation of land, water, air and catastrophic (events) impacts of climate change, global warming, ozone layer depletion, marine, radioactive pollution etc to inculcate the students about environmental awareness and safe transfer of our mother earth and its natural resources to the next generation
4. Safe guard against industrial accidents particularly nuclear accidents
5. Constitutional provisions for the protection of natural resources

SYLLABUS

UNIT – I**Global Environmental Crisis:**

Environmental Studies - Definition, Scope and importance, Need for public awareness. Global Environmental Crisis

Ecosystems:

Basic Concepts - Structure and Functions of Ecosystems: Producers, Consumers and Decomposers. Types of Ecosystems: Forest Ecosystems, Grassland Ecosystems Desert Ecosystems and Aquatic Ecosystems

UNIT-II**Biodiversity:**

Introduction to Biodiversity, Values of Bio-diversity, Bio-geographical classification of India, India as a Mega-diversity habitat, Threats to biodiversity, Hotspots of Biodiversity, Conservation of Biodiversity: In-situ and Ex-situ conservation of Biodiversity.

UNIT-III

Environmental and Natural Resources Management:

Land Resources: Land degradation, soil erosion and desertification, Effects of modern agriculture. **Forest Resources:** Use and over exploitation-Mining and Dams-their effects on forest and tribal people. **Water resources:** Use and over utilization of surface and ground water, Floods, droughts, conflict over water, water logging and salinity, dams – benefits and problems. **Energy Resources:** Renewable and non-renewable energy sources, use of alternate energy sources-impact of energy use on environment.

UNIT-IV

Environmental Pollution:

Causes, Effects and Control measures of - Air pollution, Water pollution, Soil pollution, Marine Pollution, Thermal pollution, Noise pollution, Nuclear Hazards; Climate change and Global warming, Acid rain and Ozone layer depletion. Solid Waste Management: Composting, Vermiculture, Urban and Industrial Wastes, Recycling and Reuse.

Environmental Problems in India:

Drinking water, Sanitation and Public health, Population growth and Environment; Water Scarcity and Ground Water Depletion; Rain water harvesting, Cloud seeding and Watershed management.

UNIT-V

Institutions and Governance:

Regulations by Government- Environmental Protection Act, Air (Prevention & Control of Pollution) Act, Water (Prevention and control of Pollution) Act, Wildlife Protection Act, Forest Conservation Act. Environmental Impact Assessment (EIA)

Case Studies:

Chipko Movement, Narmada Bachao Andolan, Silent Valley Project, Mathura Refinery and Taj Mahal, Industrialization of Patancheru, Nuclear reactor at Nagarjuna Sagar, Tehri Dam, Ralegaon Siddhi (Anna Hazare), Kolleru lake – Aquaculture, Fluorosis in Andhra Pradesh & Telangana.

Field Work:

Visit to a local area to document and mapping environmental assets. Visits to Industries, Water Treatment Plants, Effluent Treatment Plants.

Text Books:

1. Environmental Studies, K. V. S. G. Murali Krishna, VGS Publishers, Vijayawada
2. Environmental Studies, R. Rajagopalan, 2nd Edition, 2011, Oxford University Press.
3. Environmental Studies, Dr. M. Sita Rama Reddy, Dr. K. Anji Reddy, Silicon Publications, ASR Nagar, Bhimavaram
4. Environmental Studies, P. N. Palanisamy, P. Manikandan, A. Geetha, and K. Manjula Rani; Pearson Education, Chennai.

Reference Books:

1. Text Book of Environmental Studies, Deeshita Dave & P. Udaya Bhaskar, Cengage Learning.
2. A Textbook of Environmental Studies, Shaashi Chawla, TMH, New Delhi
3. Environmental Studies, Benny Joseph, Tata McGraw Hill Co, New Delhi
4. Perspectives in Environment Studies, Anubha Kaushik, C P Kaushik, New Age International Publishers, 2014

**ENGINEERING PHYSICS LAB
(Common to CSE, ECE& IT)****Lab : 3 Periods**
Exam : 3 Hrs.**Int. Marks : 50**
Ext. Marks : 50
Credits : 2

Course Objectives:

Training the field oriented Engineering graduates to handle instruments and their design methods to improve the accuracy of measurements.

1. To impart hands-on experience to the students entering engineering/Technology education about handling sophisticated equipment/ instruments.
2. To make the students understand the theoretical aspects of various phenomena experimentally.

Course Outcomes:

Physics lab curriculum gives fundamental understanding of design of an instrument with targeted accuracy for physical measurements.

1. Students get hands on experience in setting up experiments and using the instruments/equipment individually.
2. Get introduced to using new/ advanced technologies and understand their significance.

**LIST OF EXPERIMENTS
(Any 10 of the following listed experiments)**

1. Determination of wavelength of a source-Diffraction Grating-Normal incidence.
2. Newton's rings – Radius of Curvature of Plano - Convex Lens.
3. Determination of thickness of a spacer using wedge film and parallel interference fringes.
4. Determination of Rigidity modulus of a material- Torsional Pendulum.
5. Determination of Acceleration due to Gravity and Radius of Gyration-Compound Pendulum.
6. Melde's experiment – Transverse and Longitudinal modes.
7. Verification of laws of vibrations in stretched strings – Sonometer.
8. Determination of velocity of sound – Volume Resonator.
9. L- C- R Series Resonance Circuit.
10. Study of I/V Characteristics of Semiconductor diode.
11. I/V characteristics of Zener diode.
12. Characteristics of Thermistor – Temperature Coefficients.
13. Magnetic field along the axis of a current carrying coil – Stewart and Gee's apparatus.
14. Energy Band gap of a Semiconductor p - n junction.
15. Hall Effect in semiconductors.
16. Time constant of CR circuit.
17. Determination of wavelength of laser source using diffraction grating.
18. Determination of Young's modulus by method of single cantilever oscillations.
19. Determination of lattice constant – lattice dimensions kit.
20. Determination of Planck's constant using photocell.
21. Determination of surface tension of liquid by capillary rise method.

Reference Book:

1. Advanced Practical Physics –Vol 1 & 2 By S.Venkata Raman, S.Chand & Co.

ENGLISH COMMUNICATIONSKILLS LAB- I
(Common to All Branches)

Lab	: 3 Periods	Int.Marks	: 50
Exam	: 3 Hrs.	Ext. Marks	: 50
		Credits	: 2

Course Objectives:

1. To enable the students to learn through practice the communication skills of listening, speaking, reading and writing.
2. To make students recognize the sounds of English through Audio- Visual aids.
3. To familiarize the students with stress and intonation.
4. To help the students build their confidence in speaking skills.

Course Outcomes:

1. A study of the communicative items in the laboratory will help the students become successful in the competitive world.
2. Students improve their speaking skills in real contexts.
3. Students learn standard pronunciation and practice it daily discourse.
4. Students give up their communicative barriers.

SYLLABUS

- ❖ WHY study Spoken English?
- ❖ Making Inquiries on the phone, thanking and responding to Thanks - Practice work.
- ❖ Responding to Requests and asking for Directions - Practice work.
- ❖ Asking for Clarifications, Inviting, Expressing Sympathy, Congratulating
- ❖ Apologising, Advising, Suggesting, Agreeing and Disagreeing - Practice work.
- ❖ Letters and Sounds-Practice work.
- ❖ The Sounds of English-Practice Work
- ❖ Pronunciation
- ❖ Stress and Intonation-Practice work.

Lab Manual:

1. '**INTERACT:** English Lab Manual for Undergraduate Students' Published by Orient Blackswan Pvt Ltd.

Reference Books:

1. Strengthen your communication skills by Dr M Hari Prasad, Dr Salivendra Raju and Dr G Suvarna Lakshmi, Maruti Publications.
2. English for Professionals by Prof Eliah, B.S Publications, Hyderabad.
3. Unlock, Listening and speaking skills 2, Cambridge University Press
4. Spring Board to Success, Orient BlackSwan
5. A Practical Course in effective english speaking skills, PHI
6. Word power made handy, Dr shalini verma, Schand Company
7. Let us hear them speak, Jayashree Mohanraj, Sage texts
8. Professional Communication, Aruna Koneru, Mc Grawhill Education
9. Cornerstone, Developing soft skills, Pearson Education.

**C PROGRAMMING LAB& HARDWARE FUNDAMENTALS
(Common to CSE & IT)**

Lab	: 3 Periods	Int.Marks	: 50
Exam	: 3 Hrs.	Ext. Marks	: 50
		Credits	: 2

Course Objectives:

1. Understand the basic concept of C Programming, and its different modules that includes conditional and looping expressions, Arrays, Strings, Functions, Pointers, Structures and File programming.
2. Acquire knowledge about the basic concept of writing a program.
3. Role of constants, variables, identifiers, operators, type conversion and other building blocks of C Language.
4. Use of conditional expressions and looping statements to solve problems associated with conditions and repetitions.
5. Role of Functions involving the idea of modularity.

Course Outcomes:

1. Apply and practice logical ability to solve the problems.
2. Understand C programming development environment, compiling, debugging, and linking and executing a program using the development environment.
3. Analyzing the complexity of problems, Modularize the problems into small modules and then convert them into programs.
4. Understand and apply the in-built functions and customized functions for solving the problems.
5. Understand and apply the pointers, memory allocation techniques and use of files for dealing with variety of problems.
6. Document and present the algorithms, flowcharts and programs in form of user manuals.
7. Identification of various computer components, Installation of software

List of Programs**Exercise - 1 Basics**

- a) What is an OS Command, Familiarization of Editors - vi, Emacs
- b) Using commands like mkdir, ls, cp, mv, cat, pwd, and man
- c) C Program to Perform Adding, Subtraction, Multiplication and Division of two numbers
From Command line

Exercise - 2 Basic Math

- a) Write a C Program to Simulate 3 Laws at Motion
- b) Write a C Program to convert Celsius to Fahrenheit and vice versa

Exercise - 3 Control Flow - I

- a) Write a C Program to Find Whether the Given Year is a Leap Year or not.
- b) Write a C Program to Add Digits & Multiplication of a number

Exercise – 4 Control Flow - II

- a) Write a C Program to Find Whether the Given Number is

- i) Prime Number
 - ii) Armstrong Number
- b) Write a C program to print Floyd Triangle
- c) Write a C Program to print Pascal Triangle.

Exercise – 5 Functions

- a) Write a C Program demonstrating of parameter passing in Functions and returning values.
- b) Write a C Program illustrating Fibonacci, Factorial with Recursion without Recursion.

Exercise – 6 Control Flow – III

- a) Write a C Program to make a simple Calculator to Add, Subtract, Multiply or Divide Using switch...case
- b) Write a C Program to convert decimal to binary and hex (using switch call function the function)

Exercise – 7 Functions - Continued

- a) Write a C Program to compute the values of $\sin x$ and $\cos x$ and e^x values using Series expansion. (use factorial function)

Exercise – 8 Arrays

Demonstration of arrays

- a) Search-Linear.
- b) Sorting-Bubble, Selection.
- c) Operations on Matrix.

Exercises - 9 Structures

- a) Write a C Program to Store Information of a Movie Using Structure
- b) Write a C Program to Store Information Using Structures with Dynamically Memory Allocation
- c) Write a C Program to Add Two Complex Numbers by Passing Structure to a Function

Exercise - 10 Arrays and Pointers

- a) Write a C Program to Access Elements of an Array Using Pointer
- b) Write a C Program to find the sum of numbers with arrays and pointers.

Exercise – 11 Dynamic Memory Allocations

- a) Write a C program to find sum of n elements entered by user. To perform this program, allocate memory dynamically using malloc () function.
- b) Write a C program to find sum of n elements entered by user. To perform this program, allocate memory dynamically using calloc () function.

Understand the difference between the above two programs

Exercise – 12 Strings

- a) Implementation of string manipulation operations **with** library function.
 - i) copy
 - ii) concatenate
 - iii) length
 - iv) compare
- b) Implementation of string manipulation operations **without** library function.
 - i) copy
 - ii) concatenate
 - iii) length
 - iv) compare

Exercise -13 Files

- a) Write a C programming code to open a file and to print its contents onscreen.
- b) Write a C program to copy files

Exercise - 14 Files Continued

- a) Write a C program merges two files and stores their contents in another file.
- b) Write a C program to delete a file.

Exercise - 15

- a) System Assembling, Disassembling and identification of Parts / Peripherals.
- b) Operating System Installation-Install Operating Systems like Windows, Linux along with necessary Device Drivers.

Exercise - 16

- a) MS-Office / Open Office
 - i. Word - Formatting, Page Borders, Reviewing, Equations, symbols
 - ii. Spread Sheet-Organize data, usage of formula, graphs, charts.
 - iii. Powerpoint - features of power point, guidelines for preparing an effective presentation.
- b) Network Configuration & Software Installation-Configuring TCP/IP, Proxy, and firewall settings. Installing application software, system software & tools.

Note:

- a) All the Programs must be executed in the Linux Environment. (Mandatory)
- b) The Lab record must be a print of the LATEX (.tex) Format.

**C PROGRAMMING LAB
(For ECE)**

Lab : 3 Periods
Exam : 3 Hrs.

Int.Marks : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

1. Understand the basic concept of C Programming, and its different modules that includes conditional and looping expressions, Arrays, Strings, Functions, Pointers, Structures and File programming.
2. Acquire knowledge about the basic concept of writing a program.
3. Role of constants, variables, identifiers, operators, type conversion and other building blocks of C Language.
4. Use of conditional expressions and looping statements to solve problems associated with conditions and repetitions.
5. Role of Functions involving the idea of modularity.

Course Outcomes:

1. Apply and practice logical ability to solve the problems.
2. Understand C programming development environment, compiling, debugging, and linking and executing a program using the development environment.
3. Analyzing the complexity of problems, Modularize the problems into small modules and then convert them into programs.
4. Understand and apply the in-built functions and customized functions for solving the problems.
5. Understand and apply the pointers, memory allocation techniques and use of files for dealing with variety of problems.
6. Document and present the algorithms, flowcharts and programs in form of user manuals.
7. Identification of various computer components, Installation of software

Programming

Exercise - 1 Basics

- a) What is an OS Command, Familiarization of Editors - vi, Emacs
- b) Using commands like mkdir, ls, cp, mv, cat, pwd, and man
- c) C Program to Perform Adding, Subtraction, Multiplication and Division of two numbers
From Command line

Exercise - 2 Basic Math

- a) Write a C Program to Simulate 3 Laws at Motion
- b) Write a C Program to convert Celsius to Fahrenheit and vice versa

Exercise - 3 Control Flow - I

- a) Write a C Program to Find Whether the Given Year is a Leap Year or not.
- b) Write a C Program to Add Digits & Multiplication of a number

Exercise – 4 Control Flow - II

- a) Write a C Program to Find Whether the Given Number is
 - i. Prime Number
 - ii. Armstrong Number
- b) Write a C program to print Floyd Triangle
- c) Write a C Program to print Pascal Triangle.

Exercise – 5 Functions

- a) Write a C Program demonstrating of parameter passing in Functions and returning values.
- b) Write a C Program illustrating Fibonacci, Factorial with Recursion without Recursion.

Exercise – 6 Control Flow – III)

- a) Write a C Program to make a simple Calculator to Add, Subtract, Multiply or Divide Using switch...case
- b) Write a C Program to convert decimal to binary and hex (using switch call function the function)

Exercise – 7 Functions - Continued

- a) Write a C Program to compute the values of $\sin x$ and $\cos x$ and e^x values using Series expansion. (use factorial function)

Exercise – 8 Arrays

Demonstration of arrays

- a) Search-Linear.
- b) Sorting-Bubble, Selection.
- c) Operations on Matrix.

Exercises - 9 Structures

- a) Write a C Program to Store Information of a Movie Using Structure
- b) Write a C Program to Store Information Using Structures with Dynamically Memory Allocation
- c) Write a C Program to Add Two Complex Numbers by Passing Structure to a Function

Exercise - 10 Arrays and Pointers

- a) Write a C Program to Access Elements of an Array Using Pointer
- b) Write a C Program to find the sum of numbers with arrays and pointers.

Exercise – 11 Dynamic Memory Allocations

- a) Write a C program to find sum of n elements entered by user. To perform this program, allocate memory dynamically using `malloc ()` function.
 - b) Write a C program to find sum of n elements entered by user. To perform this program, allocate memory dynamically using `calloc ()` function.
- Understand the difference between the above two programs.

Exercise – 12 Strings

- a) Implementation of string manipulation operations **with** library function.
 - i) copy
 - ii) concatenate
 - iii) length
 - iv) compare
- b) Implementation of string manipulation operations **without** library function.
 - i) copy
 - ii) concatenate
 - iii) length
 - iv) compare

Exercise -13 Files

- a) Write a C programming code to open a file and to print its contents onscreen.
- b) Write a C program to copy files

Exercise - 14 Files Continued

- a) Write a C program merges two files and stores their contents in another file.
- b) Write a C program to delete a file.

Note:

- a) All the Programs must be executed in the Linux Environment. (Mandatory)
- b) The Lab record must be a print of the LATEX (.tex) Format.

ENGINEERING PHYSICS - VIRTUAL LABS – ASSIGNMENTS
(Common to CSE, ECE & IT)

Lab : 2 Periods

Int.Marks: 5

Course Objective: Training Engineering students to prepare a technical document and improving their writing skills.

Course Outcome: Physics Virtual laboratory curriculum in the form of assignment ensures an engineering graduate to prepare a /technical/mini-project/ experimental report with scientific temper.

LIST OF EXPERIMENTS

1. Hall Effect
2. Crystal Structure
3. Hysteresis
4. Brewster's angle
5. Magnetic Levitation / SQUID
6. Numerical Aperture of Optical fiber
7. Photoelectric Effect
8. Simple Harmonic Motion
9. Damped Harmonic Motion
10. LASER – Beam Divergence and Spot size
11. B-H curve
12. Michelson's interferometer
13. Black body radiation

URL: www.vlab.co.in

(Note: Internal Marks of **Engineering Physics - Virtual Labs – Assignments** are to be considered as Assignment marks in the Internal Marks of **Engineering Physics- B17 BS 1104**)

NCC
(Common to CSE, ECE & IT)

Practice : 2 Periods

The NCC- National Integration and Awareness- Drill- Personality Development Life Skills- Leadership- Disaster Management-Social Awareness and Community Development- Health and Hygiene- Environment Awareness and Conservation.

(**Note:** It is an uncredited course. It will not be included in the Grade Memo / Certificate. The Certificate will be issued based on the performance and attendance. This course attendance will be counted in the semester overall attendance.)

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R17)

I/IV B.TECH
(With effect from **2017-2018** Admitted Batch onwards)
Under Choice Based Credit System

GROUP-B (CSE, ECE & IT)
II-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/ Week	Internal Marks	External Marks	Total Marks
B17 BS 1201	English – II*	3	3	1	--	4	30	70	100
B17 BS 1203	Mathematics – III*	3	3	1	--	4	30	70	100
B17 BS 1205	Engineering Chemistry	3	3	1	--	4	30	70	100
B17 ME 1201	Engineering Drawing	3	1	--	3	4	30	70	100
# DS 2	Department Subject	3	3	1	--	4	30	70	100
# DS 3	Department Subject	3	3	1	--	4	30	70	100
B17 BS 1207	Engineering Chemistry Lab	2	--	--	3	3	50	50	100
B17 BS 1208	English Communication Skills Lab – II*	2	--	--	3	3	50	50	100
# DL2	Department Lab	2	--	--	3	3	50	50	100
B17 BS 1212	Inner Engineering	--	--	--	2	2	--	--	--
Total		24	16	5	14	35	330	570	900

*** Common to both Group - A and Group - B**

# DS 2	CSE & IT	B17 CS 1202	Object Oriented Programming Through C++
	ECE	B17 CS 1203	Data Structures
#DS 3	CSE & IT	B17 EC 1201	Elements of Electronics Engineering
	ECE	B17 EE 1203	Elements of Electrical Engineering
#DL2	CSE & IT	B17 CS 1205	Object Oriented Programming Lab
	ECE	B17 BS 1209	Engineering Workshop & IT Workshop

Code: B17 BS 1201

ENGLISH - II
(Common to all Branches)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To improve the language proficiency of the students in English with emphasis on LSRW skills.
2. To enable the students to study and comprehend the prescribed lessons and subjects more effectively relating to their theoretical and practical components.
3. To develop spoken and written forms of the students in both formal and informal situations.
4. To enrich the vocabulary of the students through the usage of the lexis in appropriate contexts.
5. To expose them to the correct structural patterns of English.

Course Outcomes: At the end of the Course, Student will be able to:

1. To comprehend the speech of people belonging to different backgrounds and regions.
2. Understand the importance of speaking and writing for personal and professional communication and practice it in real contexts.
3. To express fluently and accurately in social discourse.
4. Participate in group activities like role-plays, discussions and debates.
5. Identify the discourse features, and improve intensive and extensive reading skills.

SYLLABUS

UNIT I:

- A. Detailed-Text: Unit 1: ' The Greatest Resource- Education'
- B. Non-Detailed Text: Lesson 1: ' A P J Abdul Kalam' from The Great Indian Scientists.

UNIT II:

- A. Detailed-Text: Unit 2: ' A Dilemma'
- B. Non-Detailed Text: Lesson 2:'C V Raman' from The Great Indian Scientists.

UNIT III:

- A. Detailed-Text: Unit 3: 'Cultural Shock': Adjustments to new Cultural Environments
- B. Non-Detailed Text: Lesson 3:'Homi Jehangir Bhabha' from The Great Indian Scientists.

UNIT IV:

- A. Detailed-Text: Unit 4: 'The Lottery'
- B. Non-Detailed Text: Lesson 4: 'Jagadish Chandra Bose' from The Great Indian Scientists.

UNIT V:

- A. Detailed-Text: Unit 5: ' The Chief Software Architect'
- B. Non-Detailed Text: Lesson 5: ' Prafulla Chandra Ray' from The Great Indian Scientists

Detailed Textbook:

1. English Encounters Published By Maruthi Publishers.

Non-Detailed Text Book:

1. The Great Indian Scientists Published by Cengage learning.

MATHEMATICS - III
(Common to all Branches)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. The course is designed to equip the students with the necessary mathematical skills and techniques that are essential for an engineering course.
2. The skills derived from the course will help the student form a necessary base to develop analytic and design concepts.
3. Understand important concepts of Linear algebra including solution of linear simultaneous equations, eigen values and eigen vectors.
4. Learn Beta, Gamma functions and how to evaluate double and triple integrals.
5. Learn fundamental concepts of vector calculus that help understand other Engineering courses.

Course Outcomes: At the end of the Course, student will be able to:

1. Determine rank, and solve a system of linear simultaneous equations numerically using various matrix methods.
2. Determine Eigen values and Eigen vectors of a given matrix, Reduce a Quadratic form to its canonical form and classify.
3. Evaluate double integrals over a region and triple integral over a volume.
4. Use the knowledge of Beta and Gamma functions in evaluation of different integrals.
5. Find gradient of a scalar function, divergence and curl of a vector function. Use vector identities for solving problems.
6. Evaluate line, surface and volume integrals by the use of Green's, Stokes' and Gauss divergence theorems.

SYLLABUS

UNIT I: Linear systems of equations:

Rank, Echelon form, Normal form, Solution of linear systems, Gauss elimination, Gauss-Jordan, Jacobi and Gauss-Seidel methods.

Applications: Finding the current in electrical circuits.

UNIT II: Eigen values - Eigen vectors and Quadratic forms:

Eigen values, Eigen vectors, Properties, Cayley-Hamilton theorem, Inverse and powers of a matrix by using Cayley-Hamilton theorem, Diagonalization, Quadratic forms, Reduction of a Quadratic form to Canonical form, Rank, Positive, Negative, Semi-Definite and indefinite forms of a Quadratic form, Index and Signature of a Quadratic form.

Applications: Free vibration of a two-mass system.

UNIT III: Multiple integrals:

Double and triple integrals, Change of variables, Change of order of integration. Application to finding Areas, Moment of Inertia and Volumes.

Beta and Gamma functions, Properties, Relation between Beta and Gamma functions, Application to evaluation of improper integrals. The error function and the complimentary error function.

UNIT IV: Vector Differentiation:

Gradient, directional derivative, Divergence, Curl, Incompressible flow, solenoidal and irrotational vector fields, second order operators, vector identities.

UNIT V: Vector Integration:

Line integral, Work done, Potential function; Area, Surface and volume integrals, Flux, Vector integral theorems: Greens, Stokes and Gauss Divergence theorems (without proof) and related Problems.

Text Books:

1. B.S.Grewal, Higher Engineering Mathematics, 43rd Edition, Khanna Publishers.
2. N.P.Bali & Manish Goyal, Engineering Mathematics, Lakshmi Publications.

Reference Books:

1. Michael Greenberg, Advanced Engineering Mathematics, 9th edition, Pearson edn.
2. Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, Wiley-India.
3. Peter O'Neil, Advanced Engineering Mathematics, 7th edition, Cengage Learning.
4. D.W. Jordan and T. Smith, Mathematical Techniques, Oxford University Press.
5. Srimanta Pal, Subodh C.Bhunia, Engineering Mathematics, Oxford University Press.
6. Dass H.K., Rajnish Verma. Er., Higher Engineering Mathematics, S. Chand Co. Pvt. Ltd, New Delhi.

ENGINEERING CHEMISTRY
(Common to CSE, ECE & IT)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Knowledge of basic concepts of chemistry for engineering students will help them as professional engineers later in design and material selection as well as utilizing the available resources.

Course Objectives:

1. To understand the physical and mechanical properties of Polymers/Plastics/elastomers helps in selecting suitable materials for different purpose.
2. To create awareness on fuels as a source of energy for industries like thermal power stations, steel industry, fertilizer industry etc.
3. To understand the concepts of galvanic cells and corrosion with theories like electro chemical theory.
4. To understand the importance of water.
5. To understand about the materials which are used in major industries like steel and metallurgical manufacturing industries, construction and electrical equipment manufacturing industries.

Course Outcomes:

1. At the end of the course the students learn the advantages and limitations of plastic materials and their use in design.
2. Fuels which are used commonly and their economics, advantages and limitations are discussed.
3. Students gained Knowledge reasons for corrosion and some methods of corrosion control.
4. Students understands the impurities present in raw water, problems associated with them and how to avoid them.
5. Similarly students understand liquid crystals and semi conductors. Students can gain the building materials , solar materials, lubricants and energy storage devices.

SYLLABUS

UNIT-I: High Polymers and Plastics; Rubbers & Elastomers:

Polymerization Definition, Types of Polymerization, Mechanism of addition polymerization, Plastics as engineering materials, Thermoplastics and Thermosetting plastics, Compounding of plastics, Fabrication of plastics (4 techniques); Preparation, Properties and applications of Polyethylene, PVC, Bakelite, Nylon - 6,6, Bullet Proof plastics -polycarbonate and Kelvar; Fiber reinforced plastics, conducting polymers, Biodegradable Polymers - PHBV, Nylon 2, Nylon 6. Natural rubber – Vulcanization – Compounding of Rubber; Preparation, properties and applications of Buna – S; Buna – N;

UNIT-II :Fuel Technology& Lubricants:

Fuels: - Introduction – Classification of fuels, Calorific value – HCV and LCV, Determination of Calorific value by bomb calorimeter; Proximate and ultimate analysis of coal, coke: manufacture of coke by Otto – Hoffmann’s by-product coke oven process; Refining of Petroleum, Knocking-octane number of gasoline, cetane number of diesel oil. Synthetic Petrol; LPG , CNG.

Lubricants:- Definition, Mechanism of Lubrication, Properties of Lubricants (Definition and significance)

UNIT-III: Electrochemical cells and Corrosion:

Galvanic cell, single electrode potential, Calomel electrode; Modern batteries: - Lead – Acid battery; Fuel cells- Hydrogen – Oxygen cell, Lithium battery Theories of corrosion (i) dry Corrosion (ii) wet corrosion. Types of corrosion - differential aeration corrosion, pitting corrosion, galvanic corrosion, stress corrosion, Factors influencing corrosion, Protection from corrosion-material selection & design, cathodic protection, Protective coatings- metallic coatings – Galvanizing, Tinning, Electroplating; Electroless plating ; Paints.

UNIT-IV: Water technology:

Sources of water – Hardness of water – Estimation of hardness of water by EDTA method; Boiler troubles – sludge and scale formation, Boiler corrosion, caustic embrittlement, Priming and foaming; Softening of water by Lime – Soda Process, Zeolite Process, Ion – Exchange Process; Municipal water treatment; Desalination of sea water by Electrodialysis and Reverse osmosis methods.

UNIT-V :Chemistry of Engineering Materials& Advanced Engineering materials

Cement:- Manufacture of Portland cement, setting and hardening of cement, Deterioration of cement concrete.

Refractories: - Definition, Characteristics, classification, Properties and failure of refractories.

Solar Energy: - Construction and working of Photovoltaic cell, applications.

Solid State Materials: Crystal imperfections, Semi Conductors, Classification and chemistry of semi conductors: Intrinsic semiconductors; Extrinsic semiconductors; Defect semiconductors, Compound Semiconductors and Organic Semiconductors.

Liquid Crystals: - Definition – Classification with examples – Applications

Text Books:

1. Engineering Chemistry by Jain and Jain, Dhanpat Rai publishing co.
2. Engineering Chemistry by Willy India Pvt Ltd.
3. Engineering Chemistry by Dr.K.Anji Reddy and Dr.M.S.R.Reddy; Silicon Publications.

Reference Books:

1. Engineering Chemistry by Shikha Agarwal; Cambridge University Press,2015 edition.
2. A text of Engineering Chemistry by S.S.Dara; S.Chand & Co Ltd.

ENGINEERING DRAWING
(Common to CSE, ECE & IT)

Lecture	: 1 Period.	Int.Marks	: 30
Practice	: 3 Periods.	Ext. Marks	: 70
Exam	: 3Hrs.	Credits	: 3

Course Objectives:

1. To highlight the significance of universal language of engineers.
2. To impart basic knowledge and skills required to prepare engineering drawings.
3. To impart knowledge and skills required to draw projections of solids in different contexts.
4. To visualize and represent the pictorial views with proper dimensioning and scaling.

Course Outcomes:

Students will be able to

1. Apply principles of drawing to represent dimensions of an object.
2. Construct polygons and engineering curves.
3. Draw projections of points, lines, planes and solids.
4. Represent the object in 3D view through isometric views.
5. Convert the isometric view to orthographic view and vice versa.

SYLLABUS

UNIT I

Polygons: Constructing regular polygons by general methods, inscribing and describing polygons on circles.

Curves: Parabola, Ellipse and Hyperbola by general methods, cycloids, involutes, tangents & normals for the curves.

UNIT II

Orthographic Projections: Horizontal plane, vertical plane, profile plane, importance of reference lines, projections of points in various quadrants, projections of lines, lines parallel either to one of the reference planes (HP, VP or PP)

Projections of straight lines inclined to both the planes, determination of true lengths, angle of inclination and traces- HT, VT

UNIT III

Projections of planes: regular planes perpendicular/parallel to one plane and inclined to the other reference plane; inclined to both the reference planes.

UNIT IV

Projections of Solids – Prisms, Pyramids, Cones and Cylinders with the axis inclined to one of the planes.

UNIT V

Conversion of isometric views to orthographic views; Conversion of orthographic views to isometric views.

Text Books:

1. Engineering Drawing by N.D. Butt, Chariot Publications.
2. Engineering Drawing by Agarwal & Agarwal, Tata McGraw Hill Publishers

Reference Books:

1. Engineering Drawing by K.L.Narayana & P. Kannaiah, Scitech Publishers.
2. Engineering Graphics for Degree by K.C. John, PHI Publishers.
3. Engineering Graphics by P. Varghese, McGrawHill Publishers.
4. Engineering Drawing + AutoCad – K Venugopal, V. Prabhu Raja, New Age

**OBJECT-ORIENTED PROGRAMMING THROUGH C++
(Common to CSE & IT)**

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. This course is designed to provide a comprehensive study of the C++ programming language. students are able to write efficient, maintainable and portable code to design algorithmic solutions to problems.
2. Expertise in object oriented principles and their implementation in C++
3. To acquire an understanding of basic object-oriented concepts and the issues involved in effective class design.
4. To Write C++ programs that use arrays, structures, pointers, object oriented concepts such as information hiding, constructors, destructors, inheritance.

Course Outcomes:

1. Write, compile and debug programs in C++ language. Use different data types in a computer program.
2. Design programs involving decision structures, loops and functions.
3. Explain classes and abstract classes and objects, abstraction and encapsulation, inheritance, polymorphism, constructors, access control and overloading.
4. Solve a given application problem by going through the basic steps of program specifications, analysis, design, implementation and testing within the context of the object-oriented paradigm.

SYLLABUS

UNIT-I: Introduction to C++, Classes and Objects.

Difference between C and C++, Disadvantage of Conventional Programming, Basic Concepts of Object Oriented Programming, Advantage of OOP, Object Oriented Languages, Functions in C++, Operators in C++. **Classes and Objects:** Declaring Objects, Access Specifiers and their Scope, Static data members, static member functions, arrays of objects, local classes, Nested classes.

UNIT-II: Constructors, Destructors and Operator Overloading.

Constructors and Destructors: Introduction- Constructors and Destructor- types of constructors, Constructors with default Arguments, Dynamic initialization of objects, Dynamic constructors. **Operator Overloading** Introduction, Overloading Unary Operators and Binary Operators, Overloading Unary Operators and Binary Operators using friend function, Overloading Assignment Operator (=), Overloading insertion(<<) and extraction(>>) operators, Manipulation of Strings using Operators, Rules for Overloading Operators, Type Conversions.

UNIT-III: Inheritance , Pointers, Virtual Functions and Polymorphism.

Inheritance: Reusability, Types of Inheritance, Virtual Base Classes, Abstract Classes, Advantages of Inheritance, Disadvantages of Inheritance, and constructors in derived classes. **Pointers Introduction:** Pointers to Objects, “this” Pointer, Pointers to Derived Classes, including Polymorphisms and Virtual Functions, Rules for Virtual Functions, pure virtual functions.

UNIT-IV: Manipulating Strings, Managing console I/O operations and Exception Handling.

Strings: Creating String Objects, Manipulating String Objects, Relational operations, String Characteristics, Accessing Characters in Strings. **C++ Stream** Classes, Unformatted I/O operations, Formatted I/O operations, managing output with Manipulators, **Exception Handling:** Principles of Exception Handling, Exception Handling Mechanism, throwing and catching Mechanism.

UNIT-V: Generic Programming with Templates, Standard Template Library and Files.

Generic Programming with Templates, Need for Templates, Definition of class Templates, Normal Function Templates, Over Loading of Template Function-Bubble Sort Using Function Templates, Difference between Templates and Macros, Overview of **Standard Template Library**, STL Programming Model, Containers, Algorithms, Iterators, Vectors, Lists, Maps. **FILES:** Introduction, File Stream Classes, File Operations, File Pointers and Manipulators, Sequential Access Files, Random File Access Operation, Detecting End-of File, Command-Line Arguments.

Text Books:

1. A complete Guide to programming in C++, Ulla Kirch-Prinz, Peter Prinz, Jones and Bartlett Publishers (2002).
2. Programming in C++, 2nd Edition, Ashok N Kamthane, and Pearson.
3. Object Oriented Programming C++, 4th Edition, Joyce Farrell, Cengage Learning.

Reference Books:

1. C++ Programming, A Problem Solving Approach, Forouzan, Gilberg, Prasad, CENGAGE
2. The Complete Reference C++, 4th Edition, Herbert Schildt, TMH.
3. Programming solving with C++, 9th Edition Walter Savitch.
4. Object Oriented Programming in C++, 4th Edition, Robert Lafore

**DATA STRUCTURES
(For ECE)**

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To be familiar with basic techniques handling problems with Data structures
2. Solve problems using data structures such as linear lists, stacks, queues, hash tables
3. Create and traverse different types of trees and graphs.
4. To practice different searching algorithms.

Course Outcomes:

Students will be able to

1. Apply advanced data structure strategies for exploring complex data structures.
2. Compare and contrast various data structures and design techniques in the area of Performance.
3. Implement all data structures like stacks, queues, trees, lists and graphs and compare their performance and tradeoffs.
4. Implement different operations on trees.
5. Apply graphs to real time applications.
6. Perform sorting using different algorithms.

SYLLABUS

UNIT-I

Arrays and Structures

Array as an Abstract Data Type, Polynomial Abstract Data Type, Introduction to Sparse Matrix, Sparse Matrix Abstract Data Type, Representation of Multidimensional Arrays, Structures and Unions, Internal Implementation of Structures, Self-Referential Structures.

Recursion, Simple Searching and Sorting Techniques

Recursive functions, Introduction to Searching, Sequential Search, Binary Search, Interpolation Search, Selection Sort, Bubble Sort, Insertion Sort, Quick Sort, Introduction to Merge Sort, Iterative Merge Sort, Recursive Merge Sort, Heap sort.

UNIT-II

Stacks and Queues

Stack Abstract Data Type, Queue Abstract Data Type, Stacks and Queues using arrays, , Introduction to Evaluation of Expressions, Evaluating Postfix Expressions, Infix to Postfix and Prefix conversion, Multiple Stacks and Queues, Circular Queues using arrays.

UNIT-III

Linked Lists

Pointers, Dynamically Allocated Storage using pointers, Singly Linked Lists, Dynamically Linked Stacks and Queues, Polynomials, Representing Polynomials as Singly Linked Lists, Adding Polynomials, Erasing Polynomials, Polynomials as Circularly Linked Lists, Additional List Operations, Operations for Singly Linked Lists, Operations for Doubly Linked Lists, Radix Sort.

UNIT-IV

Trees

Representation of Trees, Binary Trees Abstract Data Type, Properties of Binary Trees, Binary Tree Representations, Binary Tree Traversals, Additional Binary Tree Operations, Threaded Binary Trees, Heap Abstract Data Type, Priority Queues, Insertion into a max heap, Deletion from a max heap, Heap Sort, Introduction to Binary Search Trees, Searching a Binary Search Tree, Inserting an Element into a Binary Search Tree, Deleting an Element from a Binary Search Tree, Height of a Binary Search Tree, Counting Binary Trees.

UNIT-V

Graphs

Graph Abstract Data Type, Definitions, Graph Representations, Elementary Graph Operations, Depth First Search, Breadth First Search, Connected Components, Spanning Trees, Minimum Cost Spanning Trees, Prim's and Kruskal's Algorithms, Shortest Paths and Transitive Closure, Single Source All Destination - Dijkstra's Algorithm, All Pairs Shortest Paths - Floyd's Algorithm, Transitive Closure using Warshall's Algorithm.

Text Books:

1. Fundamentals of Data Structures in C, 2nd edition, Horowitz, Sahni and Anderson-Freed, Universities Press, 2008.

Reference Books:

1. Data Structures using C by Aaron M. Tenenbaum, Y. Langsam and M.J. Augenstein, Pearson Education, 2009.
2. Data Structures with C by Seymour Lipschutz, Schaum Outline series, 2010.
3. Data Structures using C by R. Krishna Moorthy G. Indirani Kumaravel, TMH, New Delhi, 2008.

ELEMENTS OF ELECTRONICS ENGINEERING
(Common to CSE & IT)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To give the exposure to the students on semiconductor physics of the intrinsic and extrinsic semiconductors and basics of P-N junction diode.
2. To give the exposure to the students on the basics of special purpose diodes like Zener diode, photo diode, LED, and tunnel diode and rectifier circuits using diodes.
3. To give the exposure to the students on basics of BJT, and transistor circuit configurations.
4. To give exposure to the students on transistor biasing and thermal stabilization.
5. To give exposure to the students on the basics of JFET, MOSFET and FET biasing.

Course Outcomes:

After completion of the course the students will be able to

1. Understand the basic concepts of transport of charge carriers in semiconductors, drift and diffusion currents, physical structure, operation, V-I characteristics of semiconductor diode.
2. Understand the basic concepts of special types of diodes like Zener Diode, LED, Photo Diode and tunnel diode, rectifier circuits with and without filters.
3. Understand the physical structure, operation, input and output characteristics of BJT in CE, CB, CC circuit configurations.
4. Understand the basic concepts of transistor biasing and thermal stabilization.
5. Understand the physical structure, operation, characteristics and circuit models of JFET's and MOSFET's.

SYLLABUS

UNIT I: Semiconductors and P-N junction diode:

Intrinsic and extrinsic semiconductors, charge densities in semiconductors, Drift and Diffusion currents, Hall Effect, Mass action law. Basic operation and V-I Characteristics of semiconductor diode, Diode current equation, Avalanche breakdown and Zener breakdown phenomenon.

UNIT II: Special Diodes and Diode Rectifiers:

Zener Diode, LED, Photo Diode and tunnel diode, Half wave and Full wave Rectifiers- with and without filters, Bridge Rectifier, Expressions - Ripple factor, Efficiency, Capacitor filters

UNIT III: Bipolar Junction Transistor:

Introduction, construction, basic operation of npn and pnp transistors, Transistor circuit configurations- CE, CB, CC- Input and output Characteristics in various configurations. h-parameter model for transistor amplifier. (*Introductory Treatment only*).

UNIT IV: Transistor Biasing and Thermal Stabilization:

Transistor Biasing, Thermal runaway, stabilization, Different methods of Biasing-Fixed Bias, collector feedback bias, self-bias, Bias compensation.

UNIT V: Field Effect Transistors: Junction field Effect Transistors (JFET)- JFET characteristics, JFET Parameters, Small Signal model of FET, Depletion and Enhancement type MOSFET's.

Text Books:

1. Electronic Device and Circuits by SanjeevGuptha, Dhanpatrai&Co.Pvt.Ltd.
2. Electronic Device and Circuits by K.Satya Prasad, VGS.

Reference Books:

1. Integrated Electronics- Millman&Halkias, TMH.

ELEMENTS OF ELECTRICAL ENGINEERING (For ECE)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To learn the fundamentals of electrical and magnetic circuits
2. To understand the principle of operation and construction details of DC machines.
3. To understand the principle of operation of Transformers.
4. To understand the principle of operation and construction details of three phase Induction motor
5. To understand the principle of operation of synchronous generator and different measuring instruments

Course Outcomes:

1. Able to understand the basics of Magnetic Circuits and Kirchhoff's laws.
2. Able to understand the operation of DC Machines and to conduct different Tests
3. Able to analyze the Performance of Transformers.
4. Able to explain the operation of three phase induction motor.
5. Able to analyze the operation of alternator and different measuring instruments.

SYLLABUS

UNIT I: Electrical and Magnetic Circuits:

Basic definitions, Types of network elements, Ohm's Law, Kirchhoff's Laws, Series and parallel Circuits and star-delta and delta-star transformations-simple problems. Magnetic flux, MMF, Reluctance, Faraday's laws, Lenz's law, statically induced EMF, dynamically induced EMF.

UNIT – II: DC Machines:

Principle of operation of DC Generator - EMF equation – Construction-Types of DC generator- OCC of DC Generator-DC motor types - Torque equation –Losses-Efficiency-speed control methods- applications

UNIT – III: Transformers:

Principle of operation of single phase transformer - EMF equation - equivalent circuit –losses - efficiency and regulation- Open circuit and Short circuit tests.

UNIT – IV: Induction Motors:

Construction-Principle of operation of induction motor-slip- rotor frequency, slip - torque characteristics - Power flow diagram-Efficiency-Applications

UNIT – V: Synchronous Generator and Measuring Instruments:

Construction-Principle of operation of alternator-EMF equation of alternator- Regulation by Synchronous impedance method.

Classification –Deflecting, controlling, damping Torque, ammeter, voltmeter, wattmeter, MI, MC instruments-Energy meter

Text Books:

1. Electrical Machinery By Dr.P.S Bhimbra,Khanna Publications, January 2011
2. Electrical Technology by Surinder Pal Bali, Pearson Publications. January 1, 2013
3. Principles of Electrical Engineering, V.K mehta, Rohit Mehta, S. Chand Publications. Revised Edition 2012
4. A Course in Electronic Measurements and Instrumentation by A.K. Sawhney Dhanpat Rai & Co. (P) Limited; 2014 edition (2015)

Reference Books:

1. Basic Electrical Engineering, by J.B Gupta, K Kataria and Sons; Reprint 2013 edition.
2. Fundamentals of Electrical Engineering and Electronics, by Theraja B.L, S Chand Multicolor edition (1 December 2006)
3. A Course in Electronics and Electrical Measurements and Instrumentation by J.B. Gupta S K Kataria and Sons; Reprint 2013 edition (2013)

**ENGINEERING CHEMISTRY LAB
(Common to CSE,ECE& IT)****Lab : 3 Periods**
Exam : 3Hrs.**Int.Marks : 50**
Ext. Marks : 50
Credits : 2

Course Objectives:

1. To investigate and understand Physical behavior in the laboratory using scientific reasoning and logic and interpret the result of simple experiments and demonstration of chemical Principle and also evaluate the impact of chemical discoveries on how we view the world.
2. Effectively communicate experimental results and solutions to application problems through oral and written reports.
3. Understand the basic concepts, definitions, characteristics and phenomena.
4. Recognize the classical ideas and chemical phenomena and also define and analyse the concepts.

Course Outcomes:

1. An understanding of Professional and develop confidence on recent trends.
2. Able to gain technical knowledge of measuring, operating and testing of chemical instruments and equipments.
3. Acquire ability to apply knowledge of chemistry.
4. Exposed to the real time working environment.
5. Demonstrate the ability to learn Principles, design and conduct experiments.
6. Ability to work on laboratory and multidisciplinary tasks.

List of Experiments**Introduction to chemistry Laboratory**

1. Estimation of HCl using standard Sodium Hydroxide.
2. Determination of total hardness of water by EDTA method.
3. Estimation of Ferrous Iron by KMnO_4 .
4. Estimation of oxalic acid by KMnO_4
5. Estimation of Mohr's salt by $\text{K}_2\text{Cr}_2\text{O}_7$
6. Estimation of Dissolved oxygen by Winkler's method.
7. Determination of pH by pH meter and universal indicator method.
8. Conductometric titration of strong acid Vs strong base
9. Conductometric titration of strong acid Vs weak base.
10. Potentiometric titration of strong acid Vs strong base
11. Potentiometric titration of strong acid Vs weak base
12. Preparation of Phenol formaldehyde resin.
13. Determination of saponification value of oils
14. Determination of pour and cloud points of lubricating oil.
15. Determination Acid value of oil.

Text Books:

1. Engineering Chemistry Lab Manual Prepared by Chemistry Faculty of S.R.K.R.Engineering College.

Reference Books:

1. Laboratory manual on Engineering Chemistry by Dr.Sudha Rani ;Dhanpat Rai Publishing Company.
2. Engineering Chemistry Laboratory manual – I & II by Dr. K.Anji Reddy; Tulip Publications.

ENGLISH COMMUNICATION SKILLS LAB- II
(Common to All Branches)

Lab : 3 Periods
Exam : 3Hrs.

Int.Marks : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

1. To enable the students to learn demonstratively the communication skills of listening, speaking, reading and writing.
2. To enable the students participate in group interactions.
3. To improve the presentation skills of the students.
4. To help the students gain their confidence in attending the interviews.

Course Outcomes:

1. A study of the communicative items in the laboratory will help the students become successful in the competitive world.
2. Students enhance their presentation skills.
3. Students participate in group discussions and improve their team skills.
4. Students confidently face the interviews.

SYLLABUS

- ❖ Debating & Practice.
- ❖ Group Discussions & Practice.
- ❖ Presentation Skills & Practice
- ❖ Interview Skills & Practice
- ❖ Email
- ❖ Curriculum Vitae & Practice
- ❖ Idiomatic Expressions
- ❖ Common Errors in English & Practice

LAB MANUAL:

1. **‘INTERACT: English Lab Manual for Undergraduate Students’** Published by Orient Blackswan Pvt Ltd.

Reference Books:

1. Strengthen your communication skills by Dr M Hari Prasad, Dr Salivendra Raju and Dr G Suvarna Lakshmi, Maruti Publications.
2. English for Professionals by Prof Eliah, B.S Publications, Hyderabad.
3. Unlock, Listening and speaking skills 2, Cambridge University Press
4. Spring Board to Success, Orient BlackSwan
5. A Practical Course in effective english speaking skills, PHI
6. Word power made handy, Dr shalini verma, Schand Company
7. Let us hear them speak, Jayashree Mohanraj, Sage texts
8. Professional Communication, Aruna Koneru, Mc Grawhill Education
9. Cornerstone, Developing soft skills, Pearson Education

OBJECT ORIENTED PROGRAMMING LAB
(Common to CSE & IT)

Lab : 3 Periods

Exam : 3Hrs.

Int.Marks : 50

Ext. Marks : 50

Credits : 2

Course Objectives:

1. To strengthen their problem solving ability by applying the characteristics of an object-oriented approach.
2. To introduce object oriented concepts in C++ and Java.

Course Outcomes:

1. Explain what constitutes an object-oriented approach to programming and identify potential benefits of object-oriented programming over other approaches.
2. Apply an object-oriented approach to developing applications of varying complexities.

LIST OF PROGRAMS

1. Write a Programme that computes the simple interest and compound interest payable on principal amount (in Rs.) of loan borrowed by the customer from a bank for a given period of time (in years) at specific rate of interest. Further determine whether the bank will benefit by charging simple interest or compound interest
2. Write a Programme to calculate the fare for the passengers traveling in a bus. When a Passenger enters the bus, the conductor asks "What distance will you travel?" On knowing distance from passenger (as an approximate integer), the conductor mentions the fare to the passenger according to following criteria.
3. Write a C++ Program to illustrate Enumeration and Function Overloading
4. Write a C++ Program to illustrate Scope and Storage class
5. Implementation of ADT such as Stack and Queues
6. Write a C++ Program to illustrate the use of Constructors and Destructors and Constructor Overloading
7. Write a Program to illustrate Static member and methods
8. Write a Program to illustrate Bit fields
9. Write a Program to overload as binary operator, friend and member function
10. Write a Program to overload unary operator in Postfix and Prefix form as member and friend function
11. Write a C++ Program to illustrate Iterators and Containers
12. Write a C++ Program to illustrate function templates
13. Write a C++ Program to illustrate template class
14. Write C++ Programs and incorporating various forms of Inheritance
15. Write a C++ Program to illustrate Virtual functions
16. To write a C++ program to find the sum for the given variables using function with default arguments.
17. To write a C++ program to find the value of a number raised to its power that demonstrates a function using call by value.
18. To write a C++ program and to implement the concept of Call by Address
19. To write a program in C++ to prepare a student Record using class and object

20. To implement the concept of unary operator overloading by creating a C++ program.
21. Write a C++ program for swapping two values using function templates
22. Write a C++ program to implement a file handling concept using sequential access.

Reference Books:

1. C++ Programming, A Problem Solving Approach, Forouzan, Gilberg, Prasad, CENGAGE
2. Programming in C++, 2nd Edition, Ashok N Kamthane, and Pearson.
3. Programming solving with C++, 9th Edition Walter Savitch.

**ENGINEERING WORKSHOP & IT WORKSHOP
(For ECE)**

Lab : 3 Periods

Exam : 3Hrs.

Int.Marks : 50

Ext. Marks : 50

Credits : 2

PART-A ENGINEERING WORKSHOP

Course Objective:

To impart hands-on practice on basic engineering trades and skills.

Course Outcomes:

Student will be able to

1. Use various tools to prepare basic carpentry and fitting joints.
2. Prepare jobs of various shapes using black smithy.
3. Make basic house wire connections.
4. Fabricate simple components using tin smithy.

SYLLABUS

Carpentry	Fitting
1. T-Lap Joint 2. Cross Lap Joint 3. Dovetail Joint 4. Mortise and Tenon Joint	1. Vee Fit 2. Square Fit 3. Half Round Fit 4. Dovetail Fit
Black Smithy	Tin Smithy
1. Round rod to Square 2. S-Hook 3. Round Rod to Flat Ring 4. Round Rod to Square headed bolt	1. Taper Tray 2. Square Box without lid 3. Open Scoop 4. Funnel
House Wiring	
1. Parallel / Series Connection of three bulbs 2. Stair Case wiring 3. Florescent Lamp Fitting 4. Measurement of Earth Resistance	

Note: At least two exercises to be done from each trade.

Reference:

1. Elements of workshop technology, Vol.1 by S. K. and H. K. Choudary.

PART B: IT WORKSHOP:

Course Objectives:

1. Understand the basic components and peripherals of a computer.
2. To become familiar in configuring a system.
3. Learn the usage of productivity tools.
4. Acquire knowledge about the netiquette and cyber hygiene.
5. Get hands on experience in trouble shooting a system

Course Outcomes:

At the end of the course the students can able to

1. Assemble and disassemble the systems
2. Use the Microsoft office tools
3. Install various software
4. Know about various search engines
5. Trouble shoot various Hardware and Software problems
6. Use MATLAB and LATEX software

LIST OF EXERCISES

1. System Assembling, Disassembling and identification of Parts / Peripherals
2. Operating System Installation-Install Operating Systems like Windows, Linux along with necessary Device Drivers.
3. MS-Office / Open Office
 - a) Word - Formatting, Page Borders, Reviewing, Equations, symbols.
 - b) Spread Sheet - organize data, usage of formula, graphs, charts.
 - c) Power point - features of power point, guidelines for preparing an effective presentation.
 - d) Access- creation of database, validate data.
4. Network Configuration & Software Installation-Configuring TCP/IP, proxy and firewall settings. Installing application software, system software & tools.
5. Internet and World Wide Web-Search Engines, Types of search engines, netiquette, cyber hygiene.
6. Trouble Shooting-Hardware trouble shooting, Software trouble shooting.
7. MATLAB- basic commands, subroutines, graph plotting.
8. LATEX-basic formatting, handling equations and images.

Reference Books:

1. Computer Hardware, Installation, Interfacing, Troubleshooting and Maintenance, K.L. James, Eastern Economy Edition.
2. Microsoft Office 2007: Introductory Concepts and Techniques, Windows XP Edition By Gary B. Shelly, Misty E. Vermaat and Thomas J. Cashman (2007, Paperback).
3. LATEX- User's Guide and Reference manual, Leslie Lamport, Pearson, LPE, 2/e.
4. Getting Started with MATLAB: A Quick Introduction for Scientists and Engineers, Rudraprathap, Oxford University Press, 2002.
5. Scott Mueller's Upgrading and Repairing PCs, 18/e, Scott. Mueller, QUE, Pearson, 2008
6. The Complete Computer upgrade and repair book, 3/e, Cheryl A Schmidt, Dreamtech.
7. Comdex Information Technology course tool kit Vikas Gupta, WILEY Dreamtech.
8. Introduction to Information Technology, ITL Education Solutions limited, Pearson Education.

INNER ENGINEERING
(Common to CSE, ECE & IT)

Practice : 2 Periods

Course Objectives:

Student should learn:

1. Human values and tools to lead a happy, stress-free life.
2. Yoga asanas, Pranayama, Sudarshan Kriya & Meditation
3. At-least two creative arts out of photography, sketching, craft-making, singing, clay molding, upcycling etc.
4. Concentration pranayama, Ego bursting process.
5. To take up responsibility for society and teach classes of their choice to school children.
6. About good food habits for good health.

Course Outcomes:

Student should be capable of

1. To improve his concentration levels and improve his public speaking abilities.
2. To balance his academic and non-academic activities (Work Life Balance).
3. To widen his vision and increase his breadth of perspective in his journey of 4 years.
4. To improve his communications skills, leadership, teamwork and decision-making abilities.
5. To inculcate creativity & innovation, planning & organizing as part of their life.
6. Taking responsibility for themselves and people around them.
7. To make their journey more fun and enjoyable.

SYLLABUS

Unit-I

YES!+ Workshop:

Yoga Postures – Seven Layers To our Existence – Puzzles – Sources Of Energy – Live in the present Moment – Importance of Breath – Ujjai Breath – Pranayama – Sudarshana Kriya.

Unit-II

YES!+ Workshop:

Yoga Postures (Suryanamaskars) – Giving 100% in everything – Time management – Happiness point – Opposite Values – Pranayama – Sudarshan kriya

Unit-III

YES!+ Workshop:

Yoga Postures – Knowledge points (Acceptance, opinions discretion and handling mistakes) – Eye Gazing Process – Dance – Life Story process – Sudarshana Kriya (short) – Eternal life – Ego Bursting – Relationships – Parents – Studies – Compliments/Praising process.

Unit-IV

Creative Arts:

Photography – Sketching – Handy-crafts – Clay molding – Singing – Upcycling – Communing with nature – Creative writing.

Unit -V

Service:

Leadership in action – Contributing to society – Take up Responsibility –Empowerment – Public Speaking – Art of Teaching.

REFERENCE BOOKS:

1. Discourse on Patanjali Yoga Sutras by H.H Sri SriRavishankar
2. Human values and professional ethics byRRGaur,RSangal,GPBagaria
3. The Art of Photography by AL Judge
4. Hand made in India : Crafts of India by Ranjan Aditi

(Note: It is an uncredited course. It will not be included in the Grade Memo / Certificate. The Certificate will be issued based on the performance and attendance. This course attendance will be counted in the semester overall attendance.)



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

II/IV B.TECH

(With effect from **2017-2018** Admitted Batch onwards)

INFORMATION TECHNOLOGY I-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Internal Marks	Exam Marks	Total Marks
B17 IT 2101	Data Structures	3	3	1	--	4	30	70	100
B17 IT 2102	Java Programming	3	3	1	--	4	30	70	100
B17 BS 2105	Mathematical Foundations of Computer Science	3	3	1	--	4	30	70	100
B17 IT 2103	Computer Graphics	3	3	1	--	4	30	70	100
B17 IT 2104	Data Communications	3	3	1	--	4	30	70	100
B17 IT 2105	Digital Logic Design	3	3	1	--	4	30	70	100
B17 IT 2106	Data Structures Lab	2	--	--	3	3	50	50	100
B17 IT 2107	Java Programming Lab	2	--	--	3	3	50	50	100
B17 IT 2108	Basic Coding	1	--	--	2	2	50	---	50
B17 BS 2107	English Proficiency-I	--	1	1	--	2	--	--	--
Total		23	19	7	8	34	330	520	850

DATA STRUCTURES

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To be familiar with basic techniques handling problems with Data structures
2. Solve problems using data structures such as linear lists, stacks, queues, hash tables
3. Create and traverse different types of trees and graphs.
4. To practice different searching algorithms.

Course Outcomes:

Students will be able to

1. Apply advanced data structure strategies for exploring complex data structures and implement data structures like stacks, queues
2. Implement data structures on single, circular and double linked lists.
3. Implement different operations on trees.
4. Apply graphs to real time applications.
5. Perform sorting and searching using different algorithms.

SYLLABUS**UNIT-I Basic Concepts:**

Arrays, Dynamically Allocated Arrays, Structures and Unions, Polynomials, Sparse Matrix, Representation of Multi-dimensional Array, Pointers and Dynamic Memory Allocation, Algorithm Specification, Data Abstraction, Performance Analysis, performance Measurement.

Stacks and Queues:

Stack Abstract Data Type, Queue Abstract Data Type, Stacks and Queues using arrays, Expressions, Evaluating Postfix Expressions, Infix to Postfix, Multiple Stacks and Queues, Circular Queues using arrays.

UNIT-II Linked Lists:

Single Linked Lists and Chains, Representing Chains in C, Linked Stack and Queue using Linked List, Representing Polynomials as Singly Linked Lists, Adding Polynomials, Erasing Polynomials, Polynomials as Circularly Linked Lists, Additional List Operations, Sparse Matrix Representation, Doubly Linked Lists.

UNIT-III Trees:

Representation of Trees, Binary Trees Abstract Data Type, Properties of Binary Trees, Binary Tree Representations, Binary Tree Traversals, Additional Binary Tree Operations, Threaded Binary Trees, Heap Abstract Data Type, Priority Queues, Insertion into a max heap, Deletion from a max heap, Binary Search Trees: Searching a Binary Search Tree, Inserting an Element into a Binary Search Tree, Deleting an Element from a Binary Search Tree, Height of a Binary Search Tree.

UNIT-IV Graphs:

The Graph Abstract Data Type: Definitions, Graph Representations, Depth First Search, Breadth First Search, Spanning Trees, Minimum Cost Spanning Trees: Prim's and Kruskal's Algorithms, Single Source All Destination - Dijkstra's Algorithm, All Pairs Shortest Paths - Floyd's Algorithm, Transitive Closure using Warshall's Algorithm.

UNIT-V Searching and Sorting

Searching : Sequential Search, Binary Search, Interpolation Search.

Sorting : Selection Sort, Bubble Sort, Insertion Sort, Quick Sort, Merge Sort, Heap sort, Radix Sort.

Text Books:

1. Fundamentals of Data Structures in C, 2nd edition, Horowitz, Sahni and Anderson-Freed, Universities Press, 2008.

Reference Books:

1. Data Structures using C by Aaron M. Tenenbaum, Y. Langsam and M.J. Augenstein, Pearson Education, 2009.
2. Data Structures with C by Seymour Lipschutz, Schaum Outline series, 2010.
3. Data Structures using C by R. Krishna Moorthy G. Indirani Kumaravel, TMH, New Delhi, 2008.

JAVA PROGRAMMING

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To introduce the object oriented programming concepts.
2. To understand object oriented programming concepts, and apply them in solving problems.
3. To introduce the implementation of packages and interfaces
4. To introduce the concepts of exception handling and multithreading.
5. To introduce the design of Graphical User Interface using applets and swing controls.

Course Outcomes: At the end of the Course, Student will be :

1. Able to solve real world problems using OOP techniques.
2. Able to understand the use of abstract classes.
3. Able to solve problems using java I/o classes.
4. Able to develop multithreaded applications.
5. Able to develop applets for web applications.
6. Able to design GUI based applications

SYLLABUS**UNIT- I**

Fundamentals: HTML, OOP Concepts, Comparing JAVA with C & C++, JAVA Programming language Syntax, Variables, Data types, statements and expressions.

UNIT -II

Control Statements: If else, for, while, and do while loops, Switch statements, break and continue.

Arrays & Structures: One Dimensional & Two Dimensional Arrays, Functions: Parameter Passing, this and super keywords.

UNIT -III

Features of JAVA: Classes and Interfaces, Threads and multithreaded programming, Exception handling.

UNIT -IV

Introduction to packages, Math package, Lang package, Util package.

Stream based I/O (java.io) – The Stream classes-Byte streams and Character streams, Reading console Input and Writing Console Output, File class, Reading and writing Files.

UNIT -V

GUI Programming with Swing–Introduction, limitations of AWT, MVC architecture, components, containers. Understanding Layout Managers, Flow Layout, Border Layout, Grid Layout, Card Layout, Grid Bag Layout.

Event Handling- The Delegation event model- Events, Event sources, Event Listeners, Event classes, Handling mouse and keyboard events, Adapter classes, Inner classes, Anonymous Inner classes. A Simple Swing Application, **Applets** – Applets and HTML, Applets and Applications, passing parameters to applets. Creating a Swing Applet, Painting in Swing, A Paint example, Exploring Swing Controls- JLabel and Image Icon, JText Field, The Swing Buttons- JButton, JToggleButton, JCheckBox, JRadioButton, JTabbedPane, JScrollPane, JList, JComboBox, Swing Menus, Dialogs.

Text Books:

1. Introduction to Java programming, a primer , Balaguruswamy.
2. Java The complete reference, 9th edition, Herbert Schildt, McGraw Hill Education (India) Pvt. Ltd.

Reference Books:

1. Introduction to Java Programming , Daneal /Young PHI
2. Object Oriented Programming through Java, P. Radha Krishna.
3. Understanding Object-Oriented Programming with Java, updated edition, T. Budd, Pearson Education.

MATHEMATICAL FOUNDATIONS OF COMPUTER SCIENCE

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To understand the mathematical arguments using logical connectives and quantifiers and verify the validity of arguments using propositional, predicate logic and truth tables.
2. To understand various types of relations and discuss various properties of the relations.
3. To know about the concepts of counting techniques
4. To know how to solve the recurrence relations
5. To understand the concepts in graphs & also in Number theory

Course Outcomes:

At the end of the course students will be able to :

1. Rewrite the mathematical arguments using logical connectives and quantifiers and verify the validity of the arguments using propositional and predicate logic.
2. Identify and give examples of various types of relations and describe various properties of relations.
3. Solve different counting problems.
4. Solve the recurrence relations which occur in many fields.
5. Utilize the concepts in graphs and Number theory in their fields.

SYLLABUS

UNIT -I: Mathematical Logic:

Propositional Calculus: Statements and Notations, Connectives, Well Formed Formulas, Truth Tables, Tautologies, Equivalence of Formulas, Duality Law, Normal Forms, Theory of Inference for Statement Calculus, Consistency of Premises. Predicate Calculus: Predicative Logic, Statement Functions, Variables and Quantifiers, Free and Bound Variables, Inference Theory for Predicate Calculus.

UNIT -II:

Relations: Definition of Relation, Properties of Binary Relations, Relation Matrix and Digraph, Operations on Relations, Partition and Covering, Transitive Closure, Equivalence, Compatibility and Partial Ordering Relations, Hasse Diagrams.

Algebraic Structures: Algebraic Systems, Examples, General Properties, Semi Groups and Monoids, Homomorphism of Semi Groups and Monoids, Group, Subgroup, Abelian Group, Homomorphism, Isomorphism.

UNIT -III Combinatorics:

Basics of Counting, Permutations, Permutations with Repetitions, Circular Permutations, Restricted Permutations, Combinations, Restricted Combinations, Generating Functions of Permutations and Combinations, Binomial and Multinomial Coefficients, Binomial and Multinomial Theorems, The Principles of Inclusion–Exclusion, Pigeonhole Principle and its Application.

UNIT -IV: Recurrence Relations:

Generating Functions, Function of Sequences, Partial Fractions, Calculating Coefficient of Generating Functions, Recurrence Relations, Formulation as Recurrence Relations, Solving Recurrence Relations by Substitution and Generating Functions, Method of Characteristic Roots, Solving Inhomogeneous Recurrence Relations

UNIT -V:

Graph Theory: Basic Concepts of Graphs, Sub graphs, Matrix Representation of Graphs: Adjacency Matrices, Incidence Matrices, Isomorphic Graphs, Paths and Circuits, Eulerian and Hamiltonian Graphs, Multigraphs, Bipartite graphs, Planar Graphs, Euler's Formula.

Number Theory : Properties of Integers, Division theorem, Greatest Common Divisor, Euclidean Algorithm, Least Common Multiple, Testing for Prime Numbers, The Fundamental Theorem of Arithmetic, Modular Arithmetic (Fermat's Theorem and Euler's Theorem).

Text Books:

1. Discrete Mathematical Structures with Applications to Computer Science, J. P. Tremblay and P. Manohar, Tata McGraw Hill.
2. Discrete Mathematics for Computer Scientists and Mathematicians, J. L. Mott, A. Kandel, T.P. Baker, 2nd Edition, Prentice Hall of India.

Reference Books:

1. Elements of Discrete Mathematics-A Computer Oriented Approach, C. L. Liu and D.P. Mahopatra, 3rd Edition, Tata McGraw Hill.
2. Discrete Mathematics and its Applications with Combinatorics and Graph Theory, K. H. Rosen, 7th Edition, Tata McGraw Hill.
3. Discrete Mathematical Structures, Bernand Kolman, Robert C. Busby, Sharon Cutler Ross, PHI.
4. Discrete Mathematics, S. K. Chakraborty and B.K. Sarkar, Oxford, 2011.

COMPUTER GRAPHICS

Lecture : 3 Periods
Tutorial : 1 Period.
Exam : 3 Hrs.

Int. Marks : 30
Ext. Marks : 70
Credits : 3

Course Objectives:

1. Provides a comprehensive introduction to computer graphics with a foundation in Graphics Applications.
2. A thorough introduction to computer graphics techniques.
3. To give the basics of Geometric Transformations and projections.
4. To introduce three dimensional concepts and object representations with color models and basics of computer animation.

Course Outcomes:

1. The students will understand graphics principles and graphics hardware.
2. The students can demonstrate geometrical transformations.
3. The students can create interactive graphics applications and demonstrate computer graphics animation.

SYLLABUS**UNIT-I Introduction:**

Computer Graphics and their applications: Computer Aided Design, Computer Art, Entertainment, Education and Training, Graphical User Interfaces; Overview of Graphics systems: Video Display Devices, Raster Scan Systems, Random Scan Systems, Graphics Monitors And Workstations, Input Devices, Hard Copy Devices, Interactive Input Methods, Windows and Icons, Virtual Reality Environments, Graphics Software.

UNIT-II Output primitives:

Points and Lines, , Line and Curve Attributes, Color and Gray scale levels, Ant aliasing, Loading the Frame buffer, Line function, Line Drawing Algorithms, Circle Generating Algorithms, Ellipse Generating Algorithms, Pixel Addressing, Area Fill Attributes, Filled Area Primitives, Filled Area Functions, Cell Array, Character Generation, Character Attributes, Bundled Attributes.

UNIT - III Two Dimensional Transformations:

Basic 2D Transformations, Matrix Representations, Homogeneous Coordinates, Composite Transformations, Other Transformations, Transformations between Coordinate Systems, Affine Transformations.

UNIT-IV Three Dimensional Transformations & Projections:

Translation, Rotation, Scaling, Other Transformations, Composite Transformations, 3D Transformation Functions, Modeling and Coordinate Transformations, Need for projections, Parallel & Perspective projections, General Projection Transformations.

UNIT-V Viewing Pipeline and Clipping operations :

Viewing Pipeline ,Viewing Coordinates &Reference frames, Window-to-Viewport Coordinate Transformation, Two Dimensional Viewing Functions, , Three Dimensional Viewing, View Volumes, Clipping and its Operations, Types of clipping operations- Point Clipping, Line Clipping, Polygon Clipping,, Curve Clipping,, Text and Exterior Clipping.

Three Dimensional Concepts and Object representations: 3D display methods, 3D Graphics, Polygon Surfaces, Curved Lines and Surfaces, Quadratic Surfaces, Super Quadrics, Blobby Objects, Spline Representations, Bézier Curves and Surfaces, BSpline Curves and Surfaces,

Text Book:

1. Computer Graphics, Donald Hearn & M.Pauline Baker, Pearson Education ,New Delhi.

Reference Books:

1. Procedural Elements for Computer Graphics, David F.Rogers, Tata McGraw Hill Book Company, New Delhi, 2003.
2. Computer Graphics: Principles & Practice in C,J.D.Foley, S.KFeiner, AVan Dam F.H John Pearson Education ,2004.
3. Computer Graphics using Open GL,Francis S Hill Jr,Pearson Education,2004.
4. Computer Vision and Image Processing: A Practical Approach using CVIP tools, S. E.Umbaugh, Prentice Hall, 1998

DATA COMMUNICATIONS

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To study basics of data communication systems.
2. To study the various types of transmission media.
3. To study the various hardware concepts related to data communications.
4. To discuss about modem and multiplexing techniques.

Course Outcomes: At the end of the Course, Student will be able to:

1. Understand basic concepts related communication systems.
2. Understand different transmission Media.
3. Understand concepts related to data communication hardware.
4. Understand basic functionality of modems.

SYLLABUS**UNIT- I**

Introduction to Data Communications: A Communications Model, Data Communications and Data Communications Networking, Protocols and Protocol Architecture, Characteristics of Data Transmission: Concepts and Terminology, Analog and Digital Data Transmission, Transmission Impairments

UNIT- II

Transmission Media: Guided Transmission Media, Wireless Transmission. Data Encoding: Digital Data-Digital Signals, Digital Data-Analog Signals, Analog Data-Digital Signals, Analog Data-Analog Signals.

UNIT- III

Data Communication Interface: Asynchronous and Synchronous Transmission, Line Configurations, Interfacing. Data Link Control Flow Control, Error Detection, Error Control

UNIT- IV

Multiplexing: Frequency-Division Multiplexing, Synchronous Time-Division Multiplexing: Characteristics, TDM Link Control, Digital Carrier Systems, Statistical Time-Division Multiplexing: Characteristics.

UNIT -V

Data Communications Hardware: Terminals: Introduction, Basic Terminal Components, Enhanced Terminal Components, General-Purpose Terminals, Remote Job Entry Terminals, Transaction Terminals, Clustering of Terminal Devices. Communication Processing Hardware: Introduction, Switching Processors, Multidrop Lines, Multiplexers, Concentrators, Front-End Processors

Text Books:

1. William Stallings, Data and Computer Communications, 7th Edition, PH/Pearson Edu.Inc.
2. Mary E.S. Loomis, Data Communications, PHI-N.J., 1983 (Unit V).

Reference Books:

1. Behrouz A. Forouzan, Data Communications and Networking, 3rd Edition TMH, 2004.
2. William A. Shay, Understanding Data Communications & Networks, 2nd Edition Thomson-Brooks/Cole –Vikas Publishing House, 1999.
3. Michale A. Miller, Data & Network Communications, Thomson/Delmar –Vikas Pub. House, 2000

DIGITAL LOGIC DESIGN

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To introduce the basic principles for design of combinational circuit and sequential circuits.
2. To learn simple digital circuits in preparation for computer engineering.

Course Outcomes:

1. An ability to define different number systems, binary addition and subtraction, 2's complement representation and operations with this representation. The different Boolean algebra theorems and apply them for logic functions.
2. An ability to define the Karnaugh map for a few variables and perform an algorithmic reduction of logic functions.
3. An ability to define the following combinational circuits: multiplexer, de-multiplexers encoders/decoders, comparators, arithmetic-logic units and to be able to build simple circuits.
4. An ability to understand asynchronous and synchronous sequential circuits, like counters and shift registers.
5. An ability to understand memories like RAM and ROM, Programmable Logic Array and Programmable Array Logic.

SYLLABUS**UNIT - I**

Binary Systems and Boolean Algebra : Digital Systems. Binary Numbers. Number Base Conversions. Octal and Hexadecimal Numbers. Complements. Signed Binary Numbers. Binary Codes. Binary Storage and Registers. Binary Logic ,Basic Definitions of Boolean algebra. Axiomatic Definition of Boolean Algebra. Basic Theorems and Properties of Boolean Algebra, Boolean Functions.

UNIT - II

Logic Gates and Gate-Level Minimization: Canonical and Standard Forms. Logic Operations.. The Map Method. Four-Variable Map. Five-Variable Map. Product of Sums Simplification. Don't-Care Conditions. Digital Logic Gates. NAND and NOR Implementation. Other Two-Level Implementations. Exclusive-OR Function

UNIT -III

Combinational Logic Design,:

Combinational Circuits: Analysis Procedure. Design Procedure. Binary Adder-Subtractor. Decimal Adder. Binary Multiplier. Magnitude Comparator. Decoders. Encoders. Multiplexers.

UNIT - IV

Sequential Logic design: Sequential Circuits .Latches. Flip-Flops. Analysis of Clocked Sequential Circuits. State Reduction and Assignment. Designs Procedure. Registers. Shift Registers. Ripple Counters. Synchronous Counters. Other Counters.

UNIT-V

Memory and Programmable Logic: Introduction. Random-Access Memory. Memory Decoding, Error Detection and Correction. Read-Only Memory. Programmable Logic Array. Programmable Array Logic.

Text Book:

1. Digital Design, 3rd Edition, M.Morris Mano, Pearson Education.

Reference Books:

1. Digital Logic Design Principles, Norman Balabanian & Bradley Carlson, John Wiley & Sons(Asia) Pvt.Ltd.,2002
2. Switching and Finite Automata Theory, 2nd Edition ZVI Kohavi Tata McGraw Hill
3. Switching Theory and Logic Design ,3rd Edition ,A.Anand Kumar PHI Learning Pvt.Ltd.

DATA STRUCTURES LAB**Lab : 3 Periods****Exam : 3 Hrs.****Int.Marks : 50****Ext. Marks : 50****Credits : 2**

Course Objectives:

1. To be familiar with basic techniques handling problems with Data structures
2. Solve problems using data structures such as linear lists, stacks, queues, hash tables
3. Create and traverse different types of trees and graphs.
4. To practice different searching algorithms.

Course Outcomes:

Students will be able to

1. Apply advanced data structure strategies for exploring complex data structures.
2. Implement data structures like stacks, queues
3. Implement data structures on single, circular and double linked lists.
4. Implement different operations on trees.
5. Apply graphs to real time applications.
6. Perform sorting and searching using different algorithms.

LIST OF PROGRAMS

1. Write a program to implement the operations on stacks using Array.
2. Write a program for evaluating a given postfix expression
3. Write a program for converting a given infix expression to postfix form
4. Write a program to implement the operations on queues using Array.
5. Write a program to implement the operations on circular queues using Array.
6. Write a program to implement the Single Linked List operations (Insertion, Deletion).
7. Write a program to implement the operations on stacks using Linked List.
8. Write a program to implement the operations on Queue using Linked List.
9. Write a program to implement the Priority Queue operations using single Linked List.
10. Write a program to add two Polynomials using Linked List.
11. Write a program to add two sparse matrices using linked list
12. Write a program to implement the Circular Single Linked List operations (Insertion, Deletion).
13. Write a program to implement the Double Linked List operations (Insertion, Deletion).
14. Write a program to implement the De-queue operations using Double Linked List .
15. Write a program to create a binary search tree and for implementing the in order, preorder, post order traversal using recursion

16. Write a program for finding the Depth First Search of a graph, and Breadth First Search of a graph
17. Write a program for sorting a list using Bubble sort and then apply binary search.
18. Write a program for quick sort
19. Write a program for Heap sort
20. Write a program for Merge sort.
21. Write a program for finding the transitive closure of a digraph

Text Books:

1. Fundamentals of Data Structures in C, 2nd edition, Horowitz, Sahni and Anderson-Freed, Universities Press, 2008.

Reference Books:

1. Data Structures using C by Aaron M. Tenenbaum, Y. Langsam and M.J. Augenstein, Pearson Education, 2009.
2. Data Structures with C by Seymour Lipschutz, Schaum Outline series, 2010.
3. Data Structures using C by R. KrishnaMoorthy G. Indirani Kumaravel, TMH, New Delhi, 2008.

JAVA PROGRAMMING LAB**Lab : 3 Periods****Exam : 3 Hrs.****Int.Marks : 50****Ext. Marks : 50****Credits : 2****Course Objectives:**

1. To demonstrate the compilation and interpretation of java programs
2. To make practice on different Object Oriented Programming Concepts
3. To write java programs to implement java concepts
4. To write java programs to implement applet programming
5. To write java programs to generate and handle event handling programs

Course Outcomes:

1. Students will be able to understand compiling and interpreting programs.
2. Students will be able to Explore features of Object Oriented Programming.
3. Students will be able to implement various java concepts
4. Students will be able to Develop java Programs to implement applets
5. Students will be able to Develop java Programs to generate and handle events.

LIST OF PROGRAMS

1. (a) Program to display the area of a rectangle.
(b) Program to find Sum of series $1+x+x^2+x^3+\dots$
2. (a) Write a class to display the area of rectangle and inherit this class into other class which is displaying perimeter of a rectangle and implement.
(b) Write a class to add three no's inherit this class into other class to add five no's and implement it.
3. (a) Write a program to print the path, filename and extension for a given path of a file.
(b) Write a program to receive two command line arguments check whether they are equal or not.
4. (a) A program to take two arguments and divide the first argument with second argument and display the result. Displays the error message if divide by zero without abnormal exit.
(b) A program to accept more than one string and arrange them in alphabetical order.
(c) Write a program to display simultaneously output of even and odd numbers starting from one to specified number.
5. Write a program to accept data from keyboard and write it into a file.

6. Write a java program to implement stack & Queue operations.
7. Write a program to draw line and circle using mouse.
8. Write an applet program for drawing the bar chart.
9. Write an applet program to design a calculator for implementing basic functions like +, -, *, /.
10. Write a program to check active ports in system.

Reference Books:

1. Introduction to Java programming, a primer II, Balaguruswamy.
2. Java Complete Reference , Herbt Schild.
3. Introduction to Java programming, Daneal/Young PHI

BASIC CODING**Lab: 2 Periods****Sessionals : 50****Exam : 3 Hrs.****Credits : 1****Course Objectives:**

1. To develop programming skills among the students.
2. To familiarize the student with Control Structures, Loop Structures.
3. To familiarize the student with Basic searching and sorting Methods.
4. To familiarize the student with Functions, Recursions and Storage Classes.
5. To familiarize the student with Structures and Unions.
6. To familiarize the student with Operating System concepts.
7. To familiarize the student with Networking concepts.

Course Outcomes:**At the end of the course students will be able to**

1. Know about Control Structures, Loop Structures and branching in programming.
2. Know about various searching and sorting methods.
3. Know about Functions, Recursions and Storage Classes.
4. Know about Structures and Unions.
5. Know different Operating System concepts.
6. Differentiate OSI Model Vs. TCP/IP suite.

SYLLABUS**UNIT I Review of Programming constructs**

Programming Environment, Expressions formation, Expression evaluation, Input and Output patterns, Control Structures, Sequential branching, Unconditional branching, Loop Structures, Coding for Pattern Display.

UNIT II Introduction to Linear Data, strings and pointers

Structure of linear data, Operation logics, Matrix forms and representations, Pattern coding, Working on character data, Compiler defined methods, Substitution coding for defined methods, Row Major representation, Column Major representation, Basic searching and sorting Methods.

UNIT III Functions, Recursions and Storage Classes

Functions – Introduction to modular programming – Function Communication - Pass by value, Pass by reference – Function pointers – Recursions – Type casting – Storage classes

Practice: programs on passing an array and catching by a pointer, function returning data, comparison between recursive and Iterative solutions.

Data referencing mechanisms: Pointing to diff. data types, Referencing to Linear data, Runtime-memory allocation, Named locations vs pointed locations, Referencing a 2D-Matrix

UNIT IV User-defined datatypes, Pre-processor Directives and standard storage

Need for user-defined data type – structure definition – Structure declaration – Array within a Structure – Array of Structures – Nested Structures - Unions – Declaration of Union data type, Struct Vs Union - Enum – Pre-processor directives , Standard storage methods, Operations on file, File handling methods, Orientation to Object oriented programming

Practice: Structure padding, user-defined data storage and retrieval programs

UNIT V Operating system principles and Database concepts

Introduction to Operating system principles, Process scheduling algorithms, Deadlock detection and avoidance, Memory management, Networking: Introduction to Networking, OSI Model Vs. TCP/IP suite, Datalink layer, Internet layer, DVR Vs. LSR, Transport Layer, Application Layer

References:

1. Computer Science, A structured programming approach using C, B.A.Forouzan and R.F.Gilberg, 3rd Edition, Thomson, 2007.
2. The C –Programming Language, B.W. Kernighan, Dennis M. Ritchie, Prentice Hall India Pvt.Ltd
3. Scientific Programming: C-Language, Algorithms and Models in Science, Luciano M. Barone (Author), Enzo Marinari (Author), Giovanni Organtini, World Scientific .
4. Object Oriented Programming in C++: N. Barkakati, PHI.
5. Object Oriented Programming through C++ by Robat Laphore.
6. <https://www.geeksforgeeks.org/>.
7. <https://www.tutorialspoint.com/>

ENGLISH PROFICIENCY-I
(Common to All Branches)

Lecture	: 1Period		
Tutorial	: 1 Periods	Int.Marks	: --
Exam	: ---	Credits	: --

AIM:

Enriching the communicative competency of the students by adopting the activity-based as well as the class-oriented instruction with a view to facilitate and enable them to enhance their language proficiency skills.

Course Objectives:

The Students will be able to

1. Communicate their ideas and views effectively.
2. Practice language skills and improve their language competency.
3. Know and perform well in real life contexts.
4. Identify and examine their self attributes which require improvement and motivation.
5. Build confidence and overcome their inhibitions, stage freight, nervousness, etc.
6. Improve their innovative practices in speaking.

Course Outcomes:

The Students will

1. Improve speaking skills.
2. Enhance their listening capabilities.
3. Learn and practice the skills of composition writing.
4. Enhance their reading and understanding of different texts.
5. Improve their inter-personal communication skills.
6. Be confident in presentation skills.

SYLLABUS

UNIT-1:LISTENING

Selected Motivational Speeches

Selected Moral Stories

UNIT-2:SPEAKING

Book Review

Skit Presentation

PowerPoint Presentations

Describing event/place/thing

Extempore

Group Discussion

Picture Perception and Describing Test

UNIT-3:READING

Speeded Reading

Reading Comprehension

UNIT-4:WRITING

Paragraph Writing

Literary Appreciation – Understanding the Language of Literature

UNIT-5:PROJECT

Ad Making

Reference Books:

1. Classic Short Stories-A Reader's Digest Selection
2. English for Colleges by Brendan J. Carroll, Macmillan Publications
3. The World's Great Speeches- edited by B.S. Sekhar, Jeet Publications
4. Fundamentals of Technical Communication by Meenakshiraman, Sangeta Sharma of OUP
5. English and Communication Skills for Students of Science and Engineering, by S.P. Dhanavel, Orient Blackswan Ltd. 2009
6. Enriching Speaking and Writing Skills, Orient Blackswan Publishers
7. The Oxford Guide to Writing and Speaking by John Seely OUP

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R17)

II/IV B.TECH
(With effect from **2017-2018** Admitted Batch onwards)

INFORMATION TECHNOLOGY

II-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/ Week	Internal Marks	Exam Marks	Total Marks
B17IT 2201	Computer Organization	3	3	1	--	4	30	70	100
B17BS 2202	Probability, Statistics & Queuing Theory	3	3	1	--	4	30	70	100
B17IT 2202	Microprocessors	3	3	1	--	4	30	70	100
B17IT 2203	File Structures	3	3	1	--	4	30	70	100
B17IT 2204	Unix & Shell Programming	3	3	1	--	4	30	70	100
B17IT 2205	Formal Language and Automata Theory	3	3	1	--	4	30	70	100
B17IT 2206	Python Programming Lab	2	--	--	3	3	50	50	100
B17IT 2207	Digital Electronics And Micro Processors Lab	2	--	--	3	3	50	50	100
B17IT 2208	Advanced Coding	1	--	--	2	2	50	---	50
B17BS 2204	Professional Ethics & Human Values	--	2	--	--	2	--	--	--
B17BS 2206	English Proficiency-II	--	1	1	--	2	--	--	--
Total		23	21	7	8	36	330	520	850

COMPUTER ORGANIZATION

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To study about structure and functional components of a computer.
2. Understanding the hierarchical organization of a computer system which consists of instruction set of commands.
3. Learn about the architecture of a computer from a programming view.
4. To design a balance system that minimizes performance and utilization of all elements.

Course Outcomes:

1. Knowledge about major components of a computer such as processor, memory and I/O modules along with their interconnections internally with outside world.
2. Detailed idea about architecture of central processing unit, functions of control unit, memory, I/O devices and their issues.
3. Simple and multiple processor organization and their issues.

SYLLABUS**UNIT-I**

Register Transfer and Micro operations: Register Transfer Language, Register Transfer, Bus and Memory Transfers, Arithmetic Micro operations, Logic Micro operations, Shift Micro operations, Arithmetic Logic Shift Unit.

UNIT-II

Basic Computer Organization and Micro programmed Control : Instruction Codes, Computer Registers, Computer Instructions, Timing and Control, Instruction Cycle, Memory-Reference Instructions, Input- Output and Interrupt, Complete Computer Description, Design of Basic Computer, Control Memory, Address Sequencing, Micro program Example, Design of Control Unit.

UNIT-III

Central Processing Unit: Introduction, General Register Organization, Stack Organization, Instruction Formats, Addressing Modes, Data Transfer and Manipulation, Program Control, Reduced Instruction Set Computer(RISC)

UNIT-IV

Input/output Organization: Peripheral Devices, I/O interface, Asynchronous data transfer, Modes of transfer, priority Interrupt, Direct memory access, Input-Output Processor (IOP), Serial Communication.

UNIT-V

Memory Organization: Memory Hierarchy, Main memory, Auxiliary memory, Associate Memory, Cache Memory, and Virtual memory, Memory Management Hardware.

Text Books:

1. Computer System Architecture, M. Morris Mano, Prentice Hall of India Pvt. Ltd., Third Edition, Sept. 2008.

Reference Books:

1. Computer Architecture and Organization, William Stallings, PHI Pvt. Ltd., Eastern Economy Edition, Sixth Edition, 2003.
2. Computer Organization and Architecture, Linda Null, Julia Lobur, Narosa Publications ISBN 81- 7319-609-5.
3. Computer System Architecture”, John. P. Hayes.

PROBABILITY, STATISTICS AND QUEUING THEORY

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To illustrate the concept of a random variable, generating functions and their properties
2. To learn different probability functions and analyse various statistical measures of a few discrete/continuous distributions.
3. To understand and compute the correlation coefficient, and estimation techniques from regression lines.
4. To fit a linear or nonlinear curves using method of least squares.
5. To develop a framework for testing of hypothesis in giving inferences about Population Parameters.
6. To study Queuing models and their Characteristics.

Course Outcomes:

At the end of the course a student will be able to

1. Identify the random variable as discrete/continuous and analyse it.
2. Predict the distribution suitable for the given data from its moments.
3. Measure the intensity of association between the variables.
4. Fit a best suitable Curve for the given data.
5. Decide the test applicable for giving inference about Population Parameter based on Sample statistic.
6. Make business decisions about the resources needed to provide a service in day-to-day life applications including telecommunication, traffic engineering, computing and the design of factories, shops, offices and hospitals.

SYLLABUS**UNIT -I**

Random Variables and Probability functions: Review on basic concepts of Probability (no questions will be set on review), Definition of a random variable, Distribution function, Properties of Distribution Function, Discrete Random Variable, Probability Mass Function, Discrete Distribution Function, Continuous Random Variable, Probability Density Function, Continuous Distribution Function.

Mathematical Expectation: Mathematical Expectation of a Random Variable, Expected Value of function of a Random Variable, Addition Theorem and Multiplication Theorem of Expectation(**without proofs**), Statistical Measures like Mean, Variance, Moments and Covariance in terms of Expectations.

Generating functions: Moment generating Function, Characteristic Function and Probability generating Function of a Random Variable.

UNIT II

Discrete Distributions: Binomial distribution and Poisson distribution - Definition, Mean, Variance, moments, m.g.f., Characteristic function, p.g.f., Fitting of distributions.

Continuous Distributions: Normal Distribution - Definition, Standard Normal Variate, Mean, Variance, m.g.f., Characteristic function Applications of Normal Distribution, Importance of Normal distribution. Exponential Distribution, Definition, Mean, Variance and Memory less property of Exponential distribution.

UNIT III

Curve fitting: Method of least Squares, fitting of a Straight line, Fitting of a Parabola.

Correlation: Definition, Karl Pearson's Coefficient of Correlation, Limits for correlation coefficient, Rank Correlation, Spearman's formula for rank correlation coefficient.

Regression Analysis: Regression Lines, Regression Coefficients and their properties(without proofs)

UNIT IV

Sampling Theory: Sample, population, statistic, parameter, Sampling distribution, standard error, point and interval estimation. Testing of Hypothesis: Formulation of Null hypothesis, Alternative hypothesis, Critical region, level of significance, Errors in sampling- Type-I-error, Type-II-error, One-tailed and Two-tailed tests.

Large Sample Theory: Test of significance of single sample proportion, Test of significance for difference of proportions.

Small Sample Theory: Degrees of freedom, Student's-t-distribution: definition, t-test for single mean, t-test for difference of means, Paired t-test for difference of means.

F-distribution: definition, F-test for equality of two population variances. Chi-square distribution: definition, Chi-square test for goodness of fit, Chi-square test for Population Variance.

UNIT V

Queueing Theory: Queue description, Birth and Death Process, Distribution of Inter-arrival times, Distribution of service times, Kendall's representation of a queueing model, Operating characteristics of a queueing model, steady-state solutions of $\{M/M/1: \infty/FCFS\}$ Model and $\{M/M/1; N/FCFS\}$ Model.

Text Books:

1. Fundamentals of Mathematical Statistics by S.C.Gupta and V.K.Kapoor, Sultan Chand & Sons Publishers.
2. Probability, Statistics and Random Processes by T.Veerarajan, Tata Mc Graw Hill Pub.

Reference Books:

1. Probability & Statistics with Reliability, Queueing and Computer Applications by Kishore.S.Trivedi, Prentice Hall of India, 1999.
2. Probability and statistics for Engineers, Miller and Freund, 7th edition, Prentice-Hall India.
3. Probability and statistics for Engineers and Scientists by Ronald E. Walpole, Raymond H. Myers, Sharon L. Myers and Keying Ye, Eighth edition, Pearson Education.

MICROPROCESSORS

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To discuss about 8085 architecture, signal description and instruction set.
2. To study different programming techniques to implement in assembly language
3. To discuss about 8086 architecture, signal description and instruction set.
4. To study different peripheral devices and learn to interface with 8085.
5. To study different programming techniques to implement in MASM.

Course Outcomes: At the end of the Course:

1. Student will able to identify microprocessor and microcomputers and will be able to describe 8085 MP architecture and classify instructions .
2. Student will able to state and illustrate 8085 programming techniques and_solve_code conversions, ISR, subroutines, operations to examine results.
3. Student will able to describe 8086 MP architecture and classify instruction set of 8086 .
4. Student will able to state and illustrate 8086 programming techniques and solve code conversions, ISR, subroutines, operations to examine results.

SYLLABUS

UNIT I

Introduction to 8085 microprocessor

Internal Architecture functional/signal description of 8085 microprocessor, Instruction set, Addressing modes and programming in 8085.

UNIT II

Programming techniques

Timing diagram, counters and delays, stacks and subroutines and Interrupts in 8085

UNIT III

Memory and I/O

Classification and interfacing semiconductor memories with 8085 MPU. Interfacing characteristics of IO devices, IO device addressing methods.

UNIT IV

Peripheral devices and interfacing with 8085

Interfacing peripherals to INTEL 8085: Parallel IO interface-8255, Serial IO Interface-8251, Timer Interface-8253. Interfacing peripherals to INTEL 8085: Keyboard/Display Interface-8279, Interrupt controller Interface-8259.

UNIT V

Introduction to 8086 microprocessor and programming

The 8086 Microprocessor architecture, Internal Architecture & functional /signal description of 8086, segmented memory, Maximum 7 Minimum mode of 8086. Introduction set and programming the 8086: Addressing modes, Instruction set and assembly language programming techniques with 8086.

Text Books:

1. Microprocessor Architecture and Applications with the 8085 , Ramesh S. Gaonkar, 4th Edition, Penram International, 1999
2. Advanced Microprocessors and Peripherals, A K RAY & K M Bhurchandi , 2nd Edition, The McGraw-Hill companies.

Reference Books:

1. The 80X86 Family , Design, Programming and Interfacing, John E. Uffenbeck, 3rd Edition, Pearson Education Inc., 2002.
2. Walter A . tribal and Avatar Singh. The 8088 and 8086 Microprocessors, Programming interfacing, software, hardware and Applications, 4th Edition Pearson education Inc., 2003
3. Microprocessors and Interfacing. Programming and hardware, 2nd Edition, Douglass V. Hall. MH Edition , 1999.

FILE STRUCTURES

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To discuss about basics about file related operations
2. To study about different storage devices & retrieval techniques
3. To discuss about file organizations & compression techniques.
4. To study different index structures for retrieval.
5. To study different hashing techniques.

Course Outcomes: At the end of the Course:

1. Student will able to identify the basic operations on a file.
2. Student will able to state and illustrate various storage & retrieval mechanisms
3. Student will able to describe various compression methods & advantages of them.
4. student will be able to describe various index structures.
5. Student will able to state and illustrate hashing methods for direct access of data from files

SYLLABUS**UNIT-I: File Processing Operations and Secondary Storage**

Physical and logical files, opening, reading & writing and closing files in C, seeking and special characters in files, physical devices and logical files, file-related header files in C. Disks – organization, tracks, sectors, blocks, capacity, non-data overhead, cost of a disk access, magnetic Tape –types, performance, organization estimation of tape length and data transmission times, disk vs tape, CD-ROM – CD-ROM as a file structure, physical organization, strengths and weakness of CD-ROMs, storage hierarchy

UNIT-II : Byte Journey and buffer Management and File Structure Concepts

File manager, I/O buffer, I/O processing, buffer strategies and bottlenecks. A stream file, field structures, reading a stream of fields, record structures and that uses a length indicator, Mixing numbers and characters – use of a hex dump, reading the variable length records from the files.

UNIT-III: Managing records in C files and Organizing files for performance

Retrieving records by keys, sequential search, direct access, choosing a record structure and record length, header records, file access and file organization. Data compression, reclaiming space – record deletion and storage compaction, deleting fixed-length records for reclaiming space dynamically, deleting variable-length records, space fragmentation, replacement strategies.

UNIT-IV: Indexing and Indexed sequential file access and prefix B+ Trees

Index, A simple index with an entry sequenced file, basic operations on an indexed, entry sequenced file, indexes that are too large to hold in memory, indexing to provide access by multiple keys, retrieval using combination of secondary keys, improving the secondary index structure – inverted lists. Indexed sequential access, maintaining a sequence set, adding a simple index to the sequence set, the B tree, simple prefix B+ content of the index: separators instead of keys, the simple prefix tree maintenance, index set block size, internal set block size, internal structure of index set blocks, loading a simple prefix

UNIT-V: Hashing and Extendable hashing

Collisions in hashing, a simple hashing algorithms, hashing functions and record distributions, memory requirements, collision resolution by progressive overflow, buckets, deletions. Working of extendable hashing, implementation, deletion, extendable hashing performance

Text Book:

1. File Structures – An Object Oriented Approach with C++ by Michael J. Folk, Bill Zoellick and Greg Riccardi,, Pearson

Reference Books:

1. File Organization & Processing by Alan I.Tharp Malloy Lithographing Inc.
2. File Systems by Thomas Harbron

UNIX AND SHELL PROGRAMMING

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. Facility with UNIX command syntax and semantics.
2. Ability to work on the File systems and Directories in Unix operating system.
3. Ability to read and understand and specifications, shell scripts and programs.
4. To become fluent with system calls provided in UNIX environment.

Course Outcomes:

1. Able to working on the basic commands of UNIX operating system.
2. File processing projects will require data organization, problem solving and research.
3. Scripts and programs will demonstrate effective use of structured programming.
4. Scripts and programs will be accompanied by printed output demonstrating completion of a test plan.
5. Able to understand and handle the process management using system calls

SYLLABUS**UNIT-I:**

Introduction to Unix-Brief History-What is Unix-Unix Components-Using Unix-Commands in Unix-Some Basic Commands-Command Substitution-Giving Multiple Commands.

UNIT-II:

The File system –The Basics of Files-What’s in a File-Directories and File Names-Permissions-I Nodes-The Directory Hierarchy, File Attributes and Permissions-The File Command knowing the File Type-The Chmod Command Changing File Permissions-The Chown Command Changing the Owner of a File-The Chgrp Command Changing the Group of a File.

UNIT-III:

Using the Shell-Command Line Structure-Met characters-Creating New Commands-Command Arguments and Parameters-Program Output as Arguments-Shell Variables- -More on I/O Redirection-Looping in Shell Programs. Filters-The Grep Family-Other Filters-The Stream Editor Sed-The AWK Pattern Scanning and processing Language-Good Files and Good Filters.

UNIT-IV:

Shell Programming-Shell Variables-The Export Command-The Profile File a Script Run During Starting-The First Shell Script-The read Command-Positional parameters-The \$? Variable knowing the exit Status-More about the Set Command-The Exit Command-Branching Control Structures-Loop Control Structures-The Continue and Break Statement-The Expr Command: Performing Integer Arithmetic-Real Arithmetic in Shell Programs-The here Document(<<)-The Sleep Command-Debugging Scripts-The Script Command-The Eval Command-The Exec Command.

UNIT-V:

The Process-The Meaning-Parent and Child Processes-Types of Processes-More about Foreground and Background processes-Internal and External Commands-Process Creation-The Trap Command-The Stty Command-The Kill Command-Job Control.

Text Books:

1. The Unix programming Environment by Brian W. Kernighan & Rob Pike, Pearson.
2. Introduction to Unix Shell Programming by M.G.Venkateshmurthy, Pearson.

Reference Books:

1. Unix and shell Programming by B.M. Harwani, OXFORD university press.
2. Unix and shell Programming by Sumitabha Das, TMH.
3. Unix and shell Programming by Forouzan, Gil Berg.

FORMAL LANGUAGE AND AUTOMATA THEORY

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

The aim of this course is,

1. To introduce the concepts in automata theory and theory of computation to design grammars and recognizers for different formal languages.
2. To Employ finite state machines to solve problems in computing.
3. To introduce finite state machines, context free grammars and Turing machines and their properties as the basic for the formal expressivity of computer languages for solving linguistic decision problems.

Course Outcomes :

1. Students will be able to on **design** Finite Automata for languages with concepts of Regular Sets and Regular Grammars [K3]
2. Students will be able to **Apply** concepts of context free Grammars and able to **design** Push Down Automata from the given CFG. [K3]
3. Students will be able to **design** a Turing Machine from the given language [K3]
4. Students will be able of **identify** different types of languages using Chomsky Hierarchy and **apply** concepts of Un-decidability on problems [K3].

SYLLABUS**UNIT – I: Finite Automata**

Why Study Automata Theory? The Central Concepts of Automata Theory, Automation, Finite Automation, Transition Systems, Acceptance of a String by a Finite Automation, DFA, Design of DFAs, NFA, Design of NFA, Equivalence of DFA and NFA, Conversion of NFA into DFA, Finite Automata with E-Transition, Minimization of Finite Automata, Mealy and Moore Machines, Applications and Limitation of Finite Automata.

UNIT – II: Regular Expressions

Regular Expressions, Regular Sets, Identity Rules, Equivalence of two Regular Expressions, Manipulations of Regular Expressions, Finite Automata, and Regular Expressions, Inter Conversion, Equivalence between Finite Automata and Regular Expressions, Pumping Lemma, Closers Properties, Applications of Regular Expressions, Finite Automata and Regular Grammars, Regular Expressions and Regular Grammars.

UNIT – III: Context Free Grammars

Formal Languages, Grammars, Classification of Grammars, Chomsky Hierarchy Theorem, Context Free Grammar, Leftmost and Rightmost Derivations, Parse Trees, Ambiguous Grammars, Simplification of Context Free Grammars-Elimination of Useless Symbols, E-Productions and Unit Productions, Normal Forms for Context Free Grammars-Chomsky Normal Form and Greibach Normal Form, Pumping Lemma, Closure Properties, Applications of Context Free Grammars.

UNIT – IV: Pushdown Automata

Pushdown Automata, Definition, Model, Graphical Notation, Instantaneous Description Language Acceptance of pushdown Automata, Design of Pushdown Automata, Deterministic and Non – Deterministic Pushdown Automata, Equivalence of Pushdown Automata and Context Free Grammars Conversion, Two Stack Pushdown Automata, Application of Pushdown Automata.

UNIT – V: Turning Machine & Computability

Turing Machine, Definition, Model, Representation of Turing Machines-Instantaneous Descriptions, Transition Tables and Transition Diagrams, Language of a Turing Machine, Design of Turing Machines, Techniques for Turing Machine Construction, Types of Turing Machines, Church's Thesis, Universal Turing Machine, Restricted Turing Machine. Decidable and Un-decidable Problems, Halting Problem of Turing Machines, Post's Correspondence Problem.

Note: Theorem proofs are eliminated.

Text Books:

1. Introduction to Automata Theory, Languages and Computation, J.E.Hopcroft, R.Motwani and J.D.Ullman, 3rd Edition, Pearson, 2008.
2. Theory of Computer Science-Automata, Languages and Computation, K.L.P.Mishra and N.Chandrasekharan, 3rd Edition, PHI, 2007.

Reference Books:

1. Formal Language and Automata Theory, K.V.N.Sunitha and N.Kalyani, Pearson, 2015.
2. Introduction to Automata Theory, Formal Languages and Computation, Shyamalendu Kandar, Pearson, 2013.
3. Theory of Computation, V.Kulkarni, Oxford University Press, 2013.
4. Theory of Automata, Languages and Computation, Rajendra Kumar, McGraw Hill, 2014.

PYTHON PROGRAMMING LAB

Lab : 3 Periods

Exam : 3 Hrs.

Int.Marks : 50

Ext. Marks : 50

Credits : 2

Course Objectives:

1. Introduction to Scripting Language
2. Exposure to various problems solving approaches of computer science

Course Outcomes: At the end of the Course:

1. Making Software easily right out of the box.
2. Experience with an interpreted Language.
3. To build software for real needs.
4. Prior Introduction to testing software

LIST OF PROGRAMS

Exercise 1 - Basics

- a) Running instructions in Interactive interpreter and a Python Script
- b) Write a program to purposefully raise Indentation Error and Correct it

Exercise 2 - Operations

- a) Write a program to compute distance between two points taking input from the user (Pythagorean Theorem)
- b) Write a program add.py that takes 2 numbers as command line arguments and prints its sum.

Exercise - 3 Control Flow

- a) Write a Program for checking whether the given number is a even number or not.
- b) Using a for loop, write a program that prints out the decimal equivalents of $1/2$, $1/3$, $1/4$, $\dots$, $1/10$
- c) Write a program using a for loop that loops over a sequence. What is sequence?
- d) Write a program using a while loop that asks the user for a number, and prints a countdown from that number to zero.

Exercise 4 - Control Flow - Continued

- a) Find the sum of all the primes below two million.
Each new term in the Fibonacci sequence is generated by adding the previous two terms. By starting with 1 and 2, the first 10 terms will be:
1, 2, 3, 5, 8, 13, 21, 34, 55, 89, ...
- b) By considering the terms in the Fibonacci sequence whose values do not exceed four million, find the sum of the even-valued terms.

Exercise - 5 - DS

- a) Write a program to count the numbers of characters in the string and store them in a dictionary data structure
- b) Write a program to use split and join methods in the string and trace a birthday with a dictionary data structure.

Exercise - 6 DS - Continued

- a) Write a program combine_lists that combines these lists into a dictionary
- b) Write a program to count frequency of characters in a given file. Can you use character frequency to tell whether the given file is a Python program file, C program file or a text file?

Exercise – 7 Files

- a) Write a program to print each line of a file in reverse order.
- b) Write a program to compute the number of characters, words and lines in a file.

Exercise - 8 Functions

- a) Write a function ball collide that takes two balls as parameters and computes if they are colliding. Your function should return a Boolean representing whether or not the balls are colliding.

Hint: Represent a ball on a plane as a tuple of (x, y, r), r being the radius

If (distance between two balls centers) $\leq$ (sum of their radii) then (they are colliding)

- b) Find mean, median, mode for the given set of numbers in a list.

Exercise - 9 Functions - Continued

- a) Write a function nearly_equal to test whether two strings are nearly equal. Two strings a and b are nearly equal when a can be generated by a single mutation on b.
- b) Write a function dups to find all duplicates in the list.
- c) Write a function unique to find all the unique elements of a list.

Exercise - 10 - Functions - Problem Solving

- a) Write a function cumulative_product to compute cumulative product of a list of numbers.
- b) Write a function reverse to reverse a list. Without using the reverse function.
- c) Write function to compute gcd, lcm of two numbers. Each function shouldn't exceed one line.

Exercise 11 - Multi-D Lists

- a) Write a program that defines a matrix and prints
- b) Write a program to perform addition of two square matrices
- c) Write a program to perform multiplication of two square matrices

Exercise - 12 - Modules

- a) Install packages requests, flask and explore them. using (pip)
- b) Write a script that imports requests and fetch content from the page. Eg. (Wiki)
- c) Write a simple script that serves a simple HTTPResponse and a simple HTML Page

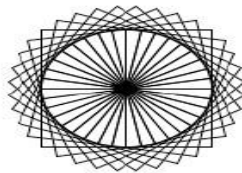
Exercise - 13 OOP

Class variables and instance variable and illustration of the self variable

- a. Robot
- b. ATM Machine

Exercise - 14 GUI, Graphics

1. Write a GUI for an Expression Calculator using tk
2. Write a program to implement the following figures using turtle



Text Books

1. Python Programming: A Modern Approach, Vamsi Kurama, Pearson
2. Learning Python, Mark Lutz, Orielly

Reference Books:

1. Think Python, Allen Downey, Green Tea Press
2. Core Python Programming, W.Chun, Pearson.
3. Introduction to Python, Kenneth A. Lambert, Cengage

DIGITAL ELECTRONICS AND MICROPROCESSORS LAB

Lab : 3 Periods

Exam : 3 Hrs.

Int.Marks : 50

Ext. Marks : 50

Credits : 2

Course Objectives:

1. To discuss about 8085 architecture, signal description and instruction set.
2. To study different programming techniques to implement in assembly language
3. To discuss about 8086 architecture, signal description and instruction set.
4. To study different peripheral devices and learn to interface with 8085.
5. To study different programming techniques to implement in MASM.

Course Outcomes: At the end of the Course:

1. Student can examine Digital trainer kit and microprocessor kit.
2. Student can calculate logical functions for coders, decoders, multiplexers and counters using digital trainer kits.
3. Student can experiment various Arithmetic and logical operations using 8085 instructions.
4. Student can experiment various Arithmetic and logical operations using 8086 instructions in MASM assembler.

SYLLABUS

Digital logic design

1. Verification of logic gates with truth tables(AND,OR,NOT,NOR,NAND,XOR)
2. NAND & NOR Implementation for basic gates
3. Design half adder & half subtractor using logic gates.
4. Design full adder & full subtractor using logic gates.
5. Design binary to gray code converter
6. Design 2to 4 line decoder
7. Design 4X1 multiplexer
8. Verification of flip flops and conversions
9. Design shift registers using flip flops.
10. Design ripple up & ripple down counters using flip flops

8085 programs

1. Write an ALP for addition of N numbers
2. Write an ALP for multiplication of two numbers.
3. Write an ALP for copying one array to another
4. Write an ALP for GCD of two numbers
5. Write an ALP for linear search
6. Write an ALP for Fibonacci series
7. Write an ALP for BCD to Binary conversion
8. Write an ALP for Binary to BCD conversion
9. Write an ALP for Bubble sort in ascending/descending order.
10. Write an ALP for division of two numbers.
11. Write an ALP for ASCII to hexadecimal conversion.
12. Write an ALP for hexadecimal to ASCII conversion.
13. Write an ALP for insertion of an element in an array
14. Write an ALP for finding largest/smallest number in an array

8086 programs

1. Write an ALP for addition of N bytes
2. Write an ALP for addition of N words
3. Write an ALP for addition of N signed bytes
4. Write an ALP for multiplication of two numbers.
5. Write an ALP for division of two numbers.
6. Write an ALP for finding largest/smallest number in an array
7. Write an ALP for GCD of two numbers
8. Write an ALP for copying a string
9. Write an ALP for string length
10. Write an ALP for finding string palindrome

Reference Books:

1. Microprocessor Architecture and Applications with the 8085 , Ramesh S. Gaonkar, 4th Edition, Penram International, 1999
2. Advanced Microprocessors and Peripherals, A K RAY & K M Bhurchandi , 2nd Edition, The McGraw-Hill companies.
3. The 80X86 Family , Design, Programming and Interfacing, John E. Uffenbeck, 3rd Edition, Pearson Education Inc., 2002.
4. Walter A. Tribal and Avatar Singh. The 8088 and 8086 Microprocessors, Programming interfacing, software, hardware and Applications, 4th Edition Pearson education Inc., 2003
5. Microprocessors and Interfacing. Programming and hardware, 2nd Edition, Douglass V. Hall. MH Edition , 1999.

ADVANCED CODING**Lab: 2 Periods****Exam : 3 Hrs.****Sessionals : 50****Credits : 1**

Course Objectives

1. To understand the basics of modular programming
2. To learn about ADT, Linked Lists and Templates.
3. To investigate different methods to find time complexities.
4. To learn about Java collections and Libraries

Course Outcomes

At the end of the course, a student should be able to:

1. Acquire coding knowledge on essential of modular programming
2. Acquire Programming knowledge on linked lists
3. Acquire coding knowledge on ADT
4. Acquire knowledge on time complexities of different methods
5. Acquire Programming skill on Java libraries and Collections

SYLLABUS**UNIT I Review Coding essentials and modular programming**

Introduction to Linear Data, Structure of linear data, Operation logics, Matrix forms and representations, Pattern coding.

Introduction to modular programming: Formation of methods, Methods: Signature and definition, Inter-method communication, Data casting & storage classes, Recursions

UNIT II Linear Linked Data

Introduction to structure pointer, Creating Links Basic problems on Linked lists, Classical problems on linked lists. Circular Linked lists, Operations on CLL, Multiple links, Operations on Doubly linked lists

UNIT III Abstract Data-structures

Stack data-structure, Operations on stack, Infix/Prefix/Post fix expression evaluations, Implementation of stack using array, Implementation of stack using linked lists.

Queue data-structure: Operations on Queues, Formation of a circular queue, Implementation of queue using stack, Implementation of stack using array, Implementation of stack using linked lists

UNIT IV Running time analysis of code and organization of linear list data

Code evaluation w.r.t running time, Loop Complexities, Recursion complexities, Searching techniques: sequential Vs. binary searching.

Organizing the list data, Significance of sorting algorithms, Basic Sorting Techniques: Bubble sort, selection sort, Classical sorting techniques: Insertion sort, Quick sort, Merge sort.

UNIT V Standard Library templates and Java collections

Introduction to C++ language features, Working on STLs, Introduction to Java as Object Oriented language, Essential Java Packages, Coding logics.

Note: This course should focus on Problems

References:

1. Computer Science, A structured programming approach using C, B.A.Forouzan and R.F.Gilberg, 3rd Edition, Thomson, 2007.
2. The C –Programming Language, B.W. Kernighan, Dennis M. Ritchie, Prentice Hall India Pvt.Ltd
3. Scientific Programming: C-Language, Algorithms and Models in Science, Luciano M. Barone (Author), Enzo Marinari (Author), Giovanni Organtini, World Scientific .
4. Object Oriented Programming in C++: N. Barkakati, PHI.
5. Object Oriented Programming through C++ by Robat Laphore.
6. <https://www.geeksforgeeks.org/>.
7. <https://www.tutorialspoint.com/>

PROFESSIONAL ETHICS & HUMAN VALUES
(Common to CSE, ECE & IT)

Tutorial : 3 Periods
Exam : ---

Int. Marks : --
Credits : --

Course Objectives:

1. To inculcate Ethics and Human Values into the young minds.
2. To develop moral responsibility and mould them as best professionals.
3. To create ethical vision and achieve harmony in life.

Course outcomes:

By the end of the course student should be able to understand the importance of ethics and values in life and society.

SYLLABUS

UNIT – I

Ethics and Human Values: Ethics and Values, Ethical Vision, Ethical Decisions, **Human Values** – Classification of Values, Universality of Values.

UNIT – II

Engineering Ethics: Nature of Engineering Ethics, Profession and Professionalism, Professional Ethics , Code of Ethics, Sample Codes – IEEE, ASCE, ASME and CSI.

UNIT – III

Engineering as Social Experimentation:

Engineering as social experimentation, Engineering Professionals – life skills, Engineers as Managers, Consultants and Leaders Role of engineers in promoting ethical climate, balanced outlook on law.

UNIT – IV

Safety Social Responsibility and Rights:

Safety and Risk, moral responsibility of engineers for safety, case studies – Bhopal gas tragedy, Chernobyl disaster, Fukushima Nuclear disaster, Professional rights, Gender discrimination, Sexual harassment at work place.

UNIT – V

Global Issues:

Globalization and MNCs, Environmental Ethics, Computer Ethics, Cyber Crimes, Ethical living, concept of Harmony in life.

Text Books:

1. Govindharajan, M., Natarajan, S. and Senthil Kumar, V.S., Engineering Ethics, Prentice Hall of India, (PHI) Delhi, 2004.
2. Subramainam, R., Professional Ethics, Oxford University Press, New Delhi, 2013.

Reference Books:

1. Charles D, Fleddermann, "Engineering Ethics", Pearson / PHI, New Jersey 2004 (Indian Reprint)

ENGLISH PROFICIENCY-II**(Common to All Branches)**

Lecture	: 1Period		
Tutorial	: 1 Periods	Int.Marks	: --
Exam	: ---	Credits	: --

AIM:

To equip the students with the components of the language required and help them gain adequate knowledge so as to become employable and competent.

Course Objectives:

Students will be able to

1. Enhance their Interpretative skills
2. Understand how to prepare a text.
3. Comprehend various types of writing discourses and respond.
4. Produce effectively different write-ups related to various business contexts.
5. Strengthen their emotional make-up.
6. Perceive various writing discourses.

Course Outcomes:

The students will

1. Develop the skills of taking and making notes
2. Interpret the pictures appropriately and effectively.
3. Read, comprehend and infer a given piece of writing effectively.
4. Learn and practice the skills of Research writing.
5. Communicate well through various forms of writing.
6. Be confident in giving presentations and dealing with people.

SYLLABUS**UNIT-1:SPEAKING**

Analyzing proverbs

Enactment of One-act play

UNIT-2:READING

Reading Comprehension

Summarizing Newspaper Article

UNIT-3:WRITING

Note Taking &Note Making

Precis Writing

Essay Writing

Letter Writing

Picture Description

Literary Appreciation– Learning the Language of Literature

UNIT-4:VOCABULARY

Indian-origin English Words

Phrasal Verbs for Day-to-Day Communication

Commonly used Idiomatic Expressions

UNIT-5:PROJECT

Research Writing

Reference Books:

1. English for Colleges by Brendan J. Carroll, Macmillan Publications
2. Effective Technical Communication by M.AshrafRizwi. Tata Mcgraw Hill
3. Enriching Speaking and Writing Skills, Orient Blackswan Publishers
4. The Oxford Guide to Writing and Speaking by John Seely OUP
5. Six Weeks to Words of Power by Wilfred Funk. W.R.Goyal Publishers
6. English for Engineers and Scientists by Sangeeta Sharma and Binod Mishra, PHI Learning



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R17)

III/IV B.TECH

(With effect from **2017-2018** Admitted Batch onwards)

INFORMATION TECHNOLOGY

I-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/ Week	Internal Marks	External Marks	Total Marks
B17 IT 3101	Computer Networks	3	3	1	-	4	30	70	100
B17 IT 3102	E-Commerce	3	3	1	-	4	30	70	100
B17 IT 3103	Compiler Design	3	3	1	-	4	30	70	100
B17 IT 3104	Operating Systems	3	3	1	-	4	30	70	100
B17 IT 3105	Database Management Systems	3	3	1	-	4	30	70	100
B17 IT 3106	Design and Analysis of Algorithms	3	3	1	-	4	30	70	100
B17 IT 3107	Database Management Systems Lab	2	-	-	3	3	50	50	100
B17 IT 3108	Unix and Operating Systems Lab	2	-	-	3	3	50	50	100
B17BS 3101	Problem Solving & Linguistic Competence	1	-	3	-	3	30	70	100
B17BS 3104	Competitive Coding-I	1	-	-	3	3	50	50	100
Total		24	18	9	9	36	360	640	1000

COMPUTER NETWORKS

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To develop an understanding of modern network architectures from a design and performance perspective.
2. To introduce the student to the major concepts involved in wide-area networks (WANs), local area networks (LANs) and Wireless LANs (WLANs).
3. To provide an opportunity to do network programming
4. To provide a WLAN measurement ideas

Course Outcomes:

1. Explain the functions of the different layer of the OSI Protocol.
2. Draw the functional block diagram of wide-area networks (WANs), local area networks (LANs) and Wireless LANs (WLANs) describe the function of each block.
3. For a given requirement (small scale) of wide-area networks (WANs), local area networks (LANs) and Wireless LANs (WLANs) design it based on the market available component
4. For a given problem related TCP/IP protocol developed the network programming.
5. Configure DNS, EMAIL, File Transfer Protocol (FTP), HTTP, Bluetooth, using open source available software and tools

SYLLABUS**UNIT-I:**

Data communication Components and Network models: Representation of data and its flow Networks , Various Connection Topology, Protocols and Standards, OSI model, Transmission Media, LAN: Wired LAN, Wireless LANs, Techniques for Bandwidth utilization: Multiplexing - Frequency division, Time division and Wave division, Concepts on spread spectrum.

UNIT-II:

Data Link Layer and Medium Access Sub Layer: Error Detection – Fundamentals, CRC; Flow Control and Error control protocols - Stop and Wait, Go back – N ARQ, Selective Repeat ARQ, Sliding Window, Multiple access protocols - CSMA/CD,CDMA/CA.

UNIT-III:

Network Layer: Switching, Logical addressing – IPV4, IPV6; Address mapping – ARP, RARP, BOOTP and DHCP–Delivery, Forwarding and Uni cast Routing protocols.

UNIT-IV:

Transport Layer: Process to Process Communication, User Datagram Protocol (UDP), Transmission Control Protocol (TCP).

UNIT-V:

Application Layer: Domain Name Space (DNS), EMAIL: SMTP, MIME, File Transfer Protocol (FTP), HTTP, Bluetooth

Text Books.

1. Data Communication and Networking, 4th Edition, Behrouz A. Forouzan, McGrawHill.
2. Data and Computer Communication, 8th Edition, William Stallings, Pearson Prentice Hall India.

Reference Books

1. Computer Networks, 8th Edition, Andrew S. Tanenbaum, Pearson New International Edition.
 2. Internetworking with TCP/IP, Volume 1, 6th Edition Douglas Comer, Prentice Hall of India.
 3. TCP/IP Illustrated, Volume 1, W. Richard Stevens, Addison-Wesley, United States of America.
- Course

E-COMMERCE**Lecture : 3 Periods****Tutorial : 1 Period.****Exam : 3 Hrs.****Int.Marks : 30****Ext. Marks : 70****Credits : 3****Course Objectives :**

1. This course introduces the concepts, vocabulary, and procedures associated with E-Commerce and the Internet.
2. To explain the students an overview of all aspects of E-Commerce.
3. To explain the Topics include development of the Internet and E-Commerce, options available for doing business on the Internet, features of Web sites and the tools used to build an E-Commerce web site, marketing issues, payment options, security issues, and customer service.

Course outcomes:

1. Ability to discuss the e-Commerce process. Describe an example of system architecture for an e-Business. List the seven major elements of web design.
2. Ability to Identify and explain fundamental web site tools including design tools, programming tools, and data processing tools. Identify the major electronic payment issues and options.
3. Ability to discuss security issues and explain procedures used to protect against security threats.
4. Ability to Identify and discuss management issues underlying e-Commerce issues including organizational structure, strategic planning, goal setting, corporate social responsibility, changing market intermediaries, resource allocation and customer service.

SYLLABUS**UNIT-I**

Electronic commerce environment and opportunities: Back ground–The Electronic commerce Environment–Electronic Market Place Technologies. Modes of electronic commerce: Overview–EDI–Migration to open EDI–Ecommerce with WWW/Internet– Commerce Net Advocacy–Web commerce going forward.

UNIT-II

Approaches to safe electronic Commerce: Overview– Source–Transport Protocols–Secure Transactions– Secure Electronic Payment Protocol–Secure Electronic Transaction–Certificates for Authentication–Security on Web Servers and enterprise networks.

UNIT-III

Electronic cash and electronic payment schemes: Internet Monetary Payment and Security requirements–payment and purchase order process–online electronic cash.

UNIT-IV

Master card/ Visa Secure electronic transaction: Introduction – Business requirements - Concepts - Payment Processing. Email and Secure Email Technologies for Electronic Commerce: Introduction –The means of Distribution –A model for Message Handling –How Does an Email Work.

UNIT-V

Internet Resources for Commerce: Introduction –Technologies for Web Servers –Internet Applications for commerce – Internet Charges –Internet Access and Architecture–Searching the Internet.

Text Books:

1. Web Commerce Technology Hand Book Daniel Minoli, Emma Minoli McGraw Hill

Reference Books:

1. Frontiers of Electronic Commerce Ravi Kalakotar, Andrew B.Whinston Addison-Wesley

COMPILER DESIGN

Lecture : 3 Periods
Tutorial : 1 Period.
Exam : 3 Hrs.

Int.Marks : 30
Ext. Marks : 70
Credits : 3

Course Objectives:

1. To understand the various phases in the design of a compiler.
2. To understand the design of top-down and bottom-up parsers.
3. To understand syntax directed translation schemes.
4. To introduce lex and yacc tools.
5. To learn to develop algorithms to generate code for a target machine.

Course Outcomes: After taking the course, students will be able to

1. Ability to design, develop, and implement a compiler for any language.
2. Able to use lex and yacc tools for developing a scanner and a parser.
3. Able to design and implement LL and LR parsers.
4. Able to design algorithms to perform code optimization in order to improve the performance of a program in terms of space and time complexity.
5. Ability to design algorithms to generate machine code

SYLLABUS**UNIT – I**

Introduction: Language Processors, the structure of a compiler, the science of building a compiler, programming language basics. Lexical Analysis: The Role of the Lexical Analyzer, Input Buffering, Recognition of Tokens, The Lexical-Analyzer Generator Lex, Finite Automata, From Regular Expressions to Automata, Design of a Lexical-Analyzer Generator, Optimization of DFA-Based Pattern Matchers.

UNIT - II

Syntax Analysis: Introduction, Context-Free Grammars, Writing a Grammar, Top-Down Parsing, Bottom-Up Parsing, Introduction to LR Parsing: Simple LR, More Powerful LR Parsers, Using Ambiguous Grammars, Parser Generators.

UNIT - III

Syntax-Directed Translation: Syntax-Directed Definitions, Evaluation Orders for SDD's, Applications of Syntax-Directed Translation, Syntax-Directed Translation Schemes, Intermediate-Code Generation: Variants of Syntax Trees, Three-Address Code, Types and Declarations, Type Checking, Control Flow, Back patching, Switch-Statements, Intermediate Code for Procedures.

UNIT – IV

Run-Time Environments: Storage organization, Stack Allocation of Space, Access to Nonlocal Data on the Stack, Heap Management.

Code Generation: Issues in the Design of a Code Generator, Basic Blocks and Flow Graphs, Optimization of Basic Blocks, A Simple Code Generator, Peephole Optimization, Register Allocation and Assignment.

UNIT - V

Machine-Independent Optimizations: The Principal Sources of Optimization, Introduction to Data-Flow Analysis, Foundations of Data-Flow Analysis, Constant Propagation, Partial Redundancy Elimination, Loops in Flow Graphs.

Text Books

1. Compilers: Principles, Techniques and Tools, Second Edition, Alfred V. Aho, Monica S. Lam, Ravi Sethi, Jeffrey D. Ullman, Pearson.

Reference Books

1. Compiler Construction-Principles and Practice, Kenneth C Loudon, Cengage Learning.
2. Modern compiler implementation in C, Andrew W Appel, Revised edition, Cambridge University Press.
3. The Theory and Practice of Compiler writing, J. P. Tremblay and P. G. Sorenson, TMH
4. Writing compilers and interpreters, R. Mak, 3rd edition, Wiley student edition.
5. lex&yacc – John R. Levine, Tony Mason, Doug Brown, O'reilly

OPERATING SYSTEMS

Lecture : 3 Periods
Tutorial : 1 Period.
Exam : 3 Hrs.

Int.Marks : 30
Ext. Marks : 70
Credits : 3

Course Objectives: The course is designed for the students to

1. Study the basic concepts of operating systems and their services.
2. Understand the structure and functions of OS.
3. Learn about Processes, Threads and Scheduling algorithms.
4. Understand the principles of concurrency and Deadlocks.
5. Learn various memory management schemes.
6. Study I/O management and File systems.
7. Learn the basics of Linux and Windows operating systems

Course Outcomes: At the end of the Course, Student will be able to

1. Evaluate and discriminate various Operating systems and Structures.
2. Explore Design issues of various Process Scheduling algorithms.
3. Apply the principles of concurrency.
4. Select suitable Deadlock handling algorithm.
5. Compare and contrast various memory management schemes.
6. Design and Implement a prototype file systems.
7. Explore Basic features of Linux and Windows Operating systems.

SYLLABUS**UNIT I :**

Introduction to Operating System Concept: Introduction to Computer System Hardware, Types of operating systems, operating systems concepts, operating systems services, Introduction to Interrupts and System calls, System call types, Operating System Structures: Monolithic, Layered, Micro Kernel and Virtual Machine (VM).

UNIT-II:

Process Management – The process, Process State Diagram, Process control block, Context Switching, Process Scheduling- Scheduling Queues, Schedulers, Operations on Processes, Scheduling-Basic Concepts, Scheduling Criteria, Scheduling Algorithms. Threads, Threading Issues, Thread Scheduling, Multiple Processors and Scheduling issues.

UNIT-III:

Memory Management: Swapping, Fragmentation, Free Space Management Techniques, Contiguous Memory Allocation, Paging, Structure of the Page Table, Segmentation.

Virtual Memory Management: Virtual Memory, Demand Paging, Page-Replacement Algorithms, Frame Allocation, Thrashing, Pre-Paging.

UNIT-IV:

Inter Process Communication (IPC), Process Synchronization, Race Condition, Critical- Section Problem, Bounded Buffer problem, Critical-Section problem Solutions: Hardware solutions, Peterson's Solution, Semaphores, Monitors, Synchronization solutions for Classic IPC Problems.

Principles of deadlock – System Model, Deadlock Characterization, Deadlock Prevention, Detection and Avoidance, Recovery from Deadlock

UNIT-V: File Concept, Access Methods, Directory & Directory structure, Disk Space Allocation methods, Protection mechanisms: Protection Domain, ACL & C-List. Overview of Mass Storage Structure, Disk Structure, Disk Scheduling, Disk Management. I/O Management, Principles of I/O Software.

Case Study: Basic concepts of LINUX, Windows Operating Systems

Text Books:

1. Operating System Concepts, Abraham Silberschatz, Peter Baer Galvin and Greg Gagne 9th Edition, John Wiley and Sons Inc., 2012.
2. Operating Systems – Internals and Design Principles, William Stallings, 7th Edition, Prentice Hall, 2011.

Reference Books:

1. Modern Operating Systems, Andrew S. Tanenbaum, Second Edition, Addison Wesley, 2001.
2. Operating Systems: A Concept-Based Approach, D M Dhamdhare, Second Edition, Tata Mc Graw-Hill Education, 2007.

DATA BASE MANAGEMENT SYSTEMS

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objective: To learn the principles of systematically designing and using large scale Database Management Systems for various applications.

Course Outcomes

1. Describe a relational database and object-oriented database.
2. Create, maintain and manipulate a relational database using SQL
3. Describe ER model and normalization for database design.
4. Examine issues in data storage and query processing and can formulate appropriate solutions.
5. Understand the role and issues in management of data such as efficiency, privacy, security, ethical responsibility, and strategic advantage.
6. Design and build database system for a given real world problem.

SYLLABUS**UNIT-I****An Overview of Database Management Systems:**

Introduction: What is Database and DBMS, File system vs. DBMS, Data models, Levels of Abstraction, Data Independence, Database system architecture, DBA, Client/Server Architecture, Three Tier Architecture.

The ER Model: Database Design and ER Diagrams-Entities Attributes, and Entity Sets-Relationship and Relationship Sets-other features of ER Model, Conceptual Design with the ER Models.

UNIT -II

The Relational Model: Integrity Constraints Over Relations- Key Constraints –Foreign Key Constraints-General Constraints, Logical Database Design: ER to Relational, Views.

Relational Algebra and Relational calculus: Operators and examples, TRC, DRC.

UNIT-III:

SQL: Queries, Constraints, Triggers: The Form of Basic SQL Query, Union, Intersect, and Except,Nested Queries, Aggregate Operators, Null Values, Triggers, Accessing Databases from applications , Introduction to JDBC.

UNIT-IV:

Schema Refinement (Normalization): Purpose of Normalization or schema refinement, concept of functional dependency, normal forms based on functional dependency(1NF, 2NF and 3 NF), concept of surrogate key, Boyce-Codd normal form(BCNF), Lossless join and dependency preserving decomposition, Fourth normal form(4NF).

UNIT-V:

Transaction Management and Concurrency Control: The ACID properties, transactions and schedules, Concurrent execution of transactions, Lock based concurrency control. Dealing with deadlocks, specialized locking techniques, Concurrency control without locking.

Database Recovery management: The LOG, Write-Ahead log protocol, checkpointing, Recovering from system crash. Introduction to ARIES,

Text Books:

1. Data base Management Systems, Raghurama Krishnan, Johannes Gehrke, TATA McGraw Hill 3rd Edition.
2. Database Systems - The Complete Book, H G Molina, J D Ullman, J Widom Pearson

References Books:

1. Introduction to Database Systems, CJ Date, Pearson
2. Data base Systems design, Implementation, and Management, Peter Rob & Carlos Coronel 7th Edn.
3. Fundamentals of Database Systems, ElmasriNavrate Pearson Education
4. Introduction to Database Systems, C.J.Date Pearson Education

DESIGN AND ANALYSIS OF ALGORITHMS

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

On completing this course student will be able to

1. Solve problems using algorithm design methods such as the greedy method, divide and conquer, dynamic programming, backtracking, and branch and bound and writing programs for these solutions
2. Analyze the asymptotic performance of algorithms.
3. Demonstrate a familiarity with major algorithms and data structures.
4. Synthesize efficient algorithms in common engineering design situations.

Course Outcomes:

1. Students will be able to understand, apply and Analyze the algorithms using asymptotic notations and Divide-and-Conquer technique on computer science problems.
2. Student will be able to understand, apply and analyze Greedy technique on computer science problems.
3. Student will be able to understand, apply and analyze Dynamic Programming on computer science problems.
4. Student will be able to understand, apply and analyze Basic Traversal and Search techniques and Backtracking on computer science problems.
5. Student will be able to understand, apply and analyze Branch-and-Bound and algebraic problems on computer science problems.

SYLLABUS**UNIT I**

Introduction: Algorithm, Algorithm specification, Performance Analysis.

Divide and Conquer: The General Method -- Binary search -- Finding Maximum and Minimum -- Merge sort -- Quick sort -- Selection -- Strassen's Matrix Multiplication -- Convex Hull

UNIT II

The Greedy Method: The General Method -- Knapsack Problem -- Job Sequencing with Deadlines -
- Optimal Storage on Tapes -- Minimum Cost Spanning Trees: Prim's Algorithm, Kruskal's Algorithm -- Optimal Merge Patterns -- Single Source Shortest Paths.

UNIT III

Dynamic Programming: The General Method -- Multistage Graphs -- All Pairs Shortest Paths -- Optimal Binary Search Trees -- 0/1 Knapsack Problem -- Reliability Design -- the Traveling Sales person Problem.

UNIT IV

Basic Traversal and Search techniques: Techniques for Binary trees – Techniques for Graphs – Connected Components and Spanning trees -- Bi-connected Components and Depth First Search.

Back Tracking: The General Method – 8-Queens problem -- Sum of Subsets -- Graph Coloring -- Hamiltonian Cycle.

UNIT V

Branch and Bound: The Method -- Least Cost (LC) Search, The 15-Puzzle: an Example -- 0/1 Knapsack Problem LC Branch-and-Bound Solution, FIFO Branch-and- Bound Solution-- Traveling sales Person Problem

Text Books:

1. Fundamentals of Computer Algorithms by Ellis Horowitz, S. Sahni et.al. Galgotia Pub. 2001.
2. The Design and Analysis of Computer Algorithms by Aho, Hopcroft, Ullman.

Reference Books:

1. Computer Algorithms: Introduction to Design and Analysis by Sara Baase, Allen Van Gelder, Pearson Education.
2. Data Structures and Algorithm Analysis in C++ by Mark Allen Weiss, Pearson Education.

DATA BASE MANAGEMENT SYSTEMS LAB

Lab	: 3 Periods	Int.Marks	: 50
Exam	: 3 Hrs.	Ext. Marks	: 50
		Credits	: 2

Course Objectives:

1. To provide a sound introduction to the discipline of database management as a subject in its own right, rather than as a compendium of techniques and product-specific tools.
2. To familiarize the participant with the nuances of database environments towards an information-oriented data-processing oriented framework
3. To give a good formal foundation on the relational model of data
4. To present SQL and procedural interfaces to SQL comprehensively
5. To give an introduction to systematic database design approaches covering conceptual design, logical design and an overview of physical design

Course Outcomes:

1. Understand, appreciate and effectively explain the underlying concepts of database technologies.
2. Design and implement a database schema for a given problem-domain normalize a database.
3. Populate and query a database using SQL DML/DDDL commands.
4. Declare and enforce integrity constraints on a database using a state-of-the-art RDBMS
5. Programming PL/SQL including stored procedures, stored functions, cursors, packages.
6. Design and build a GUI application using a 4GL

List of Experiments:

SQL: Queries to facilitate acquaintance of Built-In Functions, String Functions, Numeric Functions, Date Functions and Conversion Functions.

Queries using operators in SQL

Queries to Retrieve and Change Data: Select, Insert, Delete, and Update

Queries using Group By, Order By, and Having Clauses

Queries on Controlling Data: Commit, Rollback, and Save point

Queries to Build Report in SQL *PLUS

Queries for Creating, Dropping, and Altering Tables, Views, and Constraints

Queries on Joins and Correlated Sub-Queries

Queries on Working with Index, Sequence, Synonym, Controlling Access, and Locking Rows for Update, Creating Password and Security features.

PL/SQL

Write a PL/SQL Code using Basic Variable, Anchored Declarations, and Usage of Assignment Operation

Write a PL/SQL Code Bind and Substitution Variables. Printing in PL/SQL

Write a PL/SQL block using SQL and Control Structures in PL/SQL

Write a PL/SQL code using Triggers.

Write a PL/SQL Code using Cursors, Exceptions and Composite Data Types

Write a PL/SQL Code using Procedures, Functions.

Write a PL/SQL Code Creation of forms for any Information System such as Student Information System, Employee Information System etc.

Demonstration of database connectivity

Reference Books:

1. Oracle PL/SQL programming by Steven Feuerstein, 4thEdn, O'Reilly publications.
2. SQL, PL/SQL The programming language of Oracle by Ivan Bayross, 3rdEdn, BPB publications.
3. Oracle: The Complete Reference by Oracle Press

UNIX AND OPERATING SYSTEMS LAB**Lab : 3 Periods****Exam : 3 Hrs.****Int.Marks : 50****Ext. Marks : 50****Credits : 2****Course Objectives:**

1. To understand the design aspects of operating system.
2. To study the process management concepts & Techniques.
3. To study the storage management concepts.
4. To familiarize students with the Linux environment
5. To learn the fundamentals of shell scripting/programming
6. To familiarize students with basic Unix administration

Course Outcomes:

1. To use Unix utilities and perform basic shell control of the utilities
2. To use the Unix file system and file access control.
3. To use of an operating system to develop software
4. Work confidently in Unix/Linux environment
5. Write shell scripts to automate various tasks
6. Master the basics of Linux administration

OPERATING SYSTEMS

1. Simulate the following CPU scheduling algorithms
a) Round Robin b) SJF c) FCFS d) Priority
2. Multiprogramming-Memory management-
Implementation of fork (), wait (), exec() and exit (), System calls
3. Simulate the following
a) Multiprogramming with a fixed number of tasks (MFT)
b) Multiprogramming with a variable number of tasks (MVT)
4. Simulate Bankers Algorithm for Dead Lock Avoidance
5. Simulate Bankers Algorithm for Dead Lock Prevention.
6. Simulate the following page replacement algorithms. a) FIFO b) LRU c) LFU
7. Simulate the following File allocation strategies
a) Sequenced b) Indexed c) Linked

UNIX PROGRAMMING

1. Basic Shell Commands Shell Programs
2. Fibonacci Series
3. Designing Calculator
4. File Operations
5. Base conversion
6. Usage of cut and Filter commands (grep, awk, sed..)
7. Usage of user defined functions Administration

8. Managing User Accounts
9. User Quota Management
10. Installation of RPM software and Zipping, tar

Text Books:

1. Operating System Concepts, AbrahamSilberschatz, Peter Baer Galvin and Greg Gagne 9th Edition, John Wiley and Sons Inc., 2012.
2. Operating Systems – Internals and Design Principles, William Stallings, 7th Edition, Prentice Hall, 2011.
3. Operating Systems-S Halder, Alex A Aravind Pearson Education Second Edition 2016

Reference Books:

1. Modern Operating Systems, Andrew S. Tanenbaum, Second Edition, Addison Wesley,2001.
2. Operating Systems: A Design-Oriented Approach, Charles Crowley, Tata Mc Graw Hill Education”, 1996.
3. Operating Systems: A Concept-Based Approach, D M Dhamdhere, Second Edition, Tata Mc Graw-Hill Education, 2007.

PROBLEM SOLVING & LINGUISTIC COMPETENCE**(Common to all Branches)**

Tutorial	: 3 Periods (VA-2+QA-1)	Int.Marks	: 30
Exam	: 3 Hrs.	Ext.Marks	: 70
		Credits	: 1

Part-A: Verbal and Soft Skills-I**Course Objectives:**

1. To introduce concepts required in framing grammatically correct sentences and identifying errors while using Standard English.
2. To familiarize the learner with high frequency words as they would be used in their professional career.
3. To inculcate logical thinking in order to frame and use data as per the requirement.
4. To acquaint the learner of making a coherent and cohesive sentences and paragraphs for composing a written discourse.
5. To familiarize students with soft skills and how it influences their professional growth.

Course Outcomes:

The student will be able to

1. Detect grammatical errors in the text/sentences and rectify them while answering their competitive/ company specific tests and frame grammatically correct sentences while writing.
2. Answer questions on synonyms, antonyms and other vocabulary based exercises while attempting CAT, GRE, GATE and other related tests.
3. Use their logical thinking ability and solve questions related to analogy, syllogisms and other reasoning based exercises.
4. Choose the appropriate word/s/phrases suitable to the given context in order to make the sentence/paragraph coherent.
5. Apply soft skills in the work place and build better personal and professional relationships making informed decisions.

SYLLABUS**Grammar: (VA)**

Parts of speech(with emphasis on appropriate prepositions, co-relative conjunctions, pronouns-number and person, relative pronouns), articles(nuances while using definite and indefinite articles), tenses(with emphasis on appropriate usage according to the situation), subject – verb agreement (to differentiate between number and person) , clauses(use of the appropriate clause , conditional and relative clauses), phrases(use of the phrases, phrasal verbs) to-infinitives, gerunds, question tags, voice, direct & indirect speech, degrees of comparison, modifiers, determiners, identifying errors in a given sentence, correcting errors in sentences.

Vocabulary: (VA)

Synonyms and synonym variants(with emphasis on high frequency words), antonyms and antonym variants(with emphasis on high frequency words), contextual meanings with regard to inflections of a word, frequently confused words, words often mis-used, multiple meanings of the same word (differentiating between meanings with the help of the given context), foreign phrases, homonyms, idioms, pictorial representation of words, word roots, collocations.

Reasoning: (VA)

Critical reasoning (understanding the terminology used in CR- premise, assumption, inference, conclusion), Analogies (building relationships between a pair of words and then identifying similar relationships), Sequencing of sentences (to form a coherent paragraph, to construct a meaningful and grammatically correct sentence using the jumbled text), odd man (to use logical reasoning and eliminate the unrelated word from a group), YES-NO statements (sticking to a particular line of reasoning Syllogisms).

Usage: (VA)

Sentence completion (with emphasis on signpost words and structure of a sentence), supplying a suitable beginning/ending/middle sentence to make the paragraph coherent, idiomatic language (with emphasis on business communication), punctuation depending on the meaning of the sentence.

Soft Skills:

Introduction to Soft Skills – Significance of Inter & Intra-Personal Communication – SWOT Analysis –Creativity & Problem Solving – Leadership & Team Work - Presentation Skills Attitude – Significance – Building a positive attitude – Goal Setting – Guidelines for Goal Setting – Social Consciousness and Social Entrepreneurship – Emotional Intelligence - Stress Management, CV Making and CV Review.

Text Books:

1. Oxford Learners's Grammar – Finder by John Eastwood, Oxford Publication.
2. R S Agarwal's books on objective English and verbal reasoning
3. English Vocabulary in Use- Advanced , Cambridge University Press.
4. Collocations In Use, Cambridge University Press.
5. Soft Skills & Employability Skills by Samina Pillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
6. Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi

Reference Books:

1. English Grammar in Use by Raymond Murphy, CUP
2. Websites: Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from _IMS, Career Launcher and Time' institutes for competitive exams.
4. The Art of Public Speaking by Dale Carnegie
5. The Leader in You by Dale Carnegie
6. Emotional Intelligence by Daniel Golman
7. Stay Hungry Stay Foolish by Rashmi Bansal
8. I have a Dream by Rashmi Bansal

Part-B: Quantitative Aptitude -I

Course objectives:

The objective of introducing quantitative aptitude-1 is:

1. To familiarize students with basic problems on numbers and ratio's problems.
2. To enrich the skills of solving problems on time, work, speed, distance and also measurement of units.
3. To enable the students to work efficiently on percentage values related to shares, profit and loss problems.
4. To inculcate logical thinking by exposing the students to reasoning related questions.
5. To expose them to the practice of syllogisms and help them make right conclusions.

Course Outcomes:

1. The students will be able to perform well in calculating on number problems and various units of ratio concepts.
2. Accurate solving problems on time and distance and units related solutions.
3. The students will become adept in solving problems related to profit and loss, in specific, quantitative ability.
4. The students will present themselves well in the recruitment process using analytical and logical skills which he or she developed during the course as they are very important for any person to be placed in the industry.
5. The students will learn to apply Logical thinking to the problems of syllogisms and be able to effectively attempt competitive examinations like CAT, GRE, GATE for further studies.

SYLLABUS

Numbers, LCM and HCF, Chain Rule, Ratio and Proportion Importance of different types of numbers and uses of them: Divisibility tests, Finding remainders in various cases, Problems related to numbers, Methods to find LCM, Methods to find HCF, applications of LCM, HCF. Importance of chain rule, Problems on chain rule, Introducing the concept of ratio in three different methods, Problems related to Ratio and Proportion.

Time and work, Time and Distance Problems on man power and time related to work, Problems on alternate days, Problems on hours of working related to clock, Problems on pipes and cistern, Problems on combination of the some or all the above, Introduction of time and distance, Problems on average speed, Problems on Relative speed, Problems on trains, Problems on boats and streams, Problems on circular tracks, Problems on polygonal tracks, Problems on races.

Percentages, Profit Loss and Discount, Simple interest, Compound Interest, Partnerships, shares and dividends

Problems on percentages-Understanding of cost price, selling price, marked price, discount, percentage of profit, percentage of loss, percentage of discount, Problems on cost price, selling price, marked price, discount. Introduction of simple interest, Introduction of compound interest, Relation between simple interest and compound interest, Introduction of partnership, Sleeping partner concept and problems, Problems on shares and dividends, and stocks.

Introduction, number series, number analogy, classification, Letter series, ranking, directions Problems of how to find the next number in the series, Finding the missing number and related sums, Analogy, Sums related to number analogy, Ranking of alphabet, Sums related to Classification, Sums related to letter series, Relation between number series and letter series, Usage of directions north, south, east, west, Problems related to directions north, south, east, west.

Data sufficiency, Syllogisms Easy sums to understand data sufficiency, Frequent mistakes while doing data sufficiency, Syllogisms Problems.

Text Books:

1. Quantitative aptitude by RS Agarwal
2. Verbal and non verbal reasoning by RS Agarwal.
3. Puzzles to puzzle you by shakunataladevi

References:

1. Barron's by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2. Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from _IMS, Career Launcher and Time' institutes for competitive exams.
4. Books for cat by arunsharma
5. Elementary and Higher algebra by HS Hall and SR knight.

Websites:

1. www.m4maths.com
2. www.Indiabix.com
3. www.800score.com
4. Official GRE site
5. Official GMAT site

COMPETITIVE CODING-I

Tutorial : 3 Periods
Exam : 3 Hrs.

Int.Marks : 50
Ext. Marks : 50
Credits : 1

Course Objectives:

1. To understand the basics of Java Packages.
2. To learn about Collection Framework.
3. To investigate different methods of GUI Programming.
4. To learn about Java collections and Libraries

Course Outcomes

At the end of the course, a student should be able to:

1. Able to solve problems using java collection framework and I/o classes.
2. Able to develop multithreaded applications with synchronization.
3. Able to develop applets for web applications.
4. Able to design GUI based applications

SYLLABUS**UNIT I** Standard Library templates and Java collections

Introduction to Java as Object Oriented language, Essential Java Packages, Coding logics.

UNIT II The Collections Framework (java.util)- Collections overview, Collection Interfaces, The Collection classes- Array List, Linked List, Hash Set, Tree Set, Priority Queue, Array Deque. Accessing a Collection via an Iterator, Using an Iterator, The For-Each alternative, Map Interfaces and Classes, Comparators, Collection algorithms, Arrays, The Legacy Classes and Interfaces- Dictionary, Hashtable ,Properties, Stack, Vector.

UNIT III More Utility classes, String Tokenizer, Bit Set, Date, Calendar, Random, Formatter, Scanner.

UNIT IV GUI Programming with Swing – Introduction, limitations of AWT, MVC architecture, components, containers. Understanding Layout Managers, Flow Layout, Border Layout, Grid Layout, Card Layout, Grid Bag Layout. Event Handling- The Delegation event model- Events, Event sources, Event Listeners, Event classes, Handling mouse and keyboard events.

UNIT V Adapter classes, Inner classes, Anonymous Inner classes. A Simple Swing Application, Applets – Applets and HTML, Security Issues, Applets and Applications, passing parameters to applets. Creating a Swing Applet, Painting in Swing, A Paint example, Exploring Swing Controls- JLabel and Image Icon, JText Field, The Swing Buttons- JButton, JToggleButton, JCheckBox, JRadioButton, JTabbed Pane, JScroll Pane, JList, JComboBox, Swing Menus, Dialogs.

References:

1. Java The complete reference, 9th edition, Herbert Schildt, McGraw Hill Education (India) Pvt. Ltd. 2. Understanding Object-Oriented Programming with Java, updated edition, T. Budd, Pearson Education.
2. An Introduction to programming and OO design using Java, J. Nino and F.A. Hosch, John Wiley & sons
3. Introduction to Java programming, Y. Daniel Liang, Pearson Education.
4. Object Oriented Programming through Java, P. Radha Krishna, University Press.
5. Programming in Java, S. Malhotra, S. Chudhary, 2nd edition, Oxford Univ. Press.
6. Java Programming and Object-oriented Application Development, R. A. Johnson, Cengage Learning.
7. <https://www.geeksforgeeks.org/>.
8. <https://www.tutorialspoint.com/>

SCHEME OF INSTRUCTION & EXAMINATION
 (Regulation R17)
III/IV B.TECH
 (With effect from **2017-2018** Admitted Batch onwards)
INFORMATION TECHNOLOGY
II-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/Week	Internal Marks	External Marks	Total Marks
B17 IT 3201	Web Technologies	3	3	1	--	4	30	70	100
B17 IT 3202	Object Oriented Software Engineering	3	3	1	--	4	30	70	100
B17 IT 3203	Cryptography & Network Security	3	3	1	--	4	30	70	100
B17 IT 3204	Statistics With R Programming	3	3	1	--	4	30	70	100
B17 IT 3205	Data Ware Housing and Business Intelligence	3	3	1	--	4	30	70	100
#OE	OPEN ELECTIVE	3	3	1	--	4	30	70	100
B17 IT 3210	Web Technologies Lab	2	--	--	3	3	50	50	100
B17 IT 3211	Software Engineering and Mini Project Lab	2	--	--	3	3	50	50	100
B17BS 3201	Employability Skills	1	--	3	--	3	30	70	100
B17 BS 3205	Competitive Coding-II	1	--	--	3	3	50	50	100
B17 BS3206	IPR & PATENTS	--	--	2	--	2	--	--	--
Total		24	18	11	9	38	360	640	1000

Open Elective	B17 IT 3206	Artificial Intelligence
	B17 IT 3207	Semantic Web and Social Networks
	B17 EC 3210	Digital Signal Processing
	B17 ME 3211	Robotics
	B17 IT 3208	Image Processing
	B17 IT 3209	Operations Research

WEB TECHNOLOGIES

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

This course is designed to introduce students with no programming experience to the programming languages and techniques associated with the World Wide Web. The course will introduce web-based media-rich programming tools for creating interactive web pages.

Course Outcomes:

Analyze a web page and identify its elements and attributes.
 Create web pages using XHTML and Cascading Styles sheets.
 Build dynamic web pages.
 Build web applications using PHP.
 Programming through PERL and Ruby
 Write simple client-side scripts using AJAX

SYLLABUS

UNIT-I:**HTML, CSS**

Basic Syntax, Standard HTML Document Structure, Basic Text Markup, Images, Hypertext Links, Lists, Tables, Forms, HTML5
 CSS: Levels of Style Sheets, Style Specification Formats, Selector Forms, The Box Model, Conflict Resolution

UNIT-II:**Java script**

The Basic of Javascript: Objects, Primitives Operations and Expressions, Screen Output and Keyboard Input, Control Statements, Object Creation and Modification, Arrays, Functions, Constructors, Pattern Matching using Regular Expressions DHTML: Positioning Moving and Changing Elements

UNIT-III:

XML: Document type Definition, XML schemas, Document object model, XSLT, DOM and SAX Approaches,

AJAX A New Approach: Introduction to AJAX, Integrating PHP and AJAX.

UNIT-IV:

PHP Programming: Introducing PHP: Creating PHP script, Running PHP script.
Working with variables and constants: Using variables, Using constants, DataTypes, Operators.
Controlling program flow: Conditional statements ,Control statements, Arrays ,functions
 .Working with forms and Databases such as MySQL.

UNIT-V:

Introduction to Ruby, Variables, types, simple I/O, Control, Arrays, Hashes, Methods, Classes, Iterators, Pattern Matching. Overview of Rails.

Text Books:

1. Programming the World Wide Web, Robert W Sebesta, 7ed, Pearson.
2. Web Technologies, Uttam K Roy, Oxford
3. The Web Warrior Guide to Web Programming, Bai, Ekedahl, Farrell, Gosselin, Zak, Karparhi, MacIntyre, Morrissey, Cengage

Reference Books:

1. Ruby on Rails Up and Running, Lightning fast Web development, Bruce Tate, Curt Hibbs, O'Reilly (2006).
2. Web Technologies, HTML < JavaScript, PHP, Java, JSP, XML and AJAX, Black book, Dream Tech.
3. An Introduction to Web Design, Programming, Paul S Wang, Sanda S Katila, Cengage Learning.
4. <http://www.upriss.org.uk/perl/PerlCourse.html>

OBJECT ORIENTED SOFTWARE ENGINEERING

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course objectives:

1. To explain the importance of OOSE in Software development.
2. To explain the students the importance of Requirements Engineering.
3. To explain the role of UML and Testing in Software Development.
4. To explain the entire Software Development Process with aid of case studies.

Course Outcomes:

1. Ability to define a problem and perform Requirements Engineering.
2. Ability to draw UML diagrams for the requirements gathered.
3. Ability to implement the designed problem in Object Oriented Programming Language.
4. Test whether all the requirements specified have been achieved or not.

SYLLABUS

UNIT-I: Introduction to Object Oriented Software Engineering & Requirements Engineering: Nature of the Software, Types of Software, Software Engineering Projects, Software Engineering Activities, Introduction to Object Orientation, Software Process Models- Waterfall Model, Opportunistic Model , Phased Released Model, Spiral Model, Evolutionary Model, Concurrent Engineering Model, Domain Analysis, Problem Definition and Scope, Requirements Definition, Types of Requirements, Activities to identifying Functional and Non Functional Requirements, Techniques for Gathering and Analyzing Requirements.

UNIT-II : Unified Modeling Language & Use Case Modeling: Modeling Concepts, Types of UML Diagrams with Examples; User-Centered Design, Characteristics of Users, Developing Use-Case Models of Systems, Use-Case Diagram, Use- Case Descriptions, Usability Principles, Interaction and Behavioral Diagrams: Interaction Diagrams, State Diagrams, Case Study.

UNIT-III: Class Design and Class Diagrams: Essentials of UML Class Diagrams, Associations and Multiplicity, Other Relationships, Generalization, Instance Diagrams, Advanced Features of Class Diagrams, Component, Deployment Diagrams and Activity Diagrams,, Case Study.

UNIT-IV: Software Design and Architecture: Process of Design, Principles Leading to Good Design, Pattern Introduction, Design Patterns: Abstraction-Occurrence Pattern, General Hierarchical Pattern, Play-Role Pattern, Singleton Pattern, Observer Pattern, Delegation Pattern, Adaptor Pattern, Façade Pattern, Immutable Pattern, Read-Only Interface Pattern and The Proxy Pattern; Software Architecture Contents of Architecture Model, Architectural Patterns:

Multilayer, Client-Server, Broker, Transaction Processing, Pipe & Filter and MVC Architectural Patterns.

UNIT-V: Software Testing & Software Process Management: Overview of Testing, Testing Concepts, Testing Activities, Testing Strategies, Unit Testing, Integration Testing, Function Testing, Structural Testing, Class Based Testing Strategies, Use Case/Scenario Based Testing, Regression Testing, Performance Testing, System Testing, Acceptance Testing, Installation Testing, OO Test Design Issues, Test Case Design, Quality Assurance, Root Cause Analysis, Post-Mortem Analysis, Introduction to Software Project Management, Activities of Software Project Management, Structure of Project Plan, Software Cost Estimation, Project Scheduling.

CASE STUDY:

1. Simple Chat Instant Messaging System
2. GPS Based Automobile Navigation System
3. Waste Management Inspection Tracking System (WMITS)
4. Geographical Information System

Text Books:

1. Object-Oriented Software Engineering Practical software development using UML and Java by Timothy C. Lethbridge & Robert, Langanieri McGraw-Hill.
2. Object-Oriented Software Engineering: Using UML, Patterns and Java, Bernd Bruegge and Allen H. Dutoit, 2nd Edition, Pearson Education Asia.

Reference Books:

1. Software Engineering: A Practitioner's Approach, Roger S Pressman.
2. A Practical Guide to Testing Object-Oriented Software, John D. McGregor; David A. Sykes, Addison-Wesley Professional.
3. Software Engineering, K.K. Agarwal, New Age Publications 2008

CRYPTOGRAPHY & NETWORK SECURITY

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives: On completing this course student will be able to

1. Solve problems using algorithm design methods such as the RSA,DES,AES
2. Analyze the performance of algorithms.
3. Demonstrate a familiarity with major algorithms and Approaches.

Course Outcomes:

1. Students will be able to understand, apply and analyze the algorithms on security problems.
2. Student will be able to understand, apply and analyze symmetric and asymmetric approaches.
3. Student will be able to understand, apply and analyze security measurements.
4. Student will be able to understand, apply and analyze various malicious software.

SYLLABUS**UNIT-I**

INTRODUCTION: The need for security-Security approaches, principals of security , plain text and cipher Text- Types of attacks –substitution and Transportation Techniques –Encryption Techniques –Encryption and Decryption- Symmetric and Asymmetric Cryptography – Stenography-KDC

SYMMETRICKEYCRYPTOGRAPHIC ALGORITHMS: Feistel Cipher Structure, Data encryption standard, Triple DES, AES, Stream Ciphers and RC4.

UNIT-II

ASYMMETRICKEY CRYPTOGRAPHIC ALGORITHMS: Overview of asymmetric key cryptography, Diffie Hellman Key exchange, RSA algorithm-symmetric and asymmetric key cryptography together-Message Digest- MAC- HMAC- digital signatures.

UNIT-III

PUBLIC KEY INFRASTRUCTURE: Introduction-Digital certificates-Private Key management-The PKIX model.

USER AUTHENTICATION MECHANISMS: Introduction-Authentication basics-passwords-authentication tokens-certificate based authentication-biometrics authentication-Kerberos.

UNIT-IV

INTERNET SECURITY PROTOCOLS: Basic concepts -SSL-SHTTP-TSP-SET- SSL versus SET-3D secure protocol -Email security-WAP security -security in GSM – 3G Security, Introductiontofirewalls-IP security-VirtualPrivateNetworks.

UNIT-V

MALICIOUS SOFTWARE: Types of Malicious Software, Viruses, Viruses countermeasures, Worms, Bots, and Honey pots, Denial of Service Attacks and Flooding Attacks.

Text Books:

1. Cryptography and Network security, Atul Kahate, Tata McGraw-Hill Pub company Ltd., New Delhi
2. Computer Security by William Stallings and Lawrie Brown, Pearson Pub

Reference Books:

1. Network Security Private Communication in a public world, Charlie Kaufman, Radia Perlman & Mike Speciner, Prentice Hall of India Private Ltd., New Delhi.
2. Network Security: The Complete Reference by Roberta Bragg, Mark Phodes- Ousley, Keith Strassberg Tata McGraw-Hill

STATISTICS WITH R – PROGRAMMING

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To be familiar with basic techniques of R.
2. Solve problems using statistical methods present in R.
3. Use different types of functions available in R.
4. Create different types of graphs and plots.
5. To practice different Regression techniques.

Course Outcomes:

After taking the course, students will be able to

1. Use R for statistical programming, computation, graphics, and modeling.
2. Write functions and use R in an efficient way.
3. Fit some basic types of statistical models.
4. Use R in their own research.
5. Be able to expand their knowledge of R on their own.

SYLLABUS**UNIT - I**

Introduction, How to run R, R Sessions and Functions, Basic Math, Variables, Data Types, Vectors, Conclusion, Advanced Data Structures, Data Frames, Lists, Matrices, Arrays, Classes.

UNIT - II

R Programming Structures, Control Statements, Loops, - Looping Over Nonvector Sets,- If-Else, Arithmetic and Boolean Operators and values, Default Values for Argument, Return Values, Deciding Whether to explicitly call return- Returning Complex Objects, Functions are Objective, No Pointers in R, Recursion, A Quicksort Implementation-Extended Extended Example: A Binary Search Tree.

UNIT - III

Doing Math and Simulation in R, Math Function, Extended Example Calculating Probability-Cumulative Sums and Products-Minima and Maxima- Calculus, Functions for Statistical Distribution, Sorting, Linear Algebra Operation on Vectors and Matrices, Extended Example: Vector cross Product- Extended Example: Finding Stationary Distribution of Markov Chains, Set Operation, Input /out put, Accessing the Keyboard and Monitor, Reading and writer Files.

UNIT - IV

Graphics, Creating Graphs, The Workhorse of R Base Graphics, the plot() Function, Customizing Graphs, Saving Graphs to Files.

UNIT - V

Probability Distributions, Normal Distribution- Binomial Distribution- Poisson Distributions
Other Distribution, Basic Statistics, Correlation and Covariance, T-Tests,-ANOVA.

Linear Models, Simple Linear Regression, -Multiple Regression Generalized Linear Models,
Logistic Regression, - Poisson Regression- other Generalized Linear Models-Survival Analysis,
Nonlinear Models, Splines- Decision- Random Forests,

Text Books

1. The Art of R Programming, Norman Matloff, Cengage Learning
2. R for Everyone, Lander, Pearson

Reference Books

1. R Cookbook, Paul Teetor, Oreilly.
2. R in Action, Rob Kabacoff, Manning

DATA WARE HOUSING AND BUSINESS INTELLIGENCE

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. Approach business problems data-analytically by identifying opportunities to derive business value from data.
2. Know the basics of data mining techniques and how they can be applied to extract Relevant Business intelligence.

Course Outcomes:

1. Describe the scope and application of business intelligence and decision support.
2. Design systems for sourcing and structuring data to provide an integrated, non-volatile collection of data for decision support using data warehouses.
3. Design multidimensional data models and implement those using star schemas and Relational databases.
4. Communicate and foster realistic expectations of the role of OLAP technology.

SYLLABUS**UNIT – I**

Introduction to Data Mining: Motivation for Data Mining, Data Mining-Definition & Functionalities, Classification of DM systems, DM task primitives, Integration of a Data Mining system with a Database or a Data Warehouse, Major issues in Data Mining. Data Warehousing (Overview Only): Overview of concepts like star schema, fact and dimension tables, OLAP operations, From OLAP to Data Mining.

UNIT – II

Data Preprocessing: Why? Descriptive Data Summarization, Data Cleaning: Missing Values, Noisy Data, Data Integration and Transformation. Data Reduction:-Data Cube Aggregation, Dimensionality reduction, Data Compression, Numerosity Reduction, Data Discretization and Concept hierarchy generation for numerical and categorical data.

UNIT – III

Mining Frequent Patterns, Associations, and Correlations: Market Basket Analysis, Frequent Itemsets, Closed Itemsets, and Association Rules, Frequent Pattern Mining, Efficient and Scalable Frequent Itemset Mining Methods, The Apriori Algorithm for finding Frequent Item sets Using Candidate Generation, Generating Association Rules from Frequent Item sets, Improving the Efficiency of Apriori, Frequent Itemsets without Candidate Generation using FP Tree, Mining Multilevel Association Rules, Mining Multidimensional Association Rules, From Association Mining to Correlation Analysis, Constraint-Based Association Mining.

UNIT – IV

Classification & Prediction: What is it? Issues regarding Classification and prediction
Classification methods: Decision tree, Bayesian Classification, Rule based Prediction: Linear and non linear regression, Accuracy and Error measures, Evaluating the accuracy of a Classifier or Predictor.

Cluster Analysis: What is it? Types of Data in cluster analysis, Categories of clustering methods, Partitioning methods ñ K-Means, K-Medoids. Hierarchical Clustering- Agglomerative and Divisive Clustering, BIRCH and ROCK methods, DBSCAN, Outlier Analysis

Mining Stream and Sequence Data: What is stream data? Classification, Clustering Association Mining in stream data. Mining Sequence Patterns in Transactional Databases. Spatial Data and Text Mining: Spatial Data Cube Construction and Spatial OLAP, Mining Spatial Association and Co-location Patterns, Spatial Clustering Methods, Spatial Classification and Spatial Trend Analysis. Text Mining Text Data Analysis and Information Retrieval, Dimensionality Reduction for Text, Text Mining Approaches.

UNIT – V

Web Mining: Web mining introduction, Web Content Mining, Web Structure Mining, Web Usage mining, Automatic Classification of web Documents.

Data Mining for Business Intelligence Applications: Data mining for business Applications like Balanced Scorecard, Fraud Detection, Click stream Mining, Market Segmentation, retail industry, telecommunications industry, banking & finance and CRM etc.

Text Books:

1. Han, Kamber, "Data Mining Concepts and Techniques", Morgan Kaufmann 2 nd Edition
2. P. N. Tan, M. Steinbach, Vipin Kumar, introduction to Data Mining, Pearson Education

Reference Books:

1. MacLennan Jamie, Tang ZhaoHui and Crivat Bogdan, Data Mining with Microsoft SQL Server 2008, Wiley India Edition.
2. G. Shmueli, N.R. Patel, P.C. Bruce, "Data Mining for Business Intelligence: Concepts, Techniques and Applications in Microsoft Office Excel with XLMiner", Wiley India.
3. Michael Berry and Gordon Linoff "Data Mining Techniques", 2nd Edition Wiley Publications.
4. Alex Berson and Smith, "Data Mining and Data Warehousing and OLAP", McGraw Hill Publication.

ARTIFICIAL INTELLIGENCE (Open Elective)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To learn the difference between optimal reasoning vs human like reasoning
2. To understand the notions of state space representation, exhaustive search, heuristic search along with the time and space complexities
3. To learn different knowledge representation techniques
4. To understand the applications of AI: namely Game Playing, Theorem Proving, Expert Systems, Machine Learning and Natural Language Processing

Course Outcomes:

1. Possess the ability to formulate an efficient problem space for a problem expressed in English.
2. Possess the ability to select a search algorithm for a problem and characterize its time and space complexities.
3. Possess the skill for representing knowledge using the appropriate technique
4. Possess the ability to apply AI techniques to solve problems of Game Playing, Expert Systems, Machine Learning and Natural Language Processing

SYLLABUS

UNIT – I:

Introduction, History, Intelligent Systems, Foundations of AI, Sub areas of AI, Applications. Problem Solving – State-Space Search and Control Strategies: Introduction, General Problem Solving, Characteristics of Problem, Exhaustive Searches, Heuristic Search Techniques, Iterative-Deepening A*, Constraint Satisfaction. Game Playing, Bounded Look-ahead Strategy and use of Evaluation Functions, Alpha-Beta Pruning.

UNIT – II:

Logic Concepts and Logic Programming: Introduction, Propositional Calculus, Propositional Logic, Natural Deduction System, Axiomatic System, Semantic Tableau System in Propositional Logic, Resolution Refutation in Propositional Logic, Predicate Logic, Logic Programming. Knowledge Representation: Introduction, Approaches to Knowledge Representation, Knowledge Representation using Semantic Network, Extended Semantic Networks for KR, Knowledge Representation using Frames.

UNIT – III:

Expert System and Applications: Introduction, Phases in Building Expert Systems, Expert System Architecture, Expert Systems Vs Traditional Systems, Truth Maintenance Systems, Application of Expert Systems, List of Shells and Tools. Uncertainty Measure – Probability Theory: Introduction, Probability Theory, Bayesian Belief Networks, Certainty Factor Theory, Dempster-Shafer Theory

UNIT-IV:

Machine-Learning Paradigms: Introduction. Machine Learning Systems. Supervised and Unsupervised Learning. Inductive Learning. Learning Decision Trees (Text Book 2), Deductive Learning. Clustering, Support Vector Machines. Artificial Neural Networks: Introduction, Artificial Neural Networks, Single-Layer FeedForward Networks, Multi-Layer Feed-Forward Networks, Radial-Basis Function Networks, Design Issues of Artificial Neural Networks, Recurrent Networks.

UNIT-V:

Advanced Knowledge Representation Techniques: Case Grammars, Semantic Web Natural Language Processing: Introduction, Sentence Analysis Phases, Grammars and Parsers, Types of Parsers, Semantic Analysis, Universal Networking Knowledge.

Text Books:

1. Saroj Kaushik. Artificial Intelligence. Cengage Learning. 2011
2. Russell, Norvig: Artificial intelligence, A Modern Approach, Pearson Education, Second Edition. 2004

Reference Books:

1. Rich, Knight, Nair: Artificial intelligence, Tata McGraw Hill, Third Edition 2009.
2. Introduction to Artificial Intelligence by Eugene Charniak, Pearson.
3. Introduction to Artificial Intelligence and expert systems Dan W.Patterson. PHI.
4. Artificial Intelligence by George Fluger Pearson fifth edition.

SEMANTIC WEB AND SOCIAL NETWORKS (Open Elective)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives

1. To learn Web Intelligence
2. To learn Knowledge Representation for the Semantic Web
3. To learn Ontology Engineering
4. To learn Semantic Web Applications, Services and Technology
5. To learn Social Network Analysis and semantic web

Course Outcomes

1. Ability to understand and knowledge representation for the semantic web.
2. Ability to create ontology.
3. Ability to build a blogs and social networks.

SYLLABUS

UNIT- I

Thinking and Intelligent Web Applications, The Information Age, The World Wide Web, Limitations of Today's Web, The Next Generation Web. Machine Intelligence, Artificial Intelligence, Ontology, Inference engines, Software Agents, Berners-Lee www, Semantic Road Map, Logic on the semantic Web

UNIT- II

Ontologies and their role in the semantic web, Ontologies Languages for the Semantic Web — Resource Description Framework(RDF) / RDF Schema, Ontology Web Language(OWL), UML/XML Schema, Ontology Engineering, Constructing Ontology, Ontology Development Tools, Ontology Methods, Ontology Sharing and Merging, Ontology Libraries and Ontology Mapping.

UNIT- III

Logic, Rule and Inference Engines. Semantic Web applications and services, Semantic Search, e-learning, Semantic Bioinformatics, Knowledge Base.

UNIT- IV

XML Based Web Services, Creating an OWL-S Ontology for Web Services, Semantic Search Technology, Web Search Agents and Semantic Methods, What is social Networks analysis, development of the social networks analysis, Electronic Sources for Network Analysis — Electronic Discussion networks.

UNIT- V

Blogs and Online Communities, Web Based Networks. Building Semantic Web Applications with social network features.

Text Books

1. Thinking on the Web – Berners Lee, Godel and Turing, Wiley interscience,2008.
2. Social Networks and the Semantic Web, Peter Mika, Springer,2007.

Reference Books

1. Semantic Web Technologies, Trends and Research in Ontology Based Systems, J.Davies, Audi Studer, Paul Warren, John Wiley & Sons.
2. Semantic Web and Semantic Web Services -Liyang Lu Chapman and Hall/CRC Publishers,(Taylor & Francis Group)
3. Information Sharing on the semantic Web – HeinerStuckenschmidt; Frank Van Harmelen, Springer Publications.
4. Programming the Semantic Web, T.Segaran, C.Evans, J.Taylor,O'Reilly, SPD.

DIGITAL SIGNAL PROCESSING (Open Elective)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To introduce discrete Fourier transform and its applications.
2. To introduce signal processing concepts in systems having more than one sampling frequency.
3. To teach the design of infinite and finite impulse response filters for filtering undesired signals.

Course Outcomes:

Upon completion of the course, students will be able to:

1. Perform frequency transforms for the signals.
2. Design IIR and FIR filters.
3. Finite word length effects in digital filters.

SYLLABUS

UNIT- I**SIGNALS AND SYSTEMS**

Basic elements of DSP – concepts of frequency in Analog and Digital Signals – sampling theorem – Discrete – time signals, systems – Analysis of discrete time LTI systems – Z transform–Convolution–Correlation.

UNIT- II**FREQUENCY TRANSFORMATIONS**

Introduction to DFT – Properties of DFT – Circular Convolution – Filtering methods based on DFT – FFT Algorithms – Decimation – in – time Algorithms, Decimation – in – frequency Algorithms – Use of FFT in Linear Filtering – DCT – Use and Application of DCT.

UNIT- III**IIR FILTER DESIGN**

Structures of IIR – Analog filter design – Discrete time IIR filter from analog filter – IIR filter design by Impulse Invariance, Bilinear transformation, Approximation of derivatives – (LPF, HPF, BPF, BRF) filter design using frequency translation.

UNIT- IV**FIR FILTER DESIGN**

Structures of FIR – Linear phase FIR filter – Fourier Series – Filter design using windowing techniques (Rectangular Window, Hamming Window, Hanning Window), Frequency sampling techniques.

UNIT -V

FINITE WORD LENGTH EFFECTS IN DIGITAL FILTERS

Binary fixed point and floating point number representations – Comparison – Quantization noise – truncation and rounding – quantization noise power- input quantization error- coefficient quantization error – limit cycle oscillations-dead band- Overflow error-signal scaling.

Text Book:

1. John G. Proakis and Dimitris G.Manolakis, “Digital Signal Processing – Principles, Algorithms & Applications”, Fourth Edition, Pearson Education, Prentice Hall, 2007.

Reference Books:

1. Emmanuel C.Ifeachor, and Barrie.W.Jervis, “Digital Signal Processing”, Second Edition, Pearson Education, Prentice Hall, 2002.
2. Sanjit K. Mitra, “Digital Signal Processing – A Computer Based Approach”, Third Edition, Tata Mc Graw Hill, 2007.
3. A.V.Oppenheim, R.W. Schafer and J.R. Buck, Discrete-Time Signal Processing, 8th Indian Reprint, Pearson, 2004.
4. Andreas Antoniou, “Digital Signal Processing”, Tata McGraw Hill, 2006.

ROBOTICS
(Open Elective)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To acquire the knowledge on advanced algebraic tools for the description of motion.
2. To develop the ability to analyze and design the motion for articulated systems.
3. To develop an ability to use software tools for analysis and design of robotic systems.

Course Outcomes:

1. Be able to use matrix algebra and Lie algebra for computing the kinematics of robots.
2. Be able to calculate the forward kinematics and inverse kinematics of serial and parallel robots.
3. Be able to calculate the Jacobian for serial and parallel robot.
4. Be able to do the path planning for a robotic system.
5. Be proficient in the use of Maple or Matlab for the simulation of robots

SYLLABUS

UNIT-I**Introduction to Robotics:**

Classification components, Characteristics, Applications

Robot Kinematics: Position analysis-I

Robots as Mechanisms, Matrix Representation, Homogeneous Transformation matrices
Representation of pure translation, Representation of pure rotation about an axis.

UNIT-II**Robot Kinematics: Position analysis-II:**

Representation of combined Transformations, Transformations relative to the rotating inverse of Transformation matrices, forward and actuating Systems.

Fundamentals of actuating systems:

Characteristics of actuating systems, comparison of actuating systems, Hydraulic devices, Pneumatic Devices, Magnetostrictive actuating Systems.

UNIT-III:**Electric Actuators:**

Introduction to electronic actuators electronic motors, control of electric motors.

UNIT-IV**Sensors-I:**

Characteristics, position sensors, velocity and acceleration sensors, force and pressure, Torque and micro switches, light and Infrared sensors

UNIT-V

Sensors-II:

Touch and Tactile sensors, Proximity and range Finders, sniff sensors, vision systems, voice recognizes, voice Synthesizer, remote center compliance Device

Text Books:

1. Saeed b.Niku, Introduction to Robotics analysis, application, Person Education Asia 2001

Reference Books:

1. R.K Mittal and I J Nagrath, robotics and control TMH 2003
2. Computational Intelligence, Davis Poole, Alan mackwath randy coehel, Oxford University press 1998.

IMAGE PROCESSING (Open Elective)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To explain fundamentals of Image processing concepts.
2. To provide mathematical foundation of image enhancement , image compression and image segmentation.
3. To explain the students about Morphology and its applications in image processing.

Course Outcomes:

1. Ability to develop algorithms for fundamental concepts in Image processing.
2. Ability to perform image enhancement , image compression and image segmentation using various methods.
3. Ability to implement Image transformation techniques.

SYLLABUS

UNIT-I

Fundamentals of Image Processing: Image Acquisition, Image Model, Sampling, Quantization, Relationship Between Pixels, Distance Measures, Connectivity , Image Geometry, Photographic Film. Histogram: Definition, Decision Of Contrast Basing On Histogram, Operations Basing on Histograms Like Image Stretching, Image Sliding, Image Classification. Definition and Algorithm of Histogram Equalization.

UNIT-II

Image Enhancement in Spatial Domain : Arithmetic and Logical Operations, Pixel or Point Operations, Size Operations; Smoothing Filters-Mean, Median, Mode Filters – Comparative Study

UNIT-III

Edge enhancement in spatial domain: Edge enhancement filters, Directorial Filters, Sobel, Laplacian, Robert, KIRSCH Homogeneity & DIFF filters, PREWITT Filter, Contrast based edge enhancement techniques, Comparative study, Low pass filters, High pass filters, Sharpening filters, Comparative study, Color fundamentals and color model

UNIT-IV

Image Compression: Run Length Encoding, modified run length encoding, Contour Coding, Huffman Code, Compression Due to Change in Domain, Compression Due to Quantization Compression at the Time of Image Transmission. Brief Discussion on:- Image Compression Standards.

UNIT-V

Image Segmentation and Morphology: Definition of segmentation, Characteristics of Segmentation, Detection of Discontinuities, Thresholding. Pixel Based Segmentation Method. Region Based Segmentation Methods, Segmentation by Pixel Aggregation, Segmentation by Sub Region Aggregation, Histogram Based Segmentation

Text Book:

1. Digital Image Processing, Rafael C. Gonzalez And Richard E. Woods, Addison Wesley

Reference Books:

1. Fundamentals Of Electronic Image Processing By Arthyr – R – Weeks, Jr.(PHI).
2. Image Processing, Analysis, And Machine Vision By Milan SonkaVaclanHalavac Roger Boyle, Vikas Publishing House.
3. Digital Image Processing, S. Jayaraman, S. Esakkirajan& T. Veera Kumar, TMH
Fundamentals of Digital Image Processing, Chris Solomon, Tobi Breckon, Wiley-Blackwell

OPERATIONS RESEARCH (Open Elective)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives: The main objectives of student is to

1. Identify and develop operational research models from the verbal description of the real system.
2. Understand the apply mathematical tools that are needed to solve optimization problems.
3. Identify Special cases in Optimal Problems and solve them.

Course Outcomes:

After completion of the course, the student will be able to:

1. Understand the Methodology of Operations Research.
2. Formulate and Model the Linear Programming Problems, Transportation and Assignment Problems,
3. solving methods for LPP, duality, and sensitivity analysis.
4. Check for degeneracy and other special cases in above models
5. Construct Network flows and solve them under certainty/uncertainty.
6. Model Inventory control for EOQ.
7. Formulate Competitive models using Game theory and solve them for Optimal Decisions.

SYLLABUS

UNIT-I:

Introduction to Operations Research: Basics definition, scope, objectives, phases, models and limitations of Operations Research. Linear Programming Problem – Formulation of LPP, Graphical solution of LPP. Simplex Method, Artificial variables, big-M method, two-phase method, degeneracy and unbound solutions, Special cases: Unbounded, In-feasible, Alternative solutions.

UNIT-II:

Dual Simplex Method: Algorithm, Primal-Dual Problem: Relation between Primal-Dual problems, Solution for both Primal & Dual problem using Simplex methods.

Transportation Problems (TP): Standard Mathematical form of TP, Balanced Un-Balanced TP, Basic Feasible Solution: NW Corner rule, Least cost method, Vogel's approximation method. TP Algorithm: Optimal Feasible Solution (both Max and Min cases), Degeneracy in TP.

UNIT-III

Assignment Problem(AP): Standard Mathematical form of AP, Balanced Un Balanced AP, Assignment with only Zeros, AP with and without restrictions, Optimal Feasible solution Hungarian method (both for Max & Min cases).

Travelling Salesman Problem(TSP): Rules for TSP, Optimal path using Hungarian method.

UNIT-IV

Network Models: Rules for construction of Network, Time estimates, Forward pass and Backward pass computation. Critical Path Method (CPM), Project Evaluation and review Technique (PERT).

Inventory Models: Inventory, Costs in Inventory control, Economic Order Quantity (EOQ), Deterministic models: Static EOQ models, Dynamic EOQ Models.

UNIT-V

Game Theory: Game, Types of Games, Mathematical model of Game, value of Game, Two person Zero Sum Games: with Saddle point Games, Max(Min) approach, Dominance Principle, 2x2 Mixed strategy Games: without Saddle point Games: Arithmetic method, Graphical method.

Text books:

1. Operations Research / R.Pannarselvam/ PHI Publications
2. Operations Research –Theory & publications / S.D.Sharma-Kedarnath/McMillan publishers India Ltd.

Reference Books:

1. Introduction to O.R/Hiller &Libermann/TMH.
2. Operation Research /J.K.Sharma/MacMilan Publ.
3. Operations Research / Wagner/ PHI Publications.

WEB TECHNOLOGIES LAB

Lab : 3 Periods

Int.Marks : 50

Exam : 3 Hrs.

Ext. Marks : 50

Credits : 2

Course Objectives:

1. To acquire knowledge of XHTML, Java Script and XML to develop web applications
2. Ability to develop dynamic web content using Java Servlets and JSP
3. To understand JDBC connections and Java Mail API
4. To understand the design and development process of a complete web application

Course Outcomes:

Students will be able to develop static web sites using XHTML and Java Scripts

1. To implement XML and XSLT for web applications
2. Develop Dynamic web content using Java Servlets and JSP
3. To develop JDBC connections and implement a complete Dynamic webApplication

LIST OF PROGRAMS

1. Design the following static web pages required for an online book store web site.

1) HOME PAGE:

The static home page must contain three **frames**.

Top frame: Logo and the college name and links to Home page, Login page, Registration page, Catalogue page and Cart page (the description of these pages will be given below).

Left frame: At least four links for navigation, which will display the catalogue of respective links.

For e.g.: When you click the link “**MCA**” the catalogue for MCA Books should be displayed in the Right frame.

Right frame: The *pages to the links in the left frame must be loaded here*. Initially this page contains description of the web site.

Logo	Web Site Name			
Home	Login	Registration	Catalogue	Cart
mca mba BCA	Description of the Web Site			

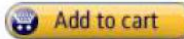
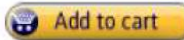
2) LOGIN PAGE

Logo	Web Site Name			
Home	Login	Registration	Catalogue	Cart
MCA MBA BCA	Login : 11a51f0003 Password: ***** Submit Reset			

3) CATALOGUE PAGE:

The catalogue page should contain the details of all the books available in the web site in a table. The details should contain the following:

1. Snap shot of Cover Page.
2. Author Name.
3. Publisher.
4. Price.
5. Add to cart button.

Logo	Web Site Name			
Home	Login	Registration	Catalogue	Cart
MCA		Book : XML Bible Author : Winston Publication : Wiely	\$ 40.5	
MBA				
BCA		Book : AI Author : S.Russel Publication : Princeton hall	\$ 63	
		Book : Java 2 Author : Watson Publication : BPB publications	\$ 35.5	
		Book : HTML in 24 hours Author : Sam Peter Publication : Sam	\$ 50	

4. REGISTRATION PAGE:

Create a “registration form” with the following fields

- 1) Name (Text field)
- 2) Password (password field)
- 3) E-mail id (text field)
- 4) Phone number (text field)
- 5) Sex (radio button)
- 6) Date of birth (3 select boxes)
- 7) Languages known (check boxes – English, Telugu, Hindi, Tamil)
- 8) Address (text area)

5. DESIGN A WEB PAGE USING CSS (Cascading Style Sheets) which includes the following:

1) Use different font, styles:

In the style definition you define how each selector should work (font, color etc.).

Then, in the body of your pages, you refer to these selectors to activate the styles

6. WRITE AN XML file which will display the Book information which includes the following:

1) Title of the book

2) Author Name

3) ISBN number

4) Publisher name

5) Edition

6) Price

Write a Document Type Definition (DTD) to validate the above XML file.

7. Write Ruby program reads a number and calculates the factorial value of it and prints the Same.

8. Write a Ruby program which counts number of lines in a text files using its regular Expressions facility.

9. Write a Ruby program that uses iterator to find out the length of a string.

10. Write simple Ruby programs that uses arrays in Ruby.

11. Write programs which uses associative arrays concept of Ruby.

12. Write Ruby program which uses Math module to find area of a triangle.

13. Write Ruby program which uses tk module to display a window

14. Define complex class in Ruby and do write methods to carry operations on complex objects.

15. Example PHP program for cotactus page.

16. User Authentication:

Assume four users user1, user2, user3 and user4 having the passwords pwd1, pwd2, pwd3 and pwd4 respectively. Write a PHP for doing the following.

1. Create a Cookie and add these four user id's and passwords to this Cookie.

2. Read the user id and passwords entered in the Login form (week1) and authenticate with the values (user id and passwords) available in the cookies.

If he is a valid user (i.e., user-name and password match) you should welcome him by name (user-name) else you should display "You are not an authenticated user ".

Use init-parameters to do this.

17. Example PHP program for registering users of a website and login.

18. Install a database (Mysql or Oracle).

Create a table which should contain at least the following fields: name, password, email-id, phone number (these should hold the data from the registration form).

Write a PHP program to connect to that database and extract data from the tables and display them. Experiment with various SQL queries.

Insert the details of the users who register with the web site, whenever a new user clicks the submit button in the registration page (week2).

19. Write a PHP which does the following job:

Insert the details of the 3 or 4 users who register with the web site (week9) by using registration form. Authenticate the user when he submits the login form using the user name and password from the database (similar to week8 instead of cookies).

20. **HTTP** is a stateless protocol. Session is required to maintain the state.

The user may add some items to cart from the catalog page. He can check the cart page for the selected items. He may visit the catalogue again and select some more items. Here our interest is the selected items should be added to the old cart rather than a new cart. Multiple users can do the same thing at a time(i.e., from different systems in the LAN using the ip-address instead of local host). This can be achieved through the use of sessions. Every user will have his own session which will be created after his successful login to the website. When the user logs out his session should get invalidated (by using the method session. Invalidate ()).Modify your catalogue and cart PHP pages to achieve the above mentioned functionality using sessions.

Reference Books:

1. Programming the World Wide Web, Robert W Sebesta, 7ed, Pearson.
2. Web Technologies, Uttam K Roy, Oxford
3. The Web Warrior Guide to Web Programming, Bai, Ekedahl, Farrelll, Gosselin, Zak,Karparhi, MacIntyre, Morrissey, Cengage.
4. Ruby on Rails Up and Running, Lightning fast Web development, Bruce Tate, Curt Hibbs, Oreilly (2006).
5. Web Technologies, HTML< JavaScript, PHP, Java, JSP, XML and AJAX, Black book, Dream Tech.
6. An Introduction to Web Design, Programming, Paul S Wang, Sanda S Katila, CengageLearning.
7. <http://www.upriss.org.uk/perl/PerlCourse.html>

SOFTWARE ENGINEERING AND MINI PROJECT LAB**Lab : 3 Periods****Int.Marks : 50****Exam : 3 Hrs.****Ext. Marks : 50****Credits : 2****Course Objectives:**

The aim of this course is,

1. To design and development complex software solutions using the state of art software engineering techniques.
2. To work knowledge of UML, source control and project management.
3. To have deep knowledge of the technologies they used for implementing their project.
4. And know how to test and document software.

Course Outcomes:

1. Students will be **Construct, Design and implement** complex software solutions.
2. Students will be able to test and document the software.
3. Students will be capable of working as part of a software team and develop significant projects under a tight deadline.
4. Students will be able **apply** the deep knowledge of the technologies they used for implementing their project.
5. Students will be able to **assess** the changes required for customization in project management.

SYLLABUS

The purpose of the Software Engineering Lab course is to familiarize the students with modern software engineering methods and tools, Rational Products. The course is realized as a project-like assignment that can, in principle, by a team of three/four students working full time. Typically the assignments have been completed during the semester by each project team.

The goal of the Software Engineering Project is to have a walk through from the requirements, design to implementing and testing. An emphasis is put on proper documentation. Term projects are projects that a group student might take through from initial specification to implementation by giving equal importance to both design and implementation.

Cycle I: Practicing UML diagrams using IBM Rational Rose. 6*3 periods= 18periods

Before developing a mini-project, in this cycle, the student is acquainted with different UML diagrams using Rational Rose. The experiments should include drawing UML diagrams listed below for two demo/example applications assigned by the lab Instructor. The input for the following experiments is problem statement for any two demo projects supplied by the instructor.

1. Introduction to Rational Rose and Practicing the following diagrams

- a. Activity diagrams for the overall business process of the projects
- b. Use-case diagram for the demo projects along with Use-case descriptions and sub-diagrams for Use-cases.

2. Interaction diagrams- Sequence diagrams and Collaboration diagrams for different scenarios of the systems with all features like actors, objects and interactions.

3. Class diagram- Class diagrams including the features like classes, relationships, attributes and methods along with their visibilities.

4. Activity diagrams, State chart and other diagrams - Activity diagrams including the features like fork join and swim lanes. State diagrams including composite states and transitions. Component diagrams, Package diagrams and Deployment diagrams.

Cycle II: Mini-Project 8*3 periods= 24periods

The project deliverables include

- Problem statement
- Requirements Analysis
- Design

A Software Design Description and a System Design.

- Detailed Project Design Architecture.
- Implementation of Design Patterns in Class Diagrams

A Test specification.

- Testing strategies
- Sample test case design

Reference Books :

1. Project-based software engineering: An Object-oriented approach, Evelyn Stiller, Cathie LeBlanc, and Pearson Education.
2. Visual Modeling with Rational Rose 2002 and UML, Terry Quatrini, Pearson Education.

EMPLOYABILITY SKILLS
(Common to all Branches)

Tutorial : 3 Periods (VA-2+QA-1)
Exam : 3 Hrs.

Int.Marks : 30
Ext.Marks : 70
Credits : 1

Part-A: Verbal Aptitude and Soft Skills-II

Course objectives:

1. To expose the students to bettering sentence expressions and also forming equivalents.
2. To instill reading and analyzing techniques for better comprehension of written discourses.
3. To create awareness among the students on the various aspects of writing, organizing data, preparing reports, and applying their writing skills in their professional career.
4. To inculcate conversational skills, nuances required when interacting in different situations.
5. To build/refine the professional qualities/skills necessary for a productive career and to instill confidence through attitude building.

Course Outcomes:

The students will be able to

1. Construct coherent, cohesive and unambiguous verbal expressions in both oral and written discourses.
2. Analyze the given data/text and find out the correct responses to the questions asked based on the reading exercises; identify relationships or patterns within groups of words or sentences
3. Write paragraphs on a particular topic, essays (issues and arguments), e mails, summaries of group discussions, reports, make notes, statement of purpose(for admission into foreign universities), letters of recommendation(for professional and educational purposes).
4. Converse with ease during interactive sessions/seminars in their classrooms, compete in literary activities like elocution, debates etc., raise doubts in class, participate in JAM sessions/versant tests with confidence and convey oral information in a professional manner.
5. Participate in group discussions/group activities, exhibit team spirit, use language effectively according to the situation, respond to their interviewer/employer with a positive mind, tailor make answers to the questions asked during their technical/personal interviews, exhibit skills required for the different kinds of interviews (stress, technical, HR) that they would face during the course of their recruitment process.

SYLLABUS

UNIT -I (VA)

Sentence Improvement (finding a substitute given under the sentence as alternatives), Sentence equivalence (completing a sentence by choosing two words either of which will fit in the blank), cloze test (reading the written discourse carefully and choosing the correct options from the alternatives and filling in the blanks), summarizing and paraphrasing.

UNIT- II (VA)

Types of passages (to understand the nature of the passage), types of questions (with emphasis on inferential and analytical questions), style and tone (to comprehend the author's intention of writing a passage), strategies for quick reading(importance given to skimming, scanning), summarizing ,reading between the lines, reading beyond the lines, techniques for answering questions related to vocabulary (with emphasis on the context), supplying suitable titles to the passage, identifying the theme and central idea of the given passages.

UNIT- III (VA)

Punctuation, discourse markers, general Essay writing, writing Issues and Arguments(with emphasis on creativity and analysis of a topic), paragraph writing, preparing reports, framing a ‘Statement of purpose’, ‘Letters of Recommendation’, business letter writing, email writing, writing letters of complaints/responses. picture perception and description, book review.

UNIT-IV (VA)

Just a minute sessions, reading news clippings in the class, extempore speech, telephone etiquette, making requests/suggestions/complaints, elocutions, debates, describing incidents and developing positive non verbal communication, story narration, product description.

UNIT-V (SS)

Employability Skills – Significance — Transition from education to workplace - Preparing a road map for employment – Getting ready for the selection process, Awareness about Industry / Companies – Importance of researching your prospective workplace - Knowing about Selection process - Resume Preparation: Common resume blunders – tips, Resume Review, Group Discussion: Essential guidelines – Personal Interview: Reasons for Rejection and Selection.

Reading/ Listening material:

1. Guide to IELTS, Cambridge University Press
2. Barron’s GRE guide.
3. Newspapers like ‘The Hindu’, ‘Times of India’, ‘Economic Times’.
4. Magazines like Frontline, Outlook and Business India.
5. News channels NDTV, National News, CNN

Text Books:

1. Objective English and Verbal Reasoning by R S Agarwal.
2. Communication Skills by Sanjay Kumar and PushpaLatha, Second Edition, OUP.
3. Business Correspondence and Report Writing – A Practical Approach to Business and Technical Communication by R C Sharma and Krishna Mohan.
4. Soft Skills & Employability Skills by Samina Pillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
5. Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi

Reference Books:

1. Oxford Guide to Effective Writing and Speaking by John Seely.
2. Collins Cobuild English Grammar by Collins
3. The Art of Public Speaking by Dale Carnegie
4. The Leader in You by Dale Carnegie
5. Emotional Intelligence by Daniel Golman
6. Stay Hungry Stay Foolish by Rashmi Bansal

Part-B: Quantitative Aptitude-II

Course objectives:

The objective of introducing quantitative aptitude-II is:

1. To refine concepts related to quantitative aptitude. – SOLVING PROBLEMS OF DI and accurate values using averages, percentages.
2. To inculcate logical thinking by exposing the students to puzzles and reasoning related questions.
3. To familiarize the students with finding out accurate date and time related problems.
4. To enable the students solve the puzzles using logical thinking.
5. To expose the students to various problems based on geometry and mensuration.

Course Outcomes:

1. The students will be able to perform well in calculating different types of data interpretation problems.
2. The students will perform efficaciously on analytical and logical problems using various methods.
3. Students will find the angle measurements of clock problems with the knowledge of calendars and clock.
4. The students will skillfully solve the puzzle problems like arrangement of different positions.
5. The students will become good at solving the problems of lines, triangulars, volume of cone, cylinder and so on.

SYLLABUS

UNIT I: Averages, mixtures and allegations, Data interpretation Understanding of AM,GM,HM-Problems on averages, Problems on mixtures standard method. Importance of data interpretation: Problems of data interpretation using line graphs, Problems of data interpretation using bar graphs, Problems of data interpretation using pie charts, Problems of data interpretation using others.

UNIT II:Puzzle test, blood Relations, permutations, Combinations and probability Importance of puzzle test, Various Blood relations-Notation to relations and sex making of family Tree diagram, Problems related to blood relations, Concept of permutation and combination, Problems on permutation, Problems on combinations, Problems involving both permutations and combinations, Concept of probability-Problems on coins, Problems on dice, Problems on cards, Problems on years.

UNIT III:Periods,Clocks, Calendars, Cubes and cuboids Deriving the formula to find the angle between hands for the given time, finding the time if the angle is known, Faulty clocks, History of calendar-Define year, leap year, Finding the day for the given date, Formula and method to find the day for the given date in easy way, Cuts to cubes, Colors to cubes, Cuts to cuboids, Colors to cuboids.

UNIT IV: Puzzles Selective puzzles from previous year placement papers, sitting arrangement, problems- circular arrangement, linear arrangement, different puzzles.

UNIT V: Geometry and Mensuration Introduction and use of geometry-Lines, Line segments, Types of angles, Intersecting lines, Parallel lines, Complementary angles, supplementary angles, Types of triangles-Problems on triangles, Types of quadrilaterals-Problems on quadrilaterals, Congruent triangles and properties, Similar triangles and its applications, Understanding about circles-Theorems on circles, Problems on circles, Tangents and circles, Importance of mensuration-Introduction of cylinder, cone, sphere, hemi sphere.

Text Books:

1. Quantitative aptitude by RS Agarwal
2. Verbal and non verbal reasoning by RS Agarwal.
3. Puzzles to puzzle you by shakunataladevi
4. More puzzles by shakunataladevi
5. Puzzles by George summers.

Reference Books:

1. Barron's by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2. Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from _IMS, Career Launcher and Time' institutes for competitive exams.
4. Books for cat by arunsharma
5. Elementary and Higher algebra by HS Hall and SR knight.

COMPETITIVE CODING-II

Lab : 3 Periods
Exam : 3 Hrs.

Int.Marks : 50
Ext. Marks : 50
Credits : 1

Course Objectives:

1. To introduce PHP language for server side scripting
2. To introduce XML and processing of XML Data with Java
3. To introduce Server side programming with Java Servlets and JSP
4. To introduce Client side scripting with Javascript and AJAX.

Course Outcomes: At the end of the course, a student should be able to:

1. Gain knowledge of client side scripting, validation of forms and AJAX programming
2. Have understanding of server side scripting with PHP language
3. Have understanding of what is XML and how to parse and use XML Data with Java
4. To introduce Server side programming with Java Servlets and JSP

SYLLABUS**UNIT- I : Introduction to PHP:**

Declaring variables, data types, arrays, strings, operators, expressions, control structures, functions, Reading data from web form controls like text boxes, radio buttons, lists etc., Handling File Uploads. Connecting to database (MySQL as reference), executing simple queries, handling results, Handling sessions and cookies.

File Handling in PHP:

File operations like opening, closing, reading, writing, appending, deleting etc. on text and binary files, listing directories.

UNIT- II XML:

Introduction to XML, Defining XML tags, their attributes and values, Document Type Definition, XML Schemes, Document Object Model, XHTML Parsing XML Data– DOM and SAX Parsers in java.

UNIT- III Servlets Programming:

Life cycle of a Servlet, deploying a servlet, The Servlet API, Reading Servlet parameters, Reading Initialization parameters, Handling Http Request & Responses, Using Cookies and Sessions, connecting to a database using JDBC.

UNIT- IV JSP Programming:

The Anatomy of a JSP Page, JSP Processing, Declarations, Directives, Expressions, Code Snippets, implicit objects, Using Beans in JSP Pages, Using Cookies and session for session tracking, connecting to database in JSP.

UNIT- V Client side Scripting:

Introduction to JavaScript: JavaScript language – declaring variables, scope of variables, functions. event handlers (onclick, onsubmit etc.), Document Object Model, Form validation. Simple AJAX application.

Text Books:

1. Web Technologies, Uttam K Roy, Oxford University Press
2. The Complete Reference PHP — Steven Holzner, Tata McGraw-Hill

Reference Books:

1. Web Programming, building internet applications, Chris Bates 2nd edition, Wiley Dreamtech
2. Java Server Pages —Hans Bergsten, SPD O'Reilly,
3. Java Script, D.Flanagan
4. Beginning Web Programming-Jon Duckett WROX.
5. Programming world wide web, R.W.Sebesta, Fourth Edition, Pearson.
6. Internet and World Wide Web — How to program. Dietel and Nieto, Pearson.

IPR & PATENTS
(Common to CSE, ECE & IT)

Tutorial : 2 Periods

Credits : 0

Course Objectives:

1. To introduce the idea of tangible and intangible property and its protection.
2. To familiarize with the frameworks for protection of intellectual property.
3. To layout the procedures to claim rights over intellectual property.

Course Outcomes:

After successful completion of the course, the student shall be able to

1. Identify various types of intangible property that an engineering professional could generate in the course of his career.
2. Distinguish between various types of protection granted to Intellectual Property such as Patents, Copy Rights, Trademarks etc.,
3. List the steps involved in getting protection over various types of intellectual property and maintaining them.
4. Take precautions in writing scientific and technical reports without plagiarism.
5. Help micro, small and medium entrepreneurs in protecting their IP and respecting others IP as part of their business processes.

SYLLABUS

UNIT I

Intellectual Property Law: Basics - Types of Intellectual Property - Innovations and Inventions - Trade related Intellectual Property Rights – Agencies Responsible for Intellectual Property Registration – Infringement - Compliance and Liability Issues

UNIT II

Principles of Copyright – Subject Matters of Copyright – Rights Afforded by Copyright Law – Copyright Ownership–Copyright Formalities and Registration – Limitations – Infringement of Copyright - Plagiarism and difference between Copyright infringement and Plagiarism

UNIT III

Introduction to Trade Mark – Trade Mark Registration Process – Post registration procedures – Trade Mark maintenance– Infringement – Dilution of Ownership of Trade Mark – Likelihood of confusion – Trade Mark claims – Trade Marks Litigation – International Trade Mark Law

UNIT IV

Introduction to Patent Law – Rights and Limitations – Rights under Patent Law – Patent Requirements – Ownership and Transfer – Patent Application Process and Granting of Patent – Patent Infringement and Litigation – International Patent Law – Double Patenting

UNIT V

Introduction to Trade Secrets – Maintaining Trade Secret – Physical Security – Employee Access Limitation – Employee Confidentiality Agreement – Trade Secret Law – Unfair Competition – Trade Secret Litigation – Breach of Contract – Applying State Law.

Text Books:

1. Kompal Bansal & Parikshit Bansal "Fundamentals of Intellectual Property for Engineers", BS Publications
2. Prabhuddha Ganguli: "Intellectual Property Rights" Tata McGraw –Hill, New Delhi
3. R. Radha Krishnan, S. Balasubramanian: "Intellectual Property Rights: Text and Cases", Excel Books, New Delhi.

Reference Books:

1. Deborah E. Bouchoux: "Intellectual Property". Cengage learning, New Delhi
2. Richard Stim: "Intellectual Property", Cengage Learning, New Delhi.